Introduction to Random Signals and Noise

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Contents

Preface **xi**

1 Introduction **1**
1.1 Random Signals and Noise 1
1.2 Modelling 1
1.3 The Concept of a Stochastic Process 2
1.3.1 Continuous Stochastic Processes 4
1.3.2 Discrete-Time Processes (Continuous Random Sequences) 5
1.3.3 Discrete Stochastic Processes 6
1.3.4 Discrete Random Sequences 7
1.3.5 Deterministic Function versus Stochastic Process 8
1.4 Summary 8

2 Stochastic Processes **9**
2.1 Stationary Processes 9
2.1.1 Cumulative Distribution Function and Probability Density Function 9
2.1.2 First-Order Stationary Processes 10
2.1.3 Second-Order Stationary Processes 11
2.1.4 *N*th-Order Stationary Processes 11
2.2 Correlation Functions 11
2.2.1 The Autocorrelation Function, Wide-Sense Stationary Processes and Ergodic Processes 11
2.2.2 Cyclo-Stationary Processes 16
2.2.3 The Cross-Correlation Function 19
2.2.4 Measuring Correlation Functions 24
2.2.5 Covariance Functions 26
2.2.6 Physical Interpretation of Process Parameters 27
2.3 Gaussian Processes 27
2.4 Complex Processes 30
2.5 Discrete-Time Processes 31
2.5.1 Mean, Correlation Functions and Covariance Functions 31
2.6 Summary 33
2.7 Problems 34

3 Spectra of Stochastic Processes **39**
3.1 The Power Spectrum 39

3.2 The Bandwidth of a Stochastic Process 43
3.3 The Cross-Power Spectrum 45
3.4 Modulation of Stochastic Processes 47
3.4.1 Modulation by a Random Carrier 49
3.5 Sampling and Analogue-To-Digital Conversion 50
3.5.1 Sampling Theorems 51
3.5.2 A/D Conversion 54
3.6 Spectrum of Discrete-Time Processes 57
3.7 Summary 58
3.8 Problems 59

4. Linear Filtering of Stochastic Processes 65
4.1 Basics of Linear Time-Invariant Filtering 65
4.2 Time Domain Description of Filtering of Stochastic Processes 68
4.2.1 The Mean Value of the Filter Output 68
4.2.2 The Autocorrelations Function of the Output 69
4.2.3 Cross-Correlation of the Input and Output 70
4.3 Spectra of the Filter Output 71
4.4 Noise Bandwidth 74
4.4.1 Band-Limited Processes and Systems 74
4.4.2 Equivalent Noise Bandwidth 75
4.5 Spectrum of a Random Data Signal 77
4.6 Principles of Discrete-Time Signals and Systems 82
4.6.1 The Discrete Fourier Transform 82
4.6.2 The z-Transform 86
4.7 Discrete-Time Filtering of Random Sequences 90
4.7.1 Time Domain Description of the Filtering 90
4.7.2 Frequency Domain Description of the Filtering 91
4.8 Summary 93
4.9 Problems 94

5 Bandpass Processes 101
5.1 Description of Deterministic Bandpass Signals 101
5.2 Quadrature Components of Bandpass Processes 106
5.3 Probability Density Functions of the Envelope and Phase of Bandpass Noise 111
5.4 Measurement of Spectra 115
5.4.1 The Spectrum Analyser 115
5.4.2 Measurement of the Quadrature Components 118
5.5 Sampling of Bandpass Processes 119
5.5.1 Conversion to Baseband 119
5.5.2 Direct Sampling 119
5.6 Summary 121
5.7 Problems 121

6 Noise in Networks and Systems 129
6.1 White and Coloured Noise 129
6.2 Thermal Noise in Resistors 130
6.3 Thermal Noise in Passive Networks 131

6.4 System Noise 137
6.4.1 Noise in Amplifiers 138
6.4.2 The Noise Figure 140
6.4.3 Noise in Cascaded systems 142
6.5 Summary 146
6.6 Problems 146

7 Detection and Optimal Filtering 153
7.1 Signal Detection 154
7.1.1 Binary Signals in Noise 154
7.1.2 Detection of Binary Signals in White Gaussian Noise 158
7.1.3 Detection of *M*-ary Signals in White Gaussian Noise 161
7.1.4 Decision Rules 165
7.2 Filters that Maximize the Signal-to-Noise Ratio 165
7.3 The Correlation Receiver 171
7.4 Filters that Minimize the Mean-Squared Error 175
7.4.1 The Wiener Filter Problem 175
7.4.2 Smoothing 176
7.4.3 Prediction 179
7.4.4 Discrete-Time Wiener Filtering 183
7.5 Summary 185
7.6 Problems 185

8 Poisson Processes and Shot Noise 193
8.1 Introduction 193
8.2 The Poisson Distribution 194
8.2.1 The Characteristic Function 194
8.2.2 Cumulants 196
8.2.3 Interarrival Time and Waiting Time 197
8.3 The Homogeneous Poisson Process 198
8.3.1 Filtering of Homogeneous Poisson Processes and Shot Noise 199
8.4 Inhomogeneous Poisson Processes 204
8.5 The Random-Pulse Process 205
8.6 Summary 207
8.7 Problems 208

References 211

Further Reading 213

Appendices 215

A. Representation of Signals in a Signal Space 215
A.1 Linear Vector Spaces 215
A.2 The Signal Space Concept 216
A.3 Gram–Schmidt Orthogonalization 218
A.4 The Representation of Noise in Signal Space 219
A.4.1 Relevant and Irrelevant Noise 221
A.5 Signal Constellations 222
A.5.1 Binary Antipodal Signals 222

A.5.2 Binary Orthogonal Signals 223
A.5.3 Multiphase Signals 224
A.5.4 Multiamplitude Signals 224
A.5.5 QAM Signals 225
A.5.6 *M*-ary Orthogonal Signals 225
A.5.7 Biorthogronal Signals 225
A.5.8 Simplex Signals 226
A.6 Problems 227

B. Attenuation, Phase Shift and Decibels 229

C. Mathematical Relations 231
C.1 Trigonometric Relations 231
C.2 Derivatives 232
C.2.1 Rules fn Differentiation 232
C.2.1 Chain Rule 232
C.2.3 Stationary Points 233
C.3 Indefinite Integrals 233
C.3.1 Basic Integrals 233
C.3.2 Integration by Parts 234
C.3.3 Rational Algebraic Functions 234
C.3.4 Trigonometric Functions 235
C.3.5 Exponential Functions 236
C.4 Definite Integrals 236
C.5 Series 237
C.6 Logarithms 238

D. Summary of Probability Theory 239

E. Definition of a Few Special Functions 241

F. The Q(.) and erfc Function 243

G. Fourier Transforms 245

H. Mathematical and Physical Constants 247

Index 249

Preface

Random signals and noise are present in several engineering systems. Practical signals seldom lend themselves to a nice mathematical deterministic description. It is partly a consequence of the chaos that is produced by nature. However, chaos can also be man-made, and one can even state that chaos is a *conditio sine qua non* to be able to transfer information. Signals that are not random in time but predictable contain no information, as was concluded by Shannon in his famous communication theory.

To deal with this randomness we have to nevertheless use a characterization in deterministic terms; i.e. we employ probability theory to determine characteristic descriptions such as mean, variance, correlation, etc. Whenever chaotic behaviour is time-dependent, as is often the case for random signals, the time parameter comes into the picture. This calls for an extension of probability theory, which is the theory of stochastic processes and random signals. With the involvement of time, the phenomenon of frequency also enters the picture. Consequently, random signal theory leans heavily on both probability and Fourier theories. Combining these subjects leads to a powerful tool for dealing with random signals and noise.

In practice, random signals may be encountered as a desired signal such as video or audio, or it may be an unwanted signal that is unintentionally added to a desired (information bearing) signal thereby disturbing the latter. One often calls this unwanted signal noise. Sometimes the undesired signal carries unwanted information and does not behave like noise in the classical sense. In such cases it is termed as interference. While it is usually difficult to distinguish (at least visually) between the desired signal and noise (or interference), by means of appropriate signal processing such a distinction can be made. For example, optimum receivers are able to enhance desired signals while suppressing noise and interference at the same time. In all cases a description of the signals is required in order to be able to analyse their impact on the performance of the system under consideration. In communication theory this situation often occurs. The random time-varying character of signals is usually difficult to describe, and this is also true for associated signal processing activities such as filtering. Nevertheless, there is a need to characterize these signals using a few deterministic parameters that allow a system user to assess system performance.

This book deals with stochastic processes and noise at an introductory level. Probability theory is assumed to be known. The same holds for mathematical background in differential and integral calculus, Fourier analysis and some basic knowledge of network and linear system theory. It introduces the subject in the form of theorems, properties and examples. Theorems and important properties are placed in frames, so that the student can easily

summarize them. Examples are mostly taken from practical applications. Each chapter concludes with a summary and a set of problems that serves as practice material. The book is well suited for dealing with the subject at undergraduate level. A few subjects can be skipped if they do not fit into a certain curriculum. Besides, the book can also serve as a reference for the experienced engineer in his daily work.

In Chapter 1 the subject is introduced and the concept of a stochastic process is presented. Different types of processes are defined and elucidated by means of simple examples.

Chapter 2 gives the basic definitions of probability density functions and includes the time dependence of these functions. The approach is based on the 'ensemble' concept. Concepts such as stationarity, ergodicity, correlation functions and covariance functions are introduced. It is indicated how correlation functions can be measured. Physical interpretation of several stochastic concepts are discussed. Cyclo-stationary and Gaussian processes receive extra attention, as they are of practical importance and possess some interesting and convenient properties. Complex processes are defined analogously to complex variables. Finally, the different concepts are reconsidered for discrete-time processes.

In Chapter 3 a description of stochastic processes in the frequency domain is given. This results in the concept of power spectral density. The bandwidth of a stochastic process is defined. Such an important subject as modulation of stochastic processes is presented, as well as the synchronous demodulation. In order to be able to define and describe the spectrum of discrete-time processes, a sampling theorem for these processes is derived.

After the basic concepts and definitions treated in the first three chapters, Chapter 4 starts with applications. Filtering of stochastic processes is the main subject of this chapter. We confine ourselves to linear, time-invariant filtering and derive both the correlation functions and spectra of a two-port system. The concept of equivalent noise bandwidth has been defined in order to arrive at an even more simple description of noise filtering in the frequency domain. Next, the calculation of the spectrum of random data signals is presented. A brief resumé of the principles of discrete-time signals and systems is dealt with using the z-transform and discrete Fourier transform, based on which the filtering of discrete-time processes is described both in time and frequency domains.

Chapter 5 is devoted to bandpass processes. The description of bandpass signals and systems in terms of quadrature components is introduced. The probability density functions of envelope and phase are derived. The measurement of spectra and operation of the spectrum analyser is discussed. Finally, sampling and conversion to baseband of bandpass processes is discussed.

Thermal noise and its impact on systems is the subject of Chapter 6. After presenting the spectral densities we consider the role of thermal noise in passive networks. System noise is considered based on the thermal noise contribution of amplifiers, the noise figure and the influence of cascading of systems on noise performance.

Chapter 7 is devoted to detection and optimal filtering. The chapter starts by considering hypothesis testing, which is applied to the detection of a binary signal disturbed by white Gaussian noise. The matched filter emerges as the optimum filter for optimum detection performance. Finally, filters that minimize the mean squared error (Wiener filters) are derived. They can be used for smoothing stored data or portions of a random signal that arrived in the past. Filters that produce an optimal prediction of future signal values can also be designed.

Finally, Chapter 8 is of a more advanced nature. It presents the basics of random point processes, of which the Poisson process is the most well known. The characteristic function

plays a crucial role in analysing these processes. Starting from that process several shot noise processes are introduced: the homogeneous Poisson process, the inhomogeneous Poisson process, the Poisson impulse process and the random-pulse process. Campbell's theorem is derived. A few application areas of random point processes are indicated.

The appendices contain a few subjects that are necessary for the main material. They are: signal space representation and definitions of attenuation, phase shift and decibels. The rest of the appendices comprises basic mathematical relations, a summary of probability theory, definitions of special functions, a list and properties of Fourier transform pairs, and a few mathematical and physical constants.

Finally, I would like to thank those people who contributed in one way or another to this text. My friend Rajan Srinivasan provided me with several suggestions to improve the content. Also, Arjan Meijerink carefully read the draft and made suggestions for improvement.

Last but certainly not least, I thank my wife Kitty, who allowed me to spend so many hours of our free time to write this text.

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1
Introduction

1.1 RANDOM SIGNALS AND NOISE

In (electrical) engineering one often encounters signals that do not have a precise mathematical description, since they develop as random functions of time. Sometimes this random development is caused by a single random variable, but often it is a consequence of many random variables. In other cases the causes of randomness are not clear and a description is not possible, but the signal is characterized by means of measurements only.

A random time function may be a desired signal, such as an audio or video signal, or it may be an unwanted signal that is unintentionally added to a desired (information) signal and disturbs the desired signal. We call the desired signal a random signal and the unwanted signal noise. However, the latter often does not behave like noise in the classical sense, but it is more like interference. Then it is an information bearing signal as well, but undesired. A desired signal and noise (or interference) can, in general, not be distinguished completely; by means of well-defined signal processing in a receiver, the desired signal may be favoured in a maximal way whereas the disturbance is suppressed as much as possible. In all cases a description of the signals is required in order to be able to analyse its impact on the performance of the system under consideration. Especially in communication theory this situation often occurs. The random character as a function of time makes the signals difficult to describe and the same holds for signal processing or filtering. Nevertheless, there is a need to characterize these signals by a few deterministic parameters that enable the system user to assess the performance of the system. The tool to deal with both random signals and noise is the concept of the stochastic process, which is introduced in Section 1.3.

This book gives an elementary introduction to the methods used to describe random signals and noise. For that purpose use is made of the laws of probability, which are extensively described in textbooks [1–5].

1.2 MODELLING

When studying and analysing random signals one is mainly committed to theory, which however, can be of good predictive value. Actually, the main activity in the field of random signals is modelling of processes and systems. Many scientists and engineers have

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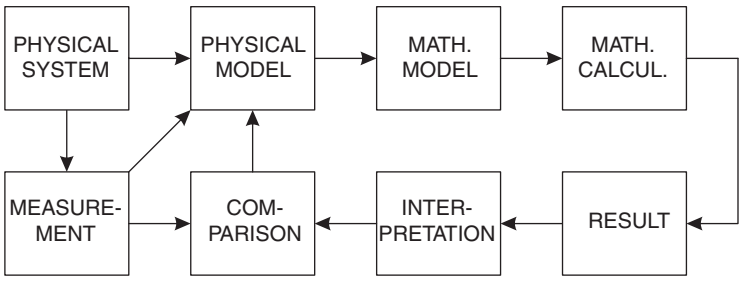


Figure 1.1 The process of modelling

contributed to that activity in the past and their results have been checked in practice. When a certain result agrees (at least to a larger extent) with practical measurements, then there is confidence in and acceptance of the result for practical application. This process of modelling has schematically been depicted in Figure 1.1.

In the upper left box of this scheme there is the important physical process. Based on our knowledge of the physics of this process we make a physical model of it. This physical model is converted into a mathematical model. Both modelling activities are typical engineer tasks. In this mathematical model the physics is no longer formally recognized, but the laws of physics will be included with their mathematical description. Once the mathematical model has been completed and the questions are clear we can forget about the physics for the time being and concentrate on doing the mathematical calculations, which may help us to find the answers to our questions. In this phase the mathematicians can help the engineer a lot. Let us suppose that the mathematical calculations give a certain outcome, or maybe several outcomes. These outcomes would then need to be interpreted in order to discover what they mean from a physical point of view. This ends the role of the mathematician, since this phase is maybe the most difficult engineering part of the process. It may happen that certain mathematical solutions have to be discarded since they contradict physical laws. Once the interpretation has been completed there is a return to the physical process, as the practical applicability of the results needs to be checked. In order to check these the quantities or functions that have been calculated are measured. The measurement is compared to the calculated result and in this way the physical model is validated. This validation may result in an adjustment of the physical model and another cycle in the loop is made. In this way the model is refined iteratively until we are satisfied about the validation. If there is a shortage of insight into the physical system, so that the physical model is not quite clear, measurements of the physical system may improve the physical model.

In the courses that are taught to students, models that have mainly been validated in this way are presented. However, it is important that students are aware of this process and the fact that the models that are presented may be a result of a difficult struggle for many years by several physicists, engineers and mathematicians. Sometimes students are given the opportunity to be involved in this process during research assignments.

1.3 THE CONCEPT OF A STOCHASTIC PROCESS

In probability theory a random variable is a rule that assigns a number to every outcome of an experiment, such as, for example, rolling a die. This random variable X is associated with a sample space S, such that according to a well-defined procedure to each event s in the

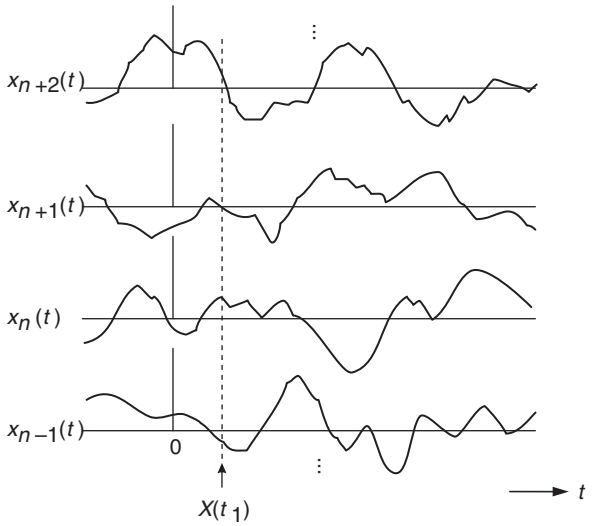


Figure 1.2 A few sample functions of a stochastic process

sample space a number is assigned to X and is denoted by $X(s)$. For stochastic processes, on the other hand, a time function $x(t, s)$ is assigned to every outcome in the sample space. Within the framework of the experiment the family (or ensemble) of all possible functions that can be realized is called the stochastic process and is denoted by $X(t, s)$. A specific waveform out of this family is denoted by $x_n(t)$ and is called a sample function or a realization of the stochastic process. When a realization in general is indicated the subscript n is omitted. Figure 1.2 shows a few sample functions that are supposed to constitute an ensemble. The figure gives an example of a finite number of possible realizations, but the ensemble may consist of an infinite number of realizations. The realizations may even be uncountable. A realization itself is sometimes called a stochastic process as well. Moreover, a stochastic process produces a random variable that arises from giving t a fixed value with s being variable. In this sense the random variable $X(t_1, s) = X(t_1)$ is found by considering the family of realizations at the fixed point in time t_1 (see Figure 1.2). Instead of $X(t_1)$ we will also use the notation X_1. The random variable X_1 describes the statistical properties of the process at the instant of time t_1. The expectation of X_1 is called the ensemble mean or the expected value or the mean of the stochastic process (at the instant of time t_1). Since t_1 may be arbitrarily chosen, the mean of the process will in general not be constant, i.e. it may have different values for different values of t. Finally, a stochastic process may represent a single number by giving both t and s fixed values. The phrase 'stochastic process' may therefore have four different interpretations. They are:

1. A family (or ensemble) of time functions. Both t and s are variables.
2. A single time function called a sample function or a realization of the stochastic process. Then t is a variable and s is fixed.
3. A random variable; t is fixed and s is variable.
4. A single number; both t and s are fixed.

Which of these four interpretations holds in a specific case should follow from the context.

Different classes of stochastic processes may be distinguished. They are classified on the basis of the characteristics of the realization values of the process x and the time parameter t. Both can be either continuous or discrete, in any combination. Based on this we have the following classes:

- Both the values of $X(t)$ and the time parameter t are continuous. Such a process is called a continuous stochastic process.
- The values of $X(t)$ are continuous, whereas time t is discrete. These processes are called discrete-time processes or continuous random sequences. In the remainder of the book we will use the term discrete-time process.
- If the values of $X(t)$ are discrete but the time axis is continuous, we call the process a discrete stochastic process.
- Finally, if both the process values and the time scale are discrete, we say that the process is a discrete random sequence.

In Table 1.1 an overview of the different classes of processes is presented. In order to get some feeling for stochastic processes we will consider a few examples.

Table 1.1 Summary of names of different processes

$X(t)$	Time	
	Continuous	Discrete
Continuous	Continuous stochastic process	Discrete-time process
Discrete	Discrete stochastic process	Discrete random sequence

1.3.1 Continuous Stochastic Processes

For this class of processes it is assumed that in principle the following holds:

$$-\infty < x(t) < \infty \quad \text{and} \quad -\infty < t < \infty \tag{1.1}$$

An example of this class was already given by Figure 1.2. This could be an ensemble of realizations of a thermal noise process as is, for instance, produced by a resistor, the characteristics of which are to be dealt with in Chapter 6. The underlying experiment is selecting a specific resistor from a collection of, let us say, 100 Ω resistors. The voltage across every selected resistor corresponds to one of the realizations in the figure.

Another example is given below.

Example 1.1:

The process we consider now is described by the equation

$$X(t) = \cos(\omega_0 t - \Theta) \tag{1.2}$$

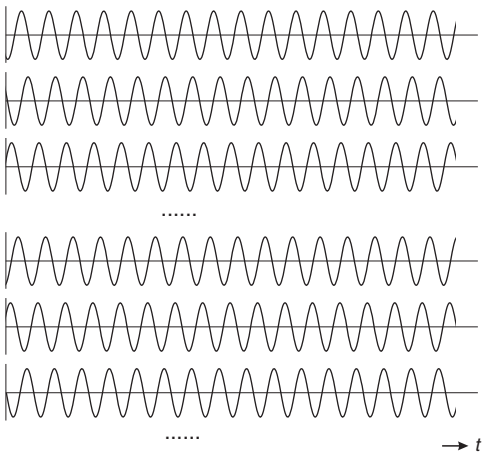


Figure 1.3 Ensemble of sample functions of the stochastic process $\cos(\omega_0 t - \Theta)$, with Θ uniformly distributed on the interval $(0, 2\pi]$

with ω_0 a constant and Θ a random variable with a uniform probability density function on the interval $(0, 2\pi]$. In this example the set of realizations is in fact uncountable, as Θ assumes continuous values. The ensemble of sample functions is depicted in Figure 1.3.

Thus each sample function consists of a cosine function with unity amplitude, but the phase of each sample function differs randomly from others. For each sample function a drawing is taken from the uniform phase distribution. We can imagine this process as follows. Consider a production process of crystal oscillators, all producing the same amplitude unity and the same radial frequency ω_0. When all those oscillators are switched on, their phases will be mutually independent. The family of all measured output waveforms can be considered as the ensemble that has been presented in Figure 1.3.

This process will get further attention in different chapters that follow.

□

1.3.2 Discrete-Time Processes (Continuous Random Sequences)

The description of this class of processes becomes more and more important due to the increasing use of modern digital signal processors which offer flexibility and increasing speed and computing power. As an example of a discrete-time process we can imagine sampling the process that was given in Figure 1.2. Let us suppose that to this process ideal sampling is applied at equidistant points in time with sampling period T_s; with ideal sampling we mean the sampling method where the values at T_s are replaced by delta functions of amplitude $X(nT_s)$ [6]. However, to indicate that it is now a discrete-time process we denote it by $X[n]$, where n is an integer running in principle from $-\infty$ to $+\infty$. We know from the sampling theorem (see Section 3.5.1 or, for instance, references [1] and [7]) that the original signal can perfectly be recovered from its samples, provided that the signals are band-limited. The process that is produced in this way is given in Figure 1.4, where the sample values are presented by means of the length of the arrows.

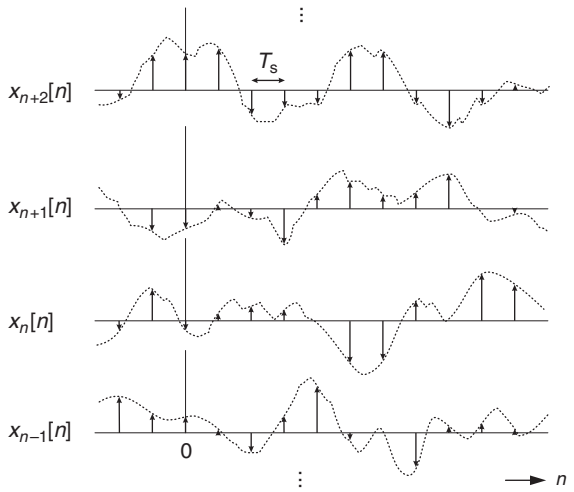


Figure 1.4 Example of a discrete-time stochastic process

Another important example of the discrete-time process is the so-called Poisson process, where there are no equidistant samples in time but the process produces 'samples' at random points in time. This process is an adequate model for shot noise and it is dealt with in Chapter 8.

1.3.3 Discrete Stochastic Processes

In this case the time is continuous and the values discrete. We present two examples of this class. The second one, the random data signal, is of great practical importance and we will consider it in further detail in Chapter 4.

Example 1.2:

This example is a very simple one. The ensemble of realizations consists of a set of constant time functions. According to the outcome of an experiment one of these constants may be chosen. This experiment can be, for example, the rolling of a die. In that case the number of realizations can be six $(n = 6)$, equal to the usual number of faces of a die. Each of the outcomes $s \in \{1, 2, 3, 4, 5, 6\}$ has a one-to-one correspondence to one of these numbered constant functions of time. The ensemble is depicted in Figure 1.5.

□

Example 1.3:

Another important stochastic process is the random data signal. It is a signal that is produced by many data sources and is described by

$$X(t) = \sum_n A_n p(t - nT - \Theta) \tag{1.3}$$

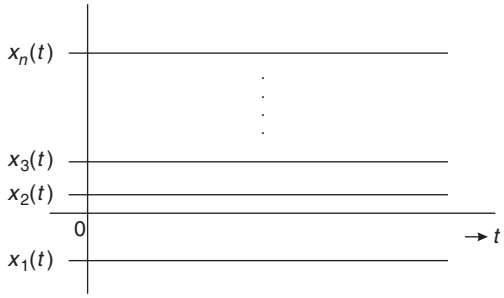


Figure 1.5 Ensemble of sample functions of the stochastic process constituted by a number of constant time functions

where $\{A_n\}$ are the data bits that are randomly chosen from the set $A_n \in \{+1, -1\}$. The rectangular pulse $p(t)$ of width T serves as the carrier of the information. Now Θ is supposed to be uniformly distributed on the bit interval $(0, T]$, so that all data sources of the family have the same bit period, but these periods are not synchronized. The ensemble is given in Figure 1.6. □

1.3.4 Discrete Random Sequences

The discrete random sequence can be imagined to result from sampling a discrete stochastic process. Figure 1.7 shows the result of sampling the random data signal from Example 1.3.

We will base the further development of the concept, description and properties of stochastic processes on the continuous stochastic process. Then we will show how these are extended to discrete-time processes. The two other classes do not get special attention, but

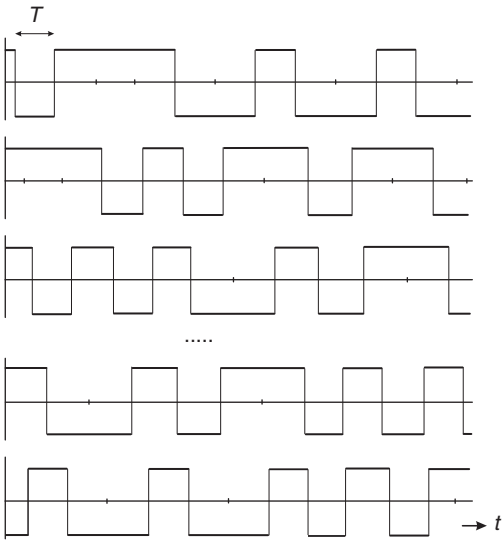


Figure 1.6 Ensemble of sample functions of the stochastic process $\sum_n A_n p(t - nT - \Theta)$, with Θ uniformly distributed on the interval $(0, T]$

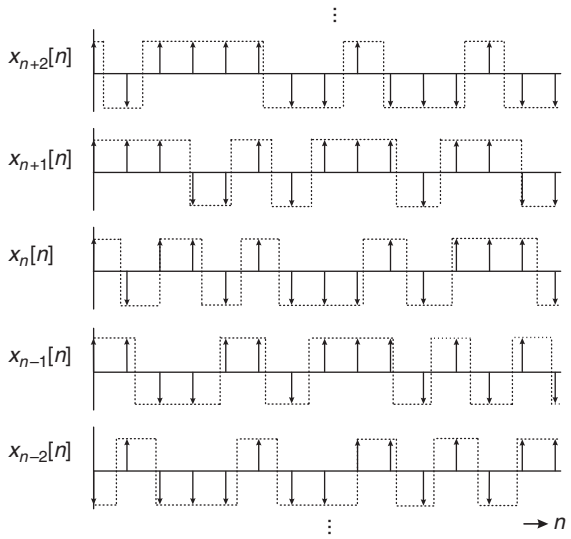


Figure 1.7 Example of a discrete random sequence

are considered as special cases of the former ones by limiting the realization values x to a discrete set.

1.3.5 Deterministic Function versus Stochastic Process

The concept of the stochastic process does not conflict with the theory of deterministic functions. It should be recognized that a deterministic function can be considered as nothing else but a special case of a stochastic process. This is elucidated by considering Example 1.1. If the random variable Θ is given the probability density function $f_\Theta(\theta) = \delta(\theta)$, then the stochastic process reduces to the function $\cos(\omega_0 t)$. The given probability density function is actually a discrete one with a single outcome. In fact, the ensemble of the process reduces in this case to a family comprising merely one member. This is a general rule; when the probability density function of the stochastic process that is governed by a single random variable consists of a single delta function, then a deterministic function results. This way of generalization avoids the often confusing discussion on the difference between a deterministic function on the one hand and a stochastic process on the other hand. In view of the consideration presented here they can actually be considered as members of the same class, namely the class of stochastic processes.

1.4 SUMMARY

Definitions of random signals and noise have been given. A random signal is, as a rule, an information carrying wanted signal that behaves randomly. Noise also behaves randomly but is unwanted and disturbs the signal. A common tool to describe both is the concept of a stochastic process. This concept has been explained and different classes of stochastic processes have been identified. They are distinguished by the behaviour of the time parameter and the values of the process. Both can either be continuous or discrete.

2
Stochastic Processes

In this chapter some basic concepts known from probability theory will be extended to include the time parameter. It is the time parameter that makes the difference between a random variable and a stochastic process. The basic concepts are: probability density function and correlation. The time dependence of the signals asks for a few new concepts, such as the correlation function, stationarity and ergodicity.

2.1 STATIONARY PROCESSES

As has been indicated in the introduction chapter we can fix the time parameter of a stochastic process. In this way we have a random variable, which can be characterized by means of a few deterministic numbers such as the mean, variance, etc. These quantities are defined using the probability density function. When fixing two time parameters we can consider two random variables simultaneously. Here also we can define joint random variables and, related to that, characterize quantities using the joint probability density function. In this way we can proceed, in general, to the case of N variables that are described by an N-dimensional joint probability density function, with N an arbitrary number.

Roughly speaking we can say that a stochastic process is stationary if its statistical properties do not depend on the time parameter. This rough definition will be elaborated in more detail in the rest of this chapter. There are several types of stationarity and for the main types we will present exact definitions in the sequel.

2.1.1 Cumulative Distribution Function and Probability Density Function

In order to be able to define stationarity, the probability distribution and density functions as they are applied to the stochastic process $X(t)$ have to be defined. For a fixed value of time parameter t_1 the cumulative probability distribution function or, for short, distribution function is defined by

$$F_X(x_1;t_1) \triangleq \mathrm{P}\{X(t_1) \leq x_1\} \tag{2.1}$$

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From this notation it follows that F_X may be a function of the value of t_1 that has been chosen.

For two random variables $X_1 = X(t_1)$ and $X_2 = X(t_2)$ we introduce the two-dimensional extension of Equation (2.1):

$$F_X(x_1, x_2; t_1, t_2) \triangleq \mathrm{P}\{X(t_1) \leq x_1, X(t_2) \leq x_2\} \tag{2.2}$$

the second-order, joint probability distribution function. In an analogous way we denote the Nth-order, joint probability distribution function

$$F_X(x_1, \ldots, x_N; t_1, \ldots, t_N) \triangleq \mathrm{P}\{X(t_1) \leq x_1, \ldots, X(t_N) \leq x_N\} \tag{2.3}$$

The corresponding (joint) probability density functions are found by taking the derivatives respectively of Equations (2.1) to (2.3):

$$f_X(x_1; t_1) \triangleq \frac{\partial F_X(x_1; t_1)}{\partial x_1} \tag{2.4}$$

$$f_X(x_1, x_2; t_1, t_2) \triangleq \frac{\partial^2 F_X(x_1, x_2; t_1, t_2)}{\partial x_1 \partial x_2} \tag{2.5}$$

$$f_X(x_1, \ldots, x_N; t_1, \ldots, t_N) \triangleq \frac{\partial^N F_X(x_1, \ldots, x_N; t_1, \ldots, t_N)}{\partial x_1 \cdots \partial x_N} \tag{2.6}$$

Two processes $X(t)$ and $Y(t)$ are called statistically independent if the set of random variables $\{X(t_1), X(t_2), \ldots, X(t_N)\}$ is independent of the set of random variables $\{Y(t'_1), Y(t'_2), \ldots, Y(t'_M)\}$, for each arbitrary choice of the time parameters $\{t_1, t_2, \ldots, t_N; t'_1, t'_2, \ldots, t'_M\}$. Independence implies that the joint probability density function can be factored in the following way:

$$\begin{aligned} &f_{X,Y}(x_1, \ldots, x_N; y_1, \ldots, y_M; t_1, \ldots, t_N; t'_1, \ldots, t'_M) \\ &\quad = f_X(x_1, \ldots, x_N; t_1, \ldots, t_N) \cdot f_Y(y_1, \ldots, y_M; t'_1, \ldots, t'_M) \end{aligned} \tag{2.7}$$

Thus, the joint probability density function of two independent processes is written as the product of the two marginal probability density functions.

2.1.2 First-Order Stationary Processes

A stochastic process is called a first-order stationary process if the first-order probability density function is independent of time. Mathematically this can be stated as

$$f_X(x_1; t_1) = f_X(x_1; t_1 + \tau) \tag{2.8}$$

holds for all τ. As a consequence of this property the mean value of such a process, denoted by $\overline{X(t)}$, is

$$\overline{X(t)} \equiv \mathrm{E}[X(t)] \triangleq \int x\, f_X(x; t)\, \mathrm{d}x = \text{constant} \tag{2.9}$$

i.e. it is independent of time.

2.1.3 Second-Order Stationary Processes

A stochastic process is called a second-order stationary process if for the two-dimensional joint probability density function

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 + \tau, t_2 + \tau) \tag{2.10}$$

for all τ. It is easy to verify that Equation (2.10) is only a function of the time difference $t_2 - t_1$ and does not depend on the absolute time. In order to gain that insight put $\tau = -t_1$.

A process that is second-order stationary is first-order stationary as well, since the second-order joint probability density function uniquely determines the lower-order (in this case first-order) probability density function.

2.1.4 *N*th-Order Stationary Processes

By extending the reasoning from the last subsection to N random variables $X_i = X(t_i)$, for $i = 1, \ldots, N$, we arrive at an Nth-order stationary process. The Nth-order joint probability density function is once more independent of a time shift; i.e.

$$f_X(x_1, \ldots, x_N; t_1, \ldots, t_N) = f_X(x_1, \ldots, x_N; t_1 + \tau, \ldots, t_N + \tau) \tag{2.11}$$

for all τ. A process that is Nth-order stationary is stationary to all orders $k \leq N$. An Nth-order stationary process where N can have an arbitrary large value is called a strict-sense stationary process.

2.2 CORRELATION FUNCTIONS

2.2.1 The Autocorrelation Function, Wide-Sense Stationary Processes and Ergodic Processes

The autocorrelation function of a stochastic process is defined as the correlation $\mathrm{E}[X_1 X_2]$ of the two random variables $X_1 = X(t_1)$ and $X_2 = X(t_2)$. These random variables are achieved by considering all realization values of the stochastic process at the instants of time t_1 and t_2 (see Figure 2.1). In general it will be a function of these two times instants. The autocorrelation function is denoted as

$$R_{XX}(t_1, t_2) \triangleq \mathrm{E}[X(t_1)X(t_2)] = \iint x_1 x_2 \, f_X(x_1, x_2; t_1, t_2) \, \mathrm{d}x_1 \, \mathrm{d}x_2 \tag{2.12}$$

Substituting $t_1 = t$ and $t_2 = t_1 + \tau$, Equation (2.12) becomes

$$R_{XX}(t, t + \tau) = \mathrm{E}[X(t) \, X(t + \tau)] \tag{2.13}$$

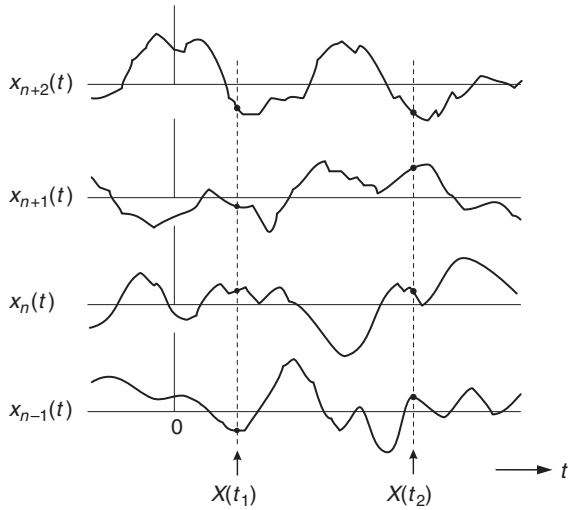


Figure 2.1 The autocorrelation of a stochastic process by considering $\mathrm{E}[X(t_1)X(t_2)]$

Since for a second-order stationary process the two-dimensional joint probability density function depends only on the time difference, the autocorrelation function will also be a function of the time difference τ. Then Equation (2.13) can be written as

$$R_{XX}(t, t+\tau) = R_{XX}(\tau) \tag{2.14}$$

The mean and autocorrelation function of a stochastic process are often its most characterizing features. Mostly, matters become easier if these two quantities do not depend on absolute time. A second-order stationary process guarantees this independence but at the same time places severe demands on the process. Therefore we define a broader class of stochastic processes, the so-called wide-sense stationary processes.

Definition

A process $X(t)$ is called wide-sense stationary if it satisfies the conditions

$$\begin{aligned} \mathrm{E}[X(t)] &= \overline{X(t)} = \text{constant} \\ \mathrm{E}[X(t)\,X(t+\tau)] &= R_{XX}(\tau) \end{aligned} \tag{2.15}$$

It will be clear that a second-order stationary process is also wide-sense stationary. The converse, however, is not necessarily true.

Properties of $R_{XX}(\tau)$

If a process is at least wide-sense stationary then its autocorrelation function exhibits the following properties:

1. $|R_{XX}(\tau)| \leq R_{XX}(0)$ (2.16)
   i.e. $|R_{XX}(\tau)|$ attains its maximum value for $\tau = 0$.
2. $R_{XX}(-\tau) = R_{XX}(\tau)$ (2.17)
   i.e. $R_{XX}(\tau)$ is an even function of τ.
3. $R_{XX}(0) = \mathrm{E}[X^2(t)]$ (2.18)
4. If $X(t)$ has no periodic component then $R_{XX}(\tau)$ comprises a constant term equal to $\overline{X(t)}^2$, i.e. $\lim_{|\tau|\to\infty} R_{XX}(\tau) = \overline{X(t)}^2$.
5. If $X(t)$ has a periodic component then $R_{XX}(\tau)$ will comprise a periodic component as well, and which has the same periodicity.

A function that does not satisfy these properties cannot be the autocorrelation function of a wide-sense stationary process. It will be clear from properties 1 and 2 that $R_{XX}(\tau)$ is not allowed to exhibit an arbitrary shape.

Proofs of the properties:

1. To prove property 1 let us consider the expression

$$\begin{aligned}\mathrm{E}[\{X(t) \pm X(t+\tau)\}^2] &= \mathrm{E}[X^2(t) + X^2(t+\tau) \pm 2X(t)\,X(t+\tau)] \\ &= 2\{R_{XX}(0) \pm R_{XX}(\tau)\} \geq 0\end{aligned} \tag{2.19}$$

   Since the expectation $\mathrm{E}[\{X(t) \pm X(t+\tau)\}^2]$ is taken over the squared value of a certain random variable, this expectation should be greater than or equal to zero. From the last line of Equation (2.19) property 1 is concluded.

2. The proof of property 2 is quite simple. In the definition of the autocorrelation function substitute $t' = t + \tau$ and the proof proceeds as follows:

$$\begin{aligned}R_{XX}(\tau) &= \mathrm{E}[X(t)\,X(t+\tau)] = \mathrm{E}[X(t'-\tau)\,X(t')] \\ &= \mathrm{E}[X(t')\,X(t'-\tau)] = R_{XX}(-\tau)\end{aligned} \tag{2.20}$$

3. Property 3 follows immediately from the definition of $R_{XX}(\tau)$ by inserting $\tau = 0$.

4. From a physical point of view most processes have the property that the random variables $X(t)$ and $X(t+\tau)$ are independent when $\tau \to \infty$. Invoking once more the definition of the

autocorrelation function it follows that

$$\lim_{\tau\to\infty} R_{XX}(\tau) = \lim_{\tau\to\infty} \mathrm{E}[X(t)\,X(t+\tau)]$$
$$= \mathrm{E}[X(t)]\,\mathrm{E}[X(t+\tau)] = \mathrm{E}^2[X(t)] = \overline{X}^2 \tag{2.21}$$

5. Periodic processes may be decomposed into cosine and sine components according to Fourier analysis. It therefore suffices to consider the autocorrelation function of one such component:

$$\mathrm{E}[\cos(\omega t - \Theta)\,\cos(\omega t + \omega\tau - \Theta)] = \tfrac{1}{2}\mathrm{E}[\cos(\omega\tau) + \cos(2\omega t + \omega\tau - 2\Theta)] \tag{2.22}$$

Since our considerations are limited to wide-sense stationary processes, the autocorrelation function should be independent of the absolute time t, and thus the expectation of the last term of the latter expression should be zero. Thus only the term comprising $\cos(\omega\tau)$ remains after taking the expectation, which proves property 5.

When talking about the mean or expectation (denoted by $\mathrm{E}[\cdot]$) the statistical average over the ensemble of realizations is meant. Since stochastic processes are time functions we can define another average, namely the time average, given by

$$\mathrm{A}[X(t)] \triangleq \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} x(t)\mathrm{d}t \tag{2.23}$$

When taking this time average only one single sample function can be involved; consequently, expressions like $\mathrm{A}[X(t)]$ and $\mathrm{A}[X(t)\,X(t+\tau)]$ will be random variables.

Definition

A wide-sense stationary process $X(t)$ satisfying the two conditions

$$\mathrm{A}[X(t)] = \mathrm{E}[X(t)] = \overline{X(t)} \tag{2.24}$$
$$\mathrm{A}[X(t)\,X(t+\tau)] = \mathrm{E}[X(t)\,X(t+\tau)] = R_{XX}(\tau) \tag{2.25}$$

is called an ergodic process.

In other words, an ergodic process has time averages $\mathrm{A}[X(t)]$ and $\mathrm{A}[X(t)\,X(t+\tau)]$ that are non-random because these time averages equal the ensemble averages $\overline{X(t)}$ and $R_{XX}(\tau)$. In the same way as several types of stationary process can be defined, several types of ergodic processes may also be introduced [1]. We will confine ourselves to the forms defined by the Equations (2.24) and (2.25). Ergodicity puts more severe demands on the process than stationarity and it is often hard to prove that indeed a process is ergodic; often it is impossible. In practice ergodicity is often just assumed without proof, unless the opposite is evident. In most cases there is no alternative, as one does not have access to the entire family

(ensemble) of sample functions, but rather just to one or a few members of it, for example one resistor, transistor or comparable noisy device is available. By assuming ergodicity a number of important statistical properties, such as the mean and the autocorrelation function of a process may be estimated from the observation of a single available realization. Fortunately, it appears that many processes are ergodic, but one should always be aware that at times one can encounter a process that is not ergodic. Later in this chapter we will develop a test for a certain class of ergodic processes.

Example 2.1:

As an example consider the process $X(t) = A\cos(\omega t - \Theta)$, with A a constant amplitude, ω a fixed but arbitrary radial frequency and Θ a random variable that is uniformly distributed on the interval $(0, 2\pi]$. The question is whether this process is ergodic in the sense as defined by Equations (2.24) and (2.25). To answer this we determine both the ensemble mean and the time average. For the time average it is found that

$$\mathrm{A}[X(t)] = \lim_{T\to\infty} \frac{1}{2T} A \int_{-T}^{T} \cos(\omega t - \Theta)\,\mathrm{d}t = \lim_{T\to\infty} \frac{1}{2T} A \frac{1}{\omega} \sin(\omega t - \Theta)\Big|_{-T}^{T} = 0 \tag{2.26}$$

The ensemble mean is

$$\mathrm{E}[X(t)] = \int f_{\Theta}(\theta)\, A\cos(\omega t - \theta)\,\mathrm{d}\theta = \frac{1}{2\pi} A \int_{0}^{2\pi} \cos(\omega t - \theta)\,\mathrm{d}\theta = 0 \tag{2.27}$$

Hence, time and ensemble averages are equal.

Let us now calculate the two autocorrelation functions. For the time-averaged autocorrelation function it is found that

$$\begin{aligned} \mathrm{A}[X(t)\,X(t+\tau)] &= \lim_{T\to\infty} \frac{1}{2T} A^2 \int_{-T}^{T} \cos(\omega t - \theta)\,\cos(\omega t + \omega\tau - \theta)\,\mathrm{d}t \\ &= \lim_{T\to\infty} \frac{1}{2T}\,\frac{1}{2} A^2 \int_{-T}^{T} [\cos(2\omega t + \omega\tau - 2\theta) + \cos(\omega\tau)]\,\mathrm{d}t \end{aligned} \tag{2.28}$$

The first term of the latter integral equals 0. The second term of the integrand does not depend on the integration variable. Hence, the autocorrelation function is given by

$$\mathrm{A}[X(t)\,X(t+\tau)] = \tfrac{1}{2} A^2 \cos\omega\tau \tag{2.29}$$

Next we consider the statistical autocorrelation function

$$\begin{aligned} \mathrm{E}[X(t)\,X(t+\tau)] &= \frac{1}{2\pi} A^2 \int_{0}^{2\pi} \cos(\omega t - \theta)\,\cos(\omega t + \omega\tau - \theta)\,\mathrm{d}\theta \\ &= \frac{1}{2\pi}\,\frac{1}{2} A^2 \int_{0}^{2\pi} [\cos(2\omega t + \omega\tau - 2\theta) + \cos(\omega\tau)]\,\mathrm{d}\theta \end{aligned} \tag{2.30}$$

Of the latter integral the first part is 0. Again, the second term of the integrand does not depend on the integration variable. The autocorrelation function is therefore

$$\mathrm{E}[X(t)\,X(t+\tau)] = \tfrac{1}{2}A^2 \cos\omega\tau \tag{2.31}$$

Hence both first-order means (time average and statistical mean) and second-order means (time-averaged and statistical autocorrelation functions) are equal. It follows that the process is ergodic.

The process $\cos(\omega t - \Theta)$ with $f_\Theta(\theta) = \delta(\theta)$ equals the deterministic function $\cos\omega t$. This process is not ergodic, since it is easily verified that the expectation (in this case the function itself) is time-dependent and thus not stationary, which is a condition for ergodicity. This example, where a probability density function that consists of a δ function reduces the process to a deterministic function, has also been mentioned in Chapter 1.

□

2.2.2 Cyclo-Stationary Processes

A process $X(t)$ is called cyclo-stationary (or periodically stationary) if the probability density function is independent of a shift in time over an integer multiple of a constant value T (the period time), so that

$$f_X(x_1,\ldots,x_N;t_1,\ldots,t_N) = f_X(x_1,\ldots,x_N;t_1+mT,\ldots,t_N+mT) \tag{2.32}$$

for each integer value of m. A cyclo-stationary process is not stationary, since Equation (2.11) is not valid for all values of τ, but only for discrete values $\tau = mT$. However, the discrete-time process $X(mT+\tau)$ is stationary for all values of τ.

A relation exists between cyclo-stationary processes and stationary processes. To see this relation it is evident from Equation (2.32) that

$$F_X(x_1,\ldots,x_N;t_1,\ldots,t_N) = F_X(x_1,\ldots,x_N;t_1+mT,\ldots,t_N+mT) \tag{2.33}$$

Next consider the modified process $\underline{X}(t) = X(t-\Theta)$, where $X(t)$ is cyclo-stationary and Θ a random variable that has a uniform probability density function on the period interval $(0,T]$. Now we define the event $\mathscr{A}$ as

$$\mathscr{A} = \{\underline{X}(t_1+\tau) \le x_1,\ldots,\underline{X}(t_N+\tau) \le x_N\} \tag{2.34}$$

The probability that this event will occur is

$$\mathrm{P}(\mathscr{A}) = \int_0^T \mathrm{P}(\mathscr{A}|\Theta=\theta) f_\Theta(\theta)\,\mathrm{d}\theta = \frac{1}{T}\int_0^T \mathrm{P}(\mathscr{A}|\Theta=\theta)\,\mathrm{d}\theta \tag{2.35}$$

For the latter integrand we write

$$\begin{aligned}\mathrm{P}(\mathscr{A}|\Theta=\theta) &= \mathrm{P}\{X(t_1+\tau-\theta) \le x_1,\ldots,X(t_N+\tau-\theta) \le x_N\} \\ &= F_X(x_1,\ldots,x_N;t_1+\tau-\theta,\ldots,t_N+\tau-\theta)\end{aligned} \tag{2.36}$$

Substituting this result in Equation (2.35) yields

$$\mathrm{P}(\mathscr{A}) = \frac{1}{T}\int_0^T F_X(x_1,\ldots,x_N;t_1+\tau-\theta,\ldots,t_N+\tau-\theta)\,\mathrm{d}\theta \tag{2.37}$$

As $X(t)$ is cyclo-stationary Equation (2.37) is independent of τ. From Equation (2.34) it follows, therefore, that $\mathrm{P}(\mathscr{A})$ represents the probability distribution function of the process $\underline{X}(t)$. Thus we have the following theorem.

Theorem 1

If $X(t)$ is a cyclo-stationary process with period time T and Θ is a random variable that is uniformly distributed on the interval $(0,T]$, then the process $\underline{X}(t) = X(t-\Theta)$ is stationary with the probability distribution function

$$F_{\underline{X}}(x_1,\ldots,x_N;t_1,\ldots,t_N) = \frac{1}{T}\int_0^T F_X(x_1,\ldots,x_N;t_1-\tau,\ldots,t_N-\tau)\,\mathrm{d}\tau \tag{2.38}$$

A special case consists of the situation where $X(t) = p(t)$ is a deterministic, periodic function with period T. Then, as far as the first-order probability distribution function $F_{\underline{X}}(x)$ is concerned, the integral from Equation (2.38) can be interpreted as the relative fraction of time during which $X(t)$ is smaller or equal to x. This is easily understood when realizing that for a deterministic function $F_X(x_1;t_1)$ is either zero or one, depending on whether $p(t_1)$ is larger or smaller than x_1.

If we take $X(t) = p(t)$, then this process $X(t)$ is strict sense cyclo-stationary and from the foregoing we have the following theorem.

Theorem 2

If $\underline{X}(t) = p(t-\Theta)$ is an arbitrary, periodic waveform with period T and Θ a random variable that is uniformly distributed on the interval $(0,T]$, then the process $\underline{X}(t)$ is strict-sense stationary and ergodic. The probability distribution function of this process reads

$$F_{\underline{X}}(x_1,\ldots,x_N;t_1,\ldots,t_N) = \frac{1}{T}\int_0^T F_p(p_1,\ldots,p_N;t_1-\tau,\ldots,t_N-\tau)\,\mathrm{d}\tau \tag{2.39}$$

The mean value of the process equals

$$\mathrm{E}[\underline{X}(t)] = \frac{1}{T}\int_0^T p(t)\,\mathrm{d}t = \mathrm{A}[p(t)] \tag{2.40}$$

and the autocorrelation function

$$R_{\underline{X}\underline{X}}(\tau) = \frac{1}{T}\int_0^T p(t)\,p(t+\tau)\,\mathrm{d}t = \mathrm{A}[p(t)\,p(t+\tau)] \tag{2.41}$$

This latter theorem is a powerful expedient when proving strict-sense stationarity and ergodicity of processes that often occur in practice. In such cases the probability distribution function is found by means of the integral given by Equation (2.39). For this integral the same interpretation is valid as for that from Equation (2.38). From the probability distribution function the probability density function can be derived using Equation (2.4). By adding a random phase Θ , with Θ uniformly distributed on the interval $(0, T]$, to a cyclo-stationary process the process can be made stationary. Although this seems to be an artificial operation, it is not so from a physical point of view. If we imagine that the ensemble of realizations originates from a set of signal generators, let us say sinusoidal wave generators, all of them tuned to the same frequency, then the waves produced by these generators will as a rule not be synchronized in phase.

Example 2.2:

The process $X(t) = \cos(\omega t)$ is not stationary, its mean value being $\cos(\omega t)$; however, it is cyclo-stationary. On the contrary, the modified process $\underline{X}(t) = \cos(\omega t - \Theta)$, with Θ uniformly distributed on the interval $(0, 2\pi]$, is strict-sense stationary and ergodic, based on Theorem 2. The ergodicity of this process was already concluded when dealing with Example 2.1. Moreover, we derived the autocorrelation function of this process as $\frac{1}{2}\cos(\omega\tau)$.

Let us now elaborate this example. We will derive the probability distribution and density functions, based on Theorem 2. For this purpose remember that the probability distribution function is given by Equation (2.39) and that this integral is interpreted as the relative fraction of time during which $X(t)$ is smaller than or equal to x. This interpretation has been further explained by means of Figure 2.2. In this figure one complete period of a cosine is presented. The constant value x is indicated. The duration that the given realization is smaller than or equal to x has been drawn by means of the bold line pieces, which are indicated by T_1 and T_2, and the complete second half of the cycle, which has a length of π. It can be seen that the line pieces T_1 and T_2 are of equal length, namely $\arcsin x$. Finally, the probability distribution function is found by adding all the bold line pieces and dividing the result by the period 2π. This leads to the probability distribution function

$$F_{\underline{X}}(x) = \mathrm{P}\{\underline{X}(t) \leq x\} = \begin{cases} \frac{1}{2} + \frac{1}{\pi}\arcsin x, & |x| \leq 1 \\ 0, & x < -1 \\ 1, & x > 1 \end{cases} \tag{2.42}$$

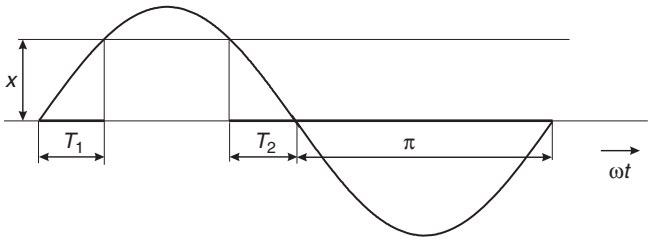


Figure 2.2 Figure to help determine the probability distribution function of the random phased cosine

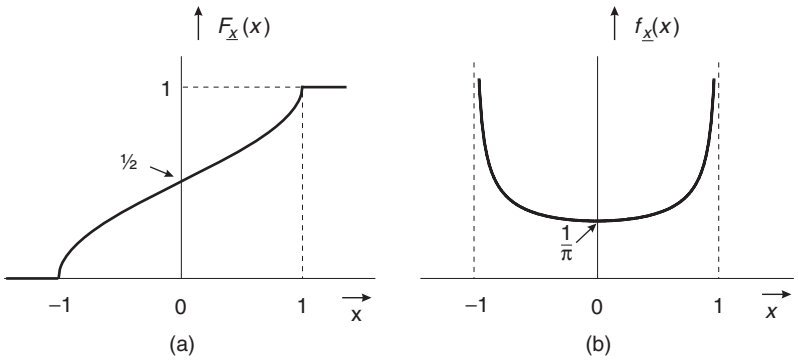


Figure 2.3 (a) The probability distribution function and (b) the probability density function of the random phased cosine

This function is depicted in Figure 2.3(a). From the probability distribution function the probability density function is easily derived by taking the derivative, i.e.

$$f_{\underline{X}}(x) \triangleq \frac{\mathrm{d}F_{\underline{X}}(x)}{\mathrm{d}x} = \begin{cases} \frac{1}{\pi}\frac{1}{\sqrt{1-x^2}}, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases} \tag{2.43}$$

This function has been plotted in Figure 2.3(b). Note the asymptotic values of the function for both $x = 1$ and $x = -1$.

□

Example 2.3:

The random data signal

$$X(t) = \sum_n A_n p(t - nT) \tag{2.44}$$

with A_n a stationary sequence of binary random variables that are selected out of the set $\{-1, +1\}$ and with autocorrelation sequence

$$\mathrm{E}[A_n A_k] = \mathrm{E}[A_n A_{n+m}] = \mathrm{E}[A_n A_{n-m}] = R_m \tag{2.45}$$

constitutes a cyclo-stationary process, where Theorems 1 and 2 can be applied. Properties of this random data signal will be derived in more detail later on (see Section 4.5).

□

2.2.3 The Cross-Correlation Function

The cross-correlation function of two stochastic processes $X(t)$ and $Y(t)$ is defined as

$$R_{XY}(t, t+\tau) \triangleq \mathrm{E}[X(t)\, Y(t+\tau)] \tag{2.46}$$

$X(t)$ and $Y(t)$ are jointly wide-sense stationary if both $X(t)$ and $Y(t)$ are wide-sense stationary and if the cross-correlation function $R_{XY}(t, t+\tau)$ is independent of the absolute time parameter, i.e.

$$R_{XY}(t, t+\tau) = \mathrm{E}[X(t)\, Y(t+\tau)] = R_{XY}(\tau) \tag{2.47}$$

Properties of $R_{XY}(\tau)$

If two processes $X(t)$ and $Y(t)$ are jointly wide-sense stationary, then the cross-correlation function has the following properties:

1. $R_{XY}(-\tau) = R_{YX}(\tau)$ (2.48)
2. $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)\, R_{YY}(0)}$ (2.49)
3. $|R_{XY}(\tau)| \leq \frac{1}{2}[R_{XX}(0) + R_{YY}(0)]$ (2.50)

A function that does not satisfy these properties cannot be the cross-correlation function of two jointly wide-sense stationary processes.

Proofs of the properties:

1. Property 1 is proved as follows. In the definition of the cross-correlation function replace $t - \tau$ by t' and do some manipulation as shown below:

$$\begin{aligned} R_{XY}(-\tau) &= \mathrm{E}[X(t)\, Y(t-\tau)] = \mathrm{E}[X(t'+\tau)\, Y(t')] \\ &= \mathrm{E}[Y(t')\, X(t'+\tau)] = R_{YX}(\tau) \end{aligned} \tag{2.51}$$

2. To prove property 2 we consider the expectation of the process $\{X(t) + cY(t+\tau)\}^2$, where c is a constant; i.e. we investigate

$$\begin{aligned} \mathrm{E}[\{X(t) + cY(t+\tau)\}^2] &= \mathrm{E}[X^2(t) + c^2 Y^2(t+\tau) + 2cX(t)\, Y(t+\tau)] \\ &= \mathrm{E}[X^2(t)] + c^2 \mathrm{E}[Y^2(t+\tau)] + 2c\mathrm{E}[X(t)\, Y(t+\tau)] \\ &= c^2 R_{YY}(0) + 2cR_{XY}(\tau) + R_{XX}(0) \end{aligned} \tag{2.52}$$

   This latter expression is a quadratic form as a function of c, and since it is the expectation of a quantity squared, this expression can never be less than zero. As a consequence the discriminant cannot be positive; i.e.

$$R_{XY}^2(\tau) - R_{XX}(0)\, R_{YY}(0) \leq 0 \tag{2.53}$$

   From this property 2 follows.

3. Property 3 is a consequence of the well-known fact that the arithmetic mean of two positive numbers is always greater than or equal to their geometric mean.

If for two processes $X(t)$ and $Y(t)$

$$R_{XY}(t, t+\tau) = 0, \quad \text{for all } t \text{ and } \tau \tag{2.54}$$

then we say that $X(t)$ and $Y(t)$ are orthogonal processes. In case two processes $X(t)$ and $Y(t)$ are statistically independent, the cross-correlation function can be written as

$$R_{XY}(t, t+\tau) = \mathrm{E}[X(t)]\,\mathrm{E}[Y(t+\tau)] \tag{2.55}$$

If, moreover, $X(t)$ and $Y(t)$ are at least wide-sense stationary, then Equation (2.55) becomes

$$R_{XY}(t, t+\tau) = \overline{X}\,\overline{Y} \tag{2.56}$$

Two stochastic processes $X(t)$ and $Y(t)$ are called jointly ergodic if the individual processes are ergodic and if the time-averaged cross-correlation function equals the statistical cross-correlation function, i.e. if

$$\mathrm{A}[X(t)\,Y(t+\tau)] = \mathrm{E}[X(t)\,Y(t+\tau)] = R_{XY}(\tau) \tag{2.57}$$

In practice one more often uses spectra, to be dealt with in the next chapter, than correlation functions, as the measurement equipment for spectra is more developed than that for correlations. In that chapter it will be shown that the correlation function acts as the basis for calculating the spectrum. However, the correlation function in itself also has interesting applications, as is concluded from the following examples.

Example 2.4:

It will be shown that based on a correlation function, by means of the system described in this example, one is able to measure a distance. Consider a system (see Figure 2.4) where a signal source produces a random signal, being a realization of a stochastic process. Let us

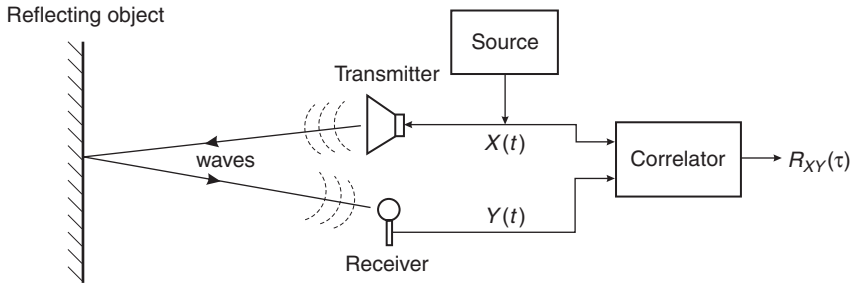


Figure 2.4 Set-up for measuring a distance based on the correlation function

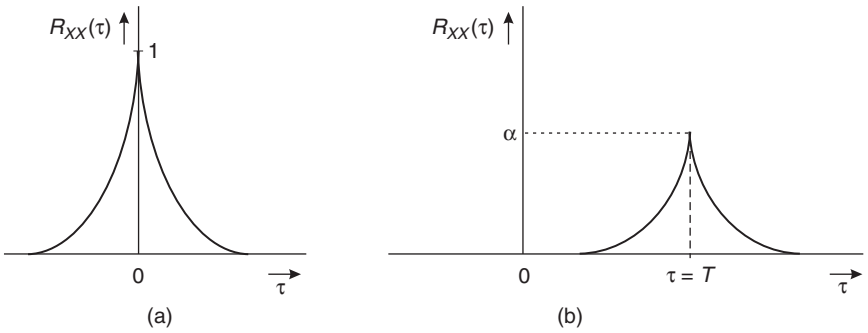


Figure 2.5 (a) The autocorrelation function of the transmitted signal and (b) the measured cross-correlation function of the distance measuring set-up

assume the process to be wide-sense stationary. The signal is applied to a transmitter that produces a wave in a transmission medium; let it be an acoustic wave or an electromagnetic wave. We denote the transmitted random wave by $X(t)$. Let us further suppose that the transmitted wave strikes a distant object and that this object (partly) reflects the wave. Then this reflected wave will travel backwards to the position of the measuring equipment. The measuring equipment comprises a receiver and the received signal is denoted as $Y(t)$. Both the transmitted signal $X(t)$ and the received signal $Y(t)$ are applied to a correlator that produces the cross-correlation function $R_{XY}(\tau)$. In the next section it will be explained how this correlation equipment operates. The reflected wave will be a delayed and attenuated version of the transmitted wave; i.e. we assume $Y(t) = \alpha X(t - T)$, where T is the total travel time. The cross-correlation function will be

$$R_{XY}(\tau) = \mathrm{E}[X(t)\, Y(t + \tau)] = \mathrm{E}[X(t)\, \alpha X(t - T + \tau)] = \alpha R_{XX}(\tau - T) \tag{2.58}$$

Most autocorrelation functions have a peak at $\tau = 0$, as shown in Figure 2.5(a). Let us normalize this peak to unity; then the cross-correlation result will be as depicted in Figure 2.5(b). From this latter picture a few conclusions may be drawn with respect to the application at hand. Firstly, when we detect the position of the peak in the cross-correlation function we will be able to establish T and if the speed of propagation of the wave in the medium is known, then the distance of the object can be derived from that. Secondly, the relative height α of the peak can be interpreted as a measure for the size of the object.

It will be clear that this method is very useful in such ranging systems as radar and underwater acoustic distance measurement. Most ranging systems use pulsed continuous wave (CW) signals for that. The advantage of the system presented here is the fact that for the transmitted signal a noise waveform is used. Such a waveform cannot easily be detected by the probed object, in contrast to the pulsed CW systems, since it has no replica available of the transmitted signal and therefore is not able to perform the correlation. The probed object only observes an increase in received noise level.

□

Example 2.5:

Yet another interesting example of the application of the correlation concept is in the field of reducing the distortion of received information signals. Let us suppose that a private subscriber has on the roof of his house an antenna for receiving TV broadcast signals. Due to a tall building near his house the TV signal is reflected, so that the subscriber receives the signal from a certain transmitter twice, once directly from the transmitter and a second time reflected from the neighbouring building. On the TV screen this produces a ghost of the original picture and spoils the picture. We call the direct signal $X(t)$ and the reflected one will then be $\alpha X(t-T)$, where T represents the difference in travel time between the direct and the reflected signal. The total received signal is therefore written as $Y(t) = X(t) + \alpha X(t-T)$. Let us consider the autocorrelation function of this process:

$$\begin{aligned} R_{YY}(\tau) &= \mathrm{E}[\{X(t) + \alpha X(t-T)\}\,\{X(t+\tau) + \alpha X(t-T+\tau)\}] \\ &= \mathrm{E}[X(t)\,X(t+\tau) + \alpha X(t)\,X(t-T+\tau) + \alpha X(t-T)\,X(t+\tau) \\ &\quad + \alpha^2 X(t-T)\,X(t-T+\tau)] \\ &= (1+\alpha^2)R_{XX}(\tau) + \alpha R_{XX}(\tau - T) + \alpha R_{XX}(\tau + T) \end{aligned} \tag{2.59}$$

The autocorrelation function $R_{YY}(\tau)$ of the received signal $Y(t)$ will consist of that of the original signal $R_{XX}(\tau)$ multiplied by $1+\alpha^2$ and besides that two shifted versions of $R_{XX}(\tau)$. These versions are multiplied by α and shifted in time over respectively T and $-T$. If it is assumed that the autocorrelation function $R_{XX}(\tau)$ shows the same peaked appearance as in the previous example, then the autocorrelation function of the received signal $Y(t)$ looks like that in Figure 2.6. Let us once again suppose that both α and T can be determined from this measurement. Then we will show that these parameters can be used to reduce the distortion caused by the reflection from the nearby building; namely we delay the received signal by an amount of T and multiply this delayed version by α. This delayed and multiplied version is subtracted from the received signal, so that after this operation we have the signal $Z(t) = Y(t) - \alpha Y(t-T)$. Inserting the undistorted signal $X(t)$ into this yields

$$\begin{aligned} Z(t) &= Y(t) - \alpha Y(t-T) \\ &= X(t) + \alpha X(t-T) - \alpha X(t-T) - \alpha^2 X(t-2T) \\ &= X(t) - \alpha^2 X(t-2T) \end{aligned} \tag{2.60}$$

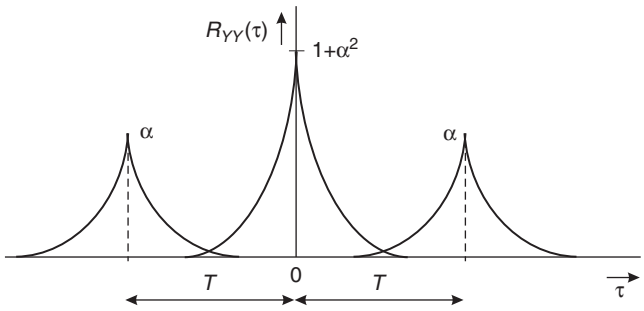


Figure 2.6 The autocorrelation function of a random signal plus its delayed and attenuated versions

From this equation it is concluded that indeed the term with a delay of T has been removed. One may argue that, instead, the term $\alpha^2 X(t-2T)$ has been introduced. That is right, but it is not unreasonable to assume that the reflection coefficient α is (much) less than unity, so that this newly introduced term is smaller by the factor of α compared to the distortion in the received signal. If this is nevertheless unacceptable then a further reduction is achieved by also adding the term $\alpha^2 Y(t-2T)$ to the received signal. This removes the distortion at $2T$ and in its turn introduces a term that is still smaller by an amount of α^3 at a delay of $3T$, etc. In this way the distortion may be reduced to an arbitrary small amount.

□

Apart from these examples there are several applications that use correlation as the basic signal processing method for extracting information from an observation.

2.2.4 Measuring Correlation Functions

In a practical situation it is impossible to measure a correlation function. This is due to the fact that we will never have available the entire ensemble of sample functions of the process in question. Even if we did have them then it would nevertheless be impossible to cope with an infinite number of sample functions. Thus we have to confine ourselves to a limited class of processes, e.g. to the class of ergodic processes. We have established before that most of the time it is difficult or even impossible to determine whether a process is ergodic or not. Unless the opposite is clear, we will assume ergodicity in practice; this greatly simplifies matters, especially for measuring correlation functions. This assumption enables the wanted correlation function based on just a single sample function to be determined, as is evident from Equation (2.25).

In Figure 2.7 a block schematic is shown for a possible set-up to measure a cross-correlation function $R_{XY}(\tau)$, where the assumption has to be made that the processes $X(t)$ and $Y(t)$ are jointly ergodic. The sample functions $x(t)$ and $y(t)$ should be applied to the inputs at least starting at $t = -T + \tau$ up until $t = T + \tau$. The signal $x(t)$ is delayed and applied to a multiplier whereas $y(t)$ is applied undelayed to a second input of the same multiplier. The multiplier's output is applied to an integrator that integrates over a period $2T$. Looking at this scheme we conclude that the measured output is

$$R_o(\tau, T) = \frac{1}{2T}\int_{-T+\tau}^{T+\tau} x(t-\tau)\, y(t)\, \mathrm{d}t = \frac{1}{2T}\int_{-T}^{T} x(t)\, y(t+\tau)\, \mathrm{d}t \tag{2.61}$$

If the integration time $2T$ is taken long enough, and remembering the assumption on the jointly ergodicity of $X(t)$ and $Y(t)$, the measured value $R_o(\tau, T)$ will approximate $R_{XY}(\tau)$. By

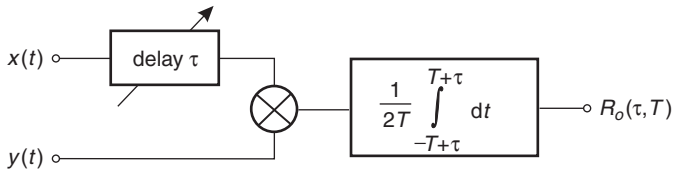


Figure 2.7 Measurement scheme for correlation functions

varying τ the function can be measured for different values of the argument. By simply short-circuiting the two inputs and applying a single signal $x(t)$ to this common input, the autocorrelation function $R_{XX}(\tau)$ is measured.

In practice only finite measuring times can be realized. In general this will introduce an error in the measured result. In the next example this point will be further elaborated.

Example 2.6:

Let us consider the example that has been subject of our studies several times before, namely the cosine waveform with amplitude A and random phase that has a uniform distribution over one period of the cosine. This process has been described in Example 2.1. Suppose we want to measure the autocorrelation function of this process using the set-up given in Figure 2.7. The inputs are short-circuited and the signal is applied to these common inputs. If the given process is substituted in Equation (2.61), we find

$$\begin{aligned} R_{\mathrm{o}}(\tau, T) &= \frac{1}{2T} A^2 \int_{-T}^{T} \cos(\omega t - \theta)\, \cos(\omega t + \omega\tau - \theta)\, \mathrm{d}t \\ &= \frac{A^2}{4T} \int_{-T}^{T} [\cos\omega\tau + \cos(2\omega t + \omega\tau - 2\theta)]\, \mathrm{d}t \end{aligned} \tag{2.62}$$

In this equation the random variable Θ has not been used, but the specific value θ that corresponds to the selected realization of the process. The first term in the integrand of this integral produces $(A^2/2)\, \cos(\omega\tau)$, the value of the autocorrelation function of this process, as was concluded in Example 2.1. The second term in the integrand must be a measurement error. The magnitude of this error is determined by evaluating the corresponding integral. This yields

$$e(\tau, T) = \frac{A^2}{2} \cos(\omega\tau - 2\theta) \frac{\sin(2\omega T)}{2\omega T} \tag{2.63}$$

The error has an oscillating character as a function of T, while the absolute value of the error decreases inversely with T. At large values of T the error approaches 0. If, for example, the autocorrelation function has to be measured with an accuracy of 1%, then the condition $1/(2\omega T) < 0.01$ should be fulfilled, or equivalently the measurement time should satisfy $2T > 100/\omega$.

Although this analysis looks nice, its applicability is limited. In practice the autocorrelation function is not known beforehand; that is why we want to measure it. Thus the above error analysis cannot be carried out. The solution to this problem consists of doing a best-effort measurement and then to make an estimate of the error in the correlation function. Looking back, it possible to decide whether the measurement time was long enough for the required accuracy. If not, the measurement can be redone using a larger (estimated) measurement time based on the error analysis. In this way accuracy can be iteratively improved.

□

2.2.5 Covariance Functions

The concept of covariance of two random variables can be extended to stochastic processes. The autocovariance function of a stochastic process is defined as

$$C_{XX}(t, t+\tau) \triangleq \mathrm{E}[\{X(t) - \mathrm{E}[X(t)]\}\,\{X(t+\tau) - \mathrm{E}[X(t+\tau)]\}] \tag{2.64}$$

This can be written as

$$C_{XX}(t, t+\tau) = R_{XX}(t, t+\tau) - \mathrm{E}[X(t)]\,\mathrm{E}[X(t+\tau)] \tag{2.65}$$

The cross-covariance function of two processes $X(t)$ and $Y(t)$ is defined as

$$C_{XY}(t, t+\tau) \triangleq \mathrm{E}[\{X(t) - \mathrm{E}[X(t)]\}\,\{Y(t+\tau) - \mathrm{E}[Y(t+\tau)]\}] \tag{2.66}$$

or

$$C_{XY}(t, t+\tau) = R_{XY}(t, t+\tau) - \mathrm{E}[X(t)]\,\mathrm{E}[Y(t+\tau)] \tag{2.67}$$

For processes that are at least jointly wide-sense stationary the second expressions in the right-hand sides of Equations (2.65) and (2.67) can be simplified, yielding

$$C_{XX}(\tau) = R_{XX}(\tau) - \overline{X}^2 \tag{2.68}$$

and

$$C_{XY}(\tau) = R_{XY}(\tau) - \overline{X}\,\overline{Y} \tag{2.69}$$

respectively. From Equation (2.68) and property 4 of the autocorrelation function in Section 2.2.1 it follows immediately that

$$\lim_{|\tau| \to \infty} C_{XX}(\tau) = 0 \tag{2.70}$$

provided the process $X(t)$ does not have a periodic component. If in Equation (2.64) the value $\tau = 0$ is used we obtain the variance of the process. In the case of wide-sense stationary processes the variance is independent of time, and using Equation (2.68) we arrive at

$$\sigma_X^2 \triangleq \mathrm{E}[\{X(t) - \mathrm{E}[X(t)]\}^2] = C_{XX}(0) = R_{XX}(0) - \overline{X}^2 \tag{2.71}$$

If for two processes

$$C_{XY}(t, t+\tau) \equiv 0 \tag{2.72}$$

then these processes are called uncorrelated processes. According to Equation (2.67) this has as a consequence

$$R_{XY}(t, t+\tau) = \mathrm{E}[X(t)]\,\mathrm{E}[Y(t+\tau)] \tag{2.73}$$

Since this latter equation is identical to Equation (2.55), it follows that independent processes are uncorrelated. The converse is not necessarily true, unless the processes are jointly Gaussian processes (see Section 2.3).

2.2.6 Physical Interpretation of Process Parameters

In the previous sections stochastic processes have been described from a mathematical point of view. In practice we want to relate these descriptions to physical concepts such as a signal, represented, for example, by a voltage or a current. In these cases the following physical interpretations are connected to the parameters of the stochastic processes:

- The mean $\overline{X(t)}$ is proportional to the d.c. component of the signal.
- The squared mean value $\overline{X(t)}^2$ is proportional to the power in the d.c. component of the signal.
- The mean squared value $\overline{X^2(t)}$ is proportional to the total average power of the signal.
- The variance $\sigma_X^2 \triangleq \overline{X^2(t)} - \overline{X(t)}^2$ is proportional to the power in the time-varying components of the signal, i.e. the a.c. power.
- The standard deviation σ_X is the square root of the mean squared value of the time-varying components, i.e. the root-mean-square (r.m.s.) value.

In Chapter 6 the proportionality factors will be deduced. Now it suffices to say that this proportionality factor becomes unity in case the load is purely resistive and equal to one.

Although the above interpretations serve to make the engineer familiar with the practical value of stochastic processes, it must be stressed that they only apply to the special case of signals that can be modelled as ergodic processes.

2.3 GAUSSIAN PROCESSES

Several processes can be modelled by what is called a Gaussian process; among these is the thermal noise process that will be presented in Chapter 6. As the name suggests, these processes are described by Gaussian distributions. Recall that the probability density function of a Gaussian random variable X is defined by [1–5]

$$f_X(x) = \frac{1}{\sigma_X\sqrt{2\pi}} \exp\left[-\frac{(x-\overline{X})^2}{2\sigma_X^2}\right] \tag{2.74}$$

The Gaussian distribution is frequently encountered in engineering and science. When considering two jointly Gaussian random variables X and Y we sometimes need the joint probability density function, as will become apparent in the sequel

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}\exp\left\{\frac{-1}{2(1-\rho^2)}\left[\frac{(x-\overline{X})^2}{\sigma_X^2} - \frac{2\rho(x-\overline{X})(y-\overline{Y})}{\sigma_X\sigma_Y} + \frac{(y-\overline{Y})^2}{\sigma_Y^2}\right]\right\} \tag{2.75}$$

where ρ is the correlation coefficient defined by

$$\rho \triangleq \frac{\mathrm{E}[(X-\overline{X})(Y-\overline{Y})]}{\sigma_X\sigma_Y} \tag{2.76}$$

For N jointly Gaussian random variables $X_1, X_2, \ldots, X_N$, the joint probability density function reads

$$f_{X_1X_2\cdots X_N}(x_1,x_2,\ldots,x_N) = \frac{|\mathbf{C}_X^{-1}|^{\frac{1}{2}}}{(2\pi)^{N/2}}\exp\left[-\frac{(\mathbf{x}-\overline{\mathbf{X}})^{\mathrm{T}}\mathbf{C}_X^{-1}(\mathbf{x}-\overline{\mathbf{X}})}{2}\right] \tag{2.77}$$

where we define the vector

$$\mathbf{x}-\overline{\mathbf{X}} \triangleq \begin{bmatrix} x_1-\overline{X_1} \\ x_2-\overline{X_2} \\ \vdots \\ x_N-\overline{X_N} \end{bmatrix} \tag{2.78}$$

and the covariance matrix

$$\mathbf{C}_X \triangleq \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1N} \\ C_{21} & C_{22} & \cdots & C_{2N} \\ \vdots & \vdots & & \vdots \\ C_{N1} & C_{N2} & \cdots & C_{NN} \end{bmatrix} \tag{2.79}$$

In the foregoing we used $\mathbf{x}^{\mathrm{T}}$ for the matrix transpose, $\mathbf{C}^{-1}$ for the matrix inverse and $|\mathbf{C}|$ for the determinant. The elements of the covariance matrix are defined by

$$C_{ij} \triangleq \mathrm{E}(X_i-\overline{X_i})(X_j-\overline{X_j})] \tag{2.80}$$

The diagonal elements of the covariance matrix equal the variances of the various random variables, i.e. $C_{ii} = \sigma_{X_i}^2$. It is easily verified that Equations (2.74) and (2.75) are special cases of Equation (2.77).

Gaussian variables as described above have a few interesting properties, which have their consequences for Gaussian processes. These properties are [1–5]:

1. Gaussian random variables are completely specified only by their first and second order moments, i.e. by their means, variances and covariances. This is immediately apparent, since these are the only quantities present in Equation (2.77).
2. When Gaussian random variables are uncorrelated, they are independent. For uncorrelated random variables (i.e. $\rho = 0$) the covariance matrix is reduced to a diagonal matrix. It is easily verified from Equation (2.77) that in such a case the probability density function of N variables can be written as the product of N functions of the type given in Equation (2.74).
3. A linear combination of Gaussian random variables produces another Gaussian variable. For the proof of this see reference [2] and Problem 8.3.

We are now able to define a Gaussian stochastic process. Referring to Equation (2.77), a process $X(t)$ is called a Gaussian process if the random variables $X_1 = X(t_1)$, $X_2 = X(t_2), \ldots, X_N = X(t_N)$ are jointly Gaussian and thus satisfy

$$f_X(x_1, \ldots, x_N; t_1, \ldots, t_N) = \frac{|\mathbf{C}_X^{-1}|^{\frac{1}{2}}}{(2\pi)^{N/2}} \exp\left[-\frac{(\mathbf{x} - \overline{\mathbf{X}})^{\mathrm{T}} \mathbf{C}_X^{-1} (\mathbf{x} - \overline{\mathbf{X}})}{2}\right] \tag{2.81}$$

for all arbitrary N and for any set of times $t_1, \ldots, t_N$. Now the mean values $\overline{X_i}$ of $X(t_i)$ are

$$\overline{X_i} = \mathrm{E}[X(t_i)] \tag{2.82}$$

and the elements of the covariance matrix are

$$\begin{aligned} C_{ij} &= \mathrm{E}[(X_i - \overline{X_i})(X_j - \overline{X_j})] \\ &= \mathrm{E}[\{X(t_i) - \mathrm{E}[X(t_i)]\}\{X(t_j) - \mathrm{E}[X(t_j)]\}] \\ &= C_{XX}(t_i, t_j) \end{aligned} \tag{2.83}$$

which is the autocovariance function as defined by Equation (2.64).

Gaussian processes have a few interesting properties.

Properties of Gaussian Processes

1. Gaussian processes are completely specified by their mean $\mathrm{E}[X(t)]$ and autocorrelation function $R_{XX}(t_i, t_j)$.
2. A wide-sense stationary Gaussian process is also strict-sense stationary.
3. If the jointly Gaussian processes $X(t)$ and $Y(t)$ are uncorrelated, then they are independent.
4. If the Gaussian process $X(t)$ is passed through a linear time-invariant system, then the corresponding output process $Y(t)$ is also a Gaussian process.

These properties are closely related to the properties of jointly Gaussian random variables previously discussed in this section. Let us briefly comment on the properties:

1. We saw before that the joint probability density function is completely determined when the mean and autocovariance are known. However, these two quantities as functions of time in their turn determine the autocorrelation function (see Equation (2.68)).
2. The nth-order probability density function of a Gaussian process only depends on the two functions $\mathrm{E}[X(t)]$ and $C_{XX}(t, t+\tau)$. When the process is wide-sense stationary then these functions do not depend on the absolute time t, and as a consequence

$$f_X(x_1, \ldots, x_N; t_1, \ldots, t_N) = f_X(x_1, \ldots, x_N; t_1 + \tau, \ldots, t_N + \tau) \tag{2.84}$$

   Since this is valid for all arbitrary N and all τ, it is concluded that the process is strict-sense stationary.
3. This property is a straightforward consequence of the property of jointly random variables discussed before.
4. Passing a process through a linear time-invariant system is described by a convolution, which may be considered as the limit of a weighted sum of samples of the input process. From the preceding we know that a linear combination of Gaussian variables produces another Gaussian variable.

2.4 COMPLEX PROCESSES

A complex stochastic process is defined by

$$Z(t) \triangleq X(t) + \mathrm{j}Y(t) \tag{2.85}$$

with $X(t)$ and $Y(t)$ real stochastic processes. Such a process is said to be stationary if $X(t)$ and $Y(t)$ are jointly stationary. Expectation and the autocorrelation function of a complex stochastic process are defined as

$$\mathrm{E}[Z(t)] \triangleq \mathrm{E}[X(t) + \mathrm{j}Y(t)] = \mathrm{E}[X(t)] + \mathrm{j}\mathrm{E}[Y(t)] \tag{2.86}$$

and

$$R_{ZZ}(t, t+\tau) \triangleq \mathrm{E}[Z^*(t)\, Z(t+\tau)] \tag{2.87}$$

where * indicates the complex conjugate.

For the autocovariance function the definition of Equation (2.87) is used, where $Z(t)$ is replaced by the stochastic process $Z(t) - \mathrm{E}[Z(t)]$. This yields

$$C_{ZZ}(t, t+\tau) = R_{ZZ}(t, t+\tau) - \mathrm{E}^*[Z(t)]\ \mathrm{E}[Z(t+\tau)] \tag{2.88}$$

The cross-correlation function of two complex processes $Z_i(t)$ and $Z_j(t)$ reads

$$R_{Z_iZ_j}(t, t+\tau) = \mathrm{E}[Z_i^*(t)\ Z_j(t+\tau)] \tag{2.89}$$

and the cross-covariance function is found from Equation (2.89) by replacing $Z_{i,j}(t)$ with $Z_{i,j}(t) - \mathrm{E}[Z_{i,j}(t)]$; this yields

$$C_{Z_iZ_j}(t, t+\tau) = R_{Z_iZ_j}(t, t+\tau) - \mathrm{E}^*[Z_i(t)]\,\mathrm{E}[Z_j(t+\tau)] \tag{2.90}$$

In the chapters that follow we will work exclusively with real processes, unless it is explicitly indicated that complex processes are considered.

One may wonder why the correlation functions of complex processes are defined in the way it has been done in Equations (2.87) and (2.89). The explanation for this arises from an engineering point of view; namely the given expressions of the correlation functions evaluated for $\tau = 0$ have to result in the expectation of the squared process for real processes. In engineering calculations real processes are replaced many times by complex processes of the form $I = \hat{I}\exp(\mathrm{j}\omega t)$ (for a current) or $V = \hat{V}\exp(\mathrm{j}\omega t)$ (for a voltage). In these cases the correlation function for $\tau = 0$ should be a quantity that is proportional to the mean power. The given definitions satisfy this requirement.

2.5 DISCRETE-TIME PROCESSES

In Chapter 1 the discrete-time process was introduced by sampling a continuous stochastic process. However, at this point we are not yet able to develop a sampling theorem for stochastic processes analogously to that for deterministic signals [1]. We will derive such a theorem in Chapter 3. This means that in this section we deal with random sequences as such, irrespective of their origin. In Chapter 1 we introduced the notation $X[n]$ for random sequences. In this section we will assume that the sequences are real. However, they can be complex valued. Extension to complex discrete-time processes is similar to what was derived in the former section.

In the next subsection we will resume the most important properties of discrete-time processes. Since such processes are actually special cases of continuous stochastic processes the properties are self-evident.

2.5.1 Mean, Correlation Functions and Covariance Functions

The mean value of a discrete-time process is found by

$$\mathrm{E}[X[n]] = \overline{X[n]} \triangleq \int_{-\infty}^{\infty} x\, f_X(x; n)\, \mathrm{d}x \tag{2.91}$$

Recall that the process is time-discrete but the x values are continuous, so that indeed the expectation (or ensemble mean) is written as an integral over a continuous probability density function. This function describes the random variable $X[n]$ by considering all

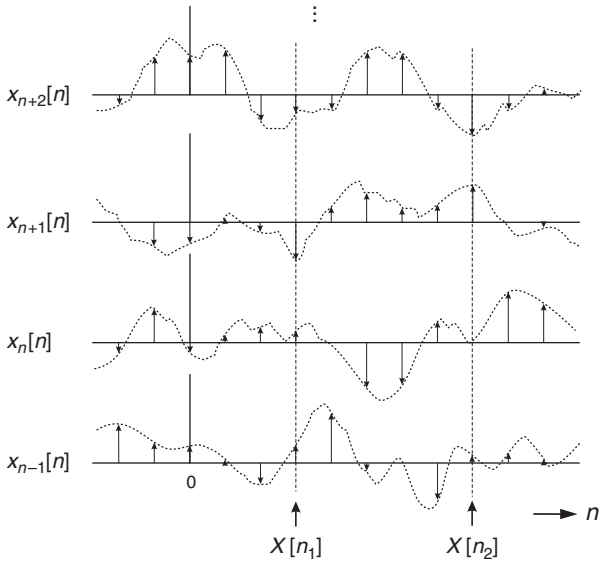


Figure 2.8 Random variables $X[n_1]$ and $X[n_2]$ that arise when considering the ensemble values of the discrete-time process $X[n]$ at fixed positions n_1 and n_2

possible ensemble realizations of the process at a fixed integer position for example n_1 (see Figure 2.8). For real processes the autocorrelation sequence is defined as

$$R_{XX}[n_1, n_2] \triangleq \mathrm{E}[X[n_1]\, X[n_2]] \triangleq \iint x_1 x_2\, f_X(x_1, x_2; n_1, n_2)\, \mathrm{d}x_1\, \mathrm{d}x_2 \tag{2.92}$$

where the process is now considered at two positions n_1 and n_2 jointly (see again Figure 2.8).

For the autocovariance sequence of this process (compare to Equation (2.65))

$$C_{XX}[n_1, n_2] = R_{XX}[n_1, n_2] - \mathrm{E}[X[n_1]]\, \mathrm{E}[X[n_2]] \tag{2.93}$$

The cross-correlation and cross-covariance sequences are defined analogously, namely respectively as

$$R_{XY}[n, n+m] \triangleq \mathrm{E}[X[n]\, Y[n+m]] \tag{2.94}$$

and (compare with Equation (2.67))

$$C_{XY}[n, n+m] \triangleq R_{XY}[n, n+m] - \mathrm{E}[X[n]]\, \mathrm{E}[Y[n+m]] \tag{2.95}$$

A discrete-time process is called wide-sense stationary if the next two conditions hold jointly:

$$\mathrm{E}[X[n]] = \text{constant} \tag{2.96}$$

$$R_{XX}[n, n+m] = R_{XX}[m] \tag{2.97}$$

i.e. the autocorrelation sequence only depends on the difference m of the integer positions.

Two discrete-time processes are jointly wide-sense stationary if they are individually wide-sense stationary and moreover

$$R_{XY}[n, n+m] = R_{XY}[m] \tag{2.98}$$

i.e. the cross-correlation sequence only depends on the difference m of the integer positions.

The time average of a discrete-time process is defined as

$$\mathrm{A}[X[n]] \triangleq \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} X[n] \tag{2.99}$$

A wide-sense stationary discrete-time process is ergodic if the two conditions are satisfied

$$\mathrm{A}[X[n]] = \mathrm{E}[X[n]] = \overline{X[n]} \tag{2.100}$$

and

$$\mathrm{A}[X[n]\,X[n+m]] = \mathrm{E}[X[n]\,X[n+m]] = R_{XX}[m] \tag{2.101}$$

2.6 SUMMARY

An ensemble is the set of all possible realizations of a stochastic process $X(t)$. A realization or sample function is provided by a random selection out of this ensemble. For the description of stochastic processes a parameter is added to the well-known definitions of the probability distribution function and the probability density function, namely the time parameter. This means that these functions in the case of a stochastic process are as a rule functions of time. When considering stationary processes certain time dependencies disappear; we thus arrive at first-order and second-order stationary processes, which are useful for practical applications.

The correlation concept is in random signal theory, analogously to probability theory, defined as the expectation of the product of two random variables. For the autocorrelation function these variables are $X(t)$ and $X(t+\tau)$, while for the cross-correlation function of two processes the quantities $X(t)$ and $Y(t+\tau)$ are used in the definition.

A wide-sense stationary process is a process where the mean value is constant and the autocorrelation function only depends on τ, not on the absolute time t. When calculating the expectations the time t is considered as a parameter; i.e. in these calculations t is given a fixed value. The random variable is the variable based on which outcome of the realization is chosen from the ensemble. When talking about 'mean' we have in mind the ensemble mean, unless it is explicitly indicated that a different definition is used (for instance the time average). In the case of an ergodic process the first- and second-order time averages equal the first- and second-order ensemble means, respectively. The theorem that has been presented on cyclo-stationary processes plays an important role in 'making stationary' certain classes of processes. The covariance functions of stochastic processes are the correlation functions of these processes minus their own process mean values.

Physical interpretations of several stochastic concepts have been presented. Gaussian processes get special attention as they are of practical importance and possess a few

interesting and convenient properties. Complex processes are defined analogously to the usual method for complex variables.

Finally, several definitions and properties of continuous stochastic processes are redefined for discrete-time processes.

2.7 PROBLEMS

2.1 All sample functions of a stochastic process are constant, i.e. $X(t) = C = \text{constant}$, where C is a discrete random variable that may assume the values $C_1 = 1$, $C_2 = 3$ and $C_3 = 4$, with probabilities of 0.5, 0.3 and 0.2, respectively.

(a) Determine the probability density function of $X(t)$.

(b) Calculate the mean and variance of $X(t)$.

2.2 Consider a stationary Gaussian process with a mean of zero.

(a) Determine and sketch the probability density function of this process after passing it through an ideal half-wave rectifier.

(b) Same question for the situation where the process is applied to a full-wave rectifier.

2.3 A stochastic process comprises four sample functions, namely $x(t, s_1) = 1$, $x(t, s_2) = t$, $x(t, s_3) = \cos t$ and $x(t, s_4) = 2 \sin t$, which occur with equal probabilities.

(a) Determine the probability density function of $X(t)$.

(b) Is the process stationary in any sense?

2.4 Consider the process

$$X(t) = \sum_{n=1}^{N} A_n \cos(\omega_n t) + B_n \sin(\omega_n t)$$

where A_n and B_n are random variables that are mutually uncorrelated, have zero mean and of which

$$\mathrm{E}[A_n^2] = \mathrm{E}[B_n^2] = \sigma^2$$

The quantities $\{\omega_n\}$ are constants.

(a) Calculate the autocorrelation function of $X(t)$.

(b) Is the process wide-sense stationary?

2.5 Consider the stochastic process $X(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$, with ω_0 a constant and A and B random variables. What are the conditions for A and B in order for $X(t)$ to be wide-sense stationary?

2.6 Consider the process $X(t) = A \cos(\omega_0 t - \Theta)$, where A and Θ are independent random variables and Θ is uniformly distributed on the interval $(0, 2\pi]$.

(a) Is this process wide-sense stationary?

(b) Is it ergodic?

2.7 Consider the two processes

$$X(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$$
$$Y(t) = A\cos(\omega_0 t) - B\sin(\omega_0 t)$$

with A and B independent random variables, both with zero mean and equal variance of σ^2. The angular frequency ω_0 is constant.

(a) Are the processes $X(t)$ and $Y(t)$ wide-sense stationary?
(b) Are they jointly wide-sense stationary?

2.8 Consider the stochastic process $X(t) = A\sin(\omega_0 t - \Theta)$, with A and ω_0 constants, and Θ a random variable that is uniformly distributed on the interval $(0, 2\pi]$. We define a new process by means of $Y(t) = X^2(t)$.

(a) Are $X(t)$ and $Y(t)$ wide-sense stationary?
(b) Calculate the autocorrelation function of $Y(t)$.
(c) Calculate the cross-correlation function of $X(t)$ and $Y(t)$.
(d) Are $X(t)$ and $Y(t)$ jointly wide-sense stationary?
(e) Calculate and sketch the probability distribution function of $Y(t)$.
(f) Calculate and sketch the probability density function of $Y(t)$.

2.9 Repeat Problem 2.8 when $X(t)$ is half-wave rectified. Use Matlab to plot the autocorrelation function.

2.10 Repeat Problem 2.8 when $X(t)$ is full-wave rectified. Use Matlab to plot the autocorrelation function.

2.11 The function $p(t)$ is defined as

$$p(t) = \begin{cases} 1, & 0 \le t \le \frac{3}{4}T \\ 0, & \text{all other values of } t \end{cases}$$

By means of this function we define the stochastic process

$$X(t) = \sum_{n=-\infty}^{\infty} p(t - nT - \Theta)$$

where Θ is a random variable that is uniformly distributed on the interval $[0, T)$.

(a) Sketch a possible realization of $X(t)$.
(b) Calculate the mean value of $X(t)$.
(c) Calculate and sketch the autocorrelation function of $X(t)$.
(d) Calculate and sketch the probability distribution function of $X(t)$.
(e) Calculate and sketch the probability density function of $X(t)$.
(f) Calculate the variance of $X(t)$.

2.12 Two functions $p_1(t)$ and $p_2(t)$ are defined as

$$p_1(t) = \begin{cases} 1, & 0 \le t \le \frac{1}{3}T \\ 0, & \text{all other values of } t \end{cases}$$

and

$$p_2(t) = \begin{cases} 1, & 0 \le t \le \frac{2}{3}T \\ 0, & \text{all other values of } t \end{cases}$$

Based on these functions the stochastic processes $X(t)$ and $Y(t)$ are defined as

$$X(t) = \sum_{n=-\infty}^{\infty} p_1(t - nT - \Theta)$$
$$Y(t) = \sum_{n=-\infty}^{\infty} p_2(t - nT - \Theta)$$

and

$$W(t) \triangleq X(t) + Y(t)$$

where Θ is a random variable that is uniformly distributed on the interval $[0, T)$.

(a) Sketch possible realizations of $X(t)$ and $Y(t)$.
(b) Calculate and sketch the autocorrelation function of $X(t)$.
(c) Calculate and sketch the autocorrelation function of $Y(t)$.
(d) Calculate and sketch the autocorrelation function of $W(t)$.
(e) Calculate the power in $W(t)$, i.e. $\mathrm{E}[W^2(t)]$.

2.13 The processes $X(t)$ and $Y(t)$ are independent with a mean value of zero and autocorrelation functions $R_{XX}(\tau) = \exp(-|\tau|)$ and $R_{YY}(\tau) = \cos(2\pi\tau)$, respectively.

(a) Derive the autocorrelation function of the sum $W_1(t) = X(t) + Y(t)$.
(b) Derive the autocorrelation function of the difference $W_2(t) = X(t) - Y(t)$.
(c) Calculate the cross-correlation function of $W_1(t)$ and $W_2(t)$.

2.14 In Figure 2.9 the autocorrelation function of a wide-sense stationary stochastic process $X(t)$ is given.

(a) Calculate the value of $\mathrm{E}[X(t)]$.
(b) Calculate the value of $\mathrm{E}[X^2(t)]$.
(c) Calculate the value of σ_X^2.

2.15 Starting from the wide-sense stationary process $X(t)$ we define a new process as $Y(t) = X(t) - X(t + T)$.

(a) Show that the mean value of $Y(t)$ is zero, even if the mean value of $X(t)$ is not zero.
(b) Show that $\sigma_Y^2 = 2\{R_{XX}(0) - R_{XX}(T)\}$.

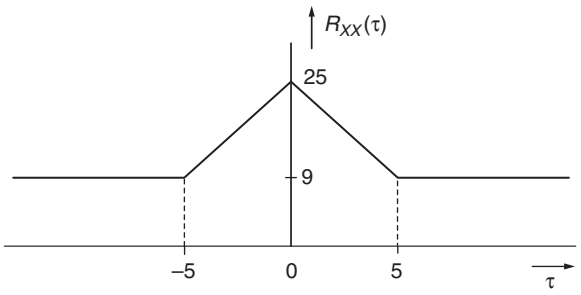


Figure 2.9

(c) If $Y(t) = X(t) + X(t+T)$ find expressions for $\mathrm{E}[Y(t)]$ and σ_Y^2. Compare these results with the answers found in (a) and (b).

2.16 Determine for each of the following functions whether it can be the autocorrelation function of a real wide-sense stationary process $X(t)$.

(a) $R_{XX}(\tau) = \mathrm{u}(\tau)\exp(-\tau)$.
(b) $R_{XX}(\tau) = 3\sin(7\tau)$.
(c) $R_{XX}(\tau) = (1+\tau^2)^{-1}$.
(d) $R_{XX}(\tau) = -\cos(2\tau)\ \exp(-|\tau|)$.
(e) $R_{XX}(\tau) = 3[\sin(4\tau)/(4\tau)]^2$.
(f) $R_{XX}(\tau) = 1 + 3\sin(8\tau)/(8\tau)$.

2.17 Consider the two processes $X(t)$ and $Y(t)$. Find expressions for the autocorrelation function of $W(t) = X(t) + Y(t)$ in the case where:

(a) $X(t)$ and $Y(t)$ are correlated;
(b) $X(t)$ and $Y(t)$ are uncorrelated;
(c) $X(t)$ and $Y(t)$ are uncorrelated and have mean values of zero.

2.18 The voltage of the output of a noise generator is measured using a d.c. voltmeter and a true root-mean-square (r.m.s.) meter that has a series capacitor at its input. The noise is known to be Gaussian and stationary. The reading of the d.c. meter is 3 V and that of the r.m.s. meter is 2 V. Derive an expression for the probability density function of the noise and make a plot of it using Matlab.

2.19 Two real jointly wide-sense stationary processes $X(t)$ and $Y(t)$ are used to define two complex processes as follows:

$$Z_1(t) = X(t) + \mathrm{j}Y(t)$$

and

$$Z_2(t) = X(t-T) - \mathrm{j}Y(t-T)$$

Calculate the cross-correlation function of the processes $Z_1(t)$ and $Z_2(t)$.

2.20 A voltage source is described as $V = 5\cos(\omega_0 t - \Theta)$, where Θ is a random variable that is uniformly distributed on $[0, 2\pi)$. This source is applied to an electric circuit and as a consequence the current flowing through the circuit is given by $I = 2\cos(\omega_0 t - \Theta + \pi/6)$.

(a) Calculate the cross-correlation function of V and I.

(b) Calculate the electrical power that is absorbed by the circuit.

(c) If in general an harmonic voltage at the terminals of a circuit is described by its complex notation $V = \hat{V}\exp[\mathrm{j}(\omega t - \Theta)]$ and the corresponding current that is flowing into the circuit by a similar notation $I = \hat{I}\exp[\mathrm{j}(\omega t - \Theta + \phi)]$, with ϕ a constant, show that the electrical power absorbed by the circuit is written as $P_{\mathrm{el}} = (\hat{V}\hat{I}\cos\phi)/2$.

2.21 Consider a discrete-time wide-sense stationary process $X[n]$. Show that for such a process

$$3R_{XX}[0] \geq |4R_{XX}[1] + 2R_{XX}[2]|$$

3

Spectra of Stochastic Processes

In Chapter 2 stochastic processes have been considered in the time domain exclusively; i.e. we used such concepts as the autocorrelation function, the cross-correlation function and the covariance function to describe the processes. When dealing with deterministic signals, we have the frequency domain at our disposal as a means to an alternative, dual description. One may wonder whether for stochastic processes a similar duality exists. This question is answered in the affirmative, but the relationship between time domain and frequency domain descriptions is different compared to deterministic signals. Hopping from one domain to the other is facilitated by the well-known Fourier transform and its inverse transform. A complicating factor is that for a random waveform (a sample function of the stochastic process) the Fourier transform generally does not exist.

3.1 THE POWER SPECTRUM

Due to the problems with the Fourier transform, a theoretical description of stochastic processes must basically start in the time domain, as given in Chapter 2. In this chapter we will confine ourselves exclusively to wide-sense stationary processes with the autocorrelation function $R_{XX}(\tau)$. Let us assume that it is allowed to apply the Fourier transform to $R_{XX}(\tau)$.

Theorem 3

The Wiener–Khinchin relations are

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) \exp(-\mathrm{j}\omega\tau)\, \mathrm{d}\tau \tag{3.1}$$

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) \exp(\mathrm{j}\omega\tau)\, \mathrm{d}\omega \tag{3.2}$$

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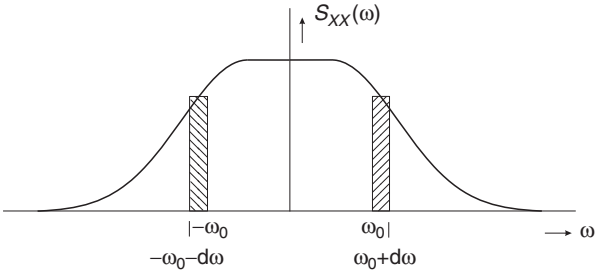


Figure 3.1 Interpretation of $S_{XX}(\omega)$

The function $S_{XX}(\omega)$ has an interesting interpretation, as will follow from the sequel. For that purpose we put the variable τ equal to zero in Equation (3.2). This yields

$$R_{XX}(0) = \mathrm{E}[X^2(t)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XX}(\omega)\,\mathrm{d}\omega \tag{3.3}$$

However, from Equation (2.18) it is concluded that $R_{XX}(0)$ equals the mean squared value of the process; this is called the mean power of the process, or just the power of the process. Now it follows from Equation (3.3) that $S_{XX}(\omega)$ represents the way in which the total power of the process is spread over the different frequency components. This is clear since integrating $S_{XX}(\omega)$ over the entire frequency axis produces the total power of the process. In other words, $2S_{XX}(\omega_0)\mathrm{d}\omega/(2\pi)$ is the power at the output of the bandpass filter with the passband transfer function

$$H(\omega) = \begin{cases} 1, & \omega_0 < |\omega| < \omega_0 + \mathrm{d}\omega \\ 0, & \text{elsewhere} \end{cases} \tag{3.4}$$

when the input of this filter consists of the process $X(t)$. This is further explained by Figure 3.1. Due to this interpretation the function $S_{XX}(\omega)$ is called the power spectral density, or briefly the power spectrum of the process $X(t)$. The Wiener–Khinchin relations state that the autocorrelation function and the power spectrum of a wide-sense stationary process are a Fourier transform pair. From the given interpretation the properties of the Fourier transform are as follows.

Properties of $S_{XX}(\omega)$

1. $S_{XX}(\omega) \geq 0$ (3.5)
2. $S_{XX}(-\omega) = S_{XX}(\omega)$, for a real process $X(t)$ (3.6)
3. $\mathrm{Im}\{S_{XX}(\omega)\} \equiv 0$ (3.7)
   where $\mathrm{Im}\{\cdot\}$ is defined as the imaginary part of the quantity between the braces
4. $\frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XX}(\omega)\,\mathrm{d}\omega = \mathrm{E}[X^2(t)] = R_{XX}(0) = P_{XX}$ (3.8)

Proofs of the properties:

1. Property 1 is connected to the interpretation of $S_{XX}(\omega)$ and a detailed proof will be given in Chapter 4.
2. Property 2 states that the power spectrum is an even function of ω. The proof of this property is based on Fourier theory and the fact that for a real process the autocorrelation function $R_{XX}(\tau)$ is real and even. The proof proceeds as follows:

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau)[\cos(\omega\tau) - \mathrm{j}\sin(\omega\tau)]\,\mathrm{d}\tau \tag{3.9}$$

   Since $R_{XX}(\tau)$ is real and even, the product of this function and a sine is odd. Therefore, this product makes no contribution to the integral, which runs over a symmetrical range of the integration variable. The remaining part is a product of $R_{XX}(\tau)$ and a cosine, both being even, resulting in an even function of ω.
3. The third property, $S_{XX}(\omega)$ being real, is proved as follows. Let us define the complex process $X(t) = R(t) + \mathrm{j}I(t)$, where $R(t)$ and $I(t)$ are real processes and represent the real and imaginary part of $X(t)$, respectively. Then after some straightforward calculations the autocorrelation function of $X(t)$ is

$$R_{XX}(\tau) = R_{RR}(\tau) + R_{II}(\tau) + \mathrm{j}[R_{RI}(\tau) - R_{IR}(\tau)] \tag{3.10}$$

   Inserting this into the Fourier integral produces the power spectrum

$$\begin{aligned} S_{XX}(\omega) = \int_{-\infty}^{\infty} &[R_{RR}(\tau) + R_{II}(\tau)][\cos(\omega\tau) - \mathrm{j}\sin(\omega\tau)] \\ &+ [R_{RI}(\tau) - R_{IR}(\tau)][\mathrm{j}\cos(\omega\tau) + \sin(\omega\tau)]\,\mathrm{d}\tau \end{aligned} \tag{3.11}$$

   The product of the sum of the two autocorrelation functions and the sine gives an odd result and consequently does not contribute to the integral. Using Equation (2.48), the difference $R_{RI}(\tau) - R_{IR}(\tau)$ can be rewritten as $R_{RI}(\tau) - R_{RI}(-\tau)$. This is an odd function and multiplied by a cosine the result remains odd. Thus, this product does not contribute to the integral either. Since all imaginary parts cancel out on integration, the resulting power spectrum will be real.
4. Property 4 follows immediately from the definition of Equation (3.2) and the definition of the autocorrelation function (see Equation (2.18)).

Example 3.1:

Consider once more the stochastic process $X(t) = A\cos(\omega_0 t - \Theta)$, with A and ω_0 constants and Θ a random variable that is uniformly distributed on the interval $(0, 2\pi]$. We know that this process is often met in practice. The autocorrelation function of this process has been

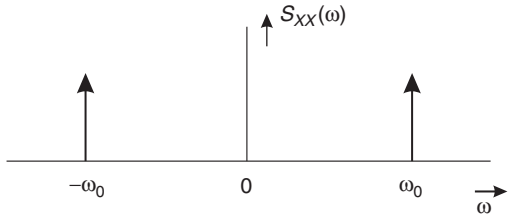


Figure 3.2 The power spectrum of a random phased cosine

shown to be $R_{XX}(\tau) = \frac{1}{2}A^2\cos(\omega_0\tau)$ (see Example 2.1). From a table of Fourier transforms (see Appendix G) it is easily revealed that

$$S_{XX}(\omega) = \frac{\pi}{2}A^2[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)] \tag{3.12}$$

This spectrum has been depicted in Figure 3.2 and consists of two δ functions, one at $\omega = \omega_0$ and another one at $\omega = -\omega_0$. Since the phase is random, introducing an extra constant phase to the cosine does not have any effect on the result. Thus, instead of the cosine a sine wave could also have been taken.

□

Example 3.2:

The second example is also important from a practical point of view, namely the spectrum of an oscillator. From physical considerations the process can be written as $X(t) = A\cos[\omega_0 t + \Psi(t)]$, with A and ω_0 constants and $\Psi(t)$ a random walk process defined by $\Psi(t) = \int_{-\infty}^{t} N(\tau)\mathrm{d}\tau$, where $N(t)$ is a so-called white noise process; i.e. the spectrum of $N(t)$ has a constant value for all frequencies. It can be shown that the autocorrelation function of the process $X(t)$ is [8]

$$R_{XX}(\tau) = \frac{A^2}{2}\exp(-\alpha|\tau|)\cos(\omega_0\tau) \tag{3.13}$$

where ω_0 is the nominal angular frequency of the oscillator and the exponential is due to random phase fluctuations. This autocorrelation function is shown in Figure 3.3(a). It will be clear that A is determined by the total power of the oscillator and from the Fourier table (see Appendix G) the power spectrum

$$S_{XX}(\omega) = \frac{\alpha A^2/2}{\alpha^2+(\omega-\omega_0)^2} + \frac{\alpha A^2/2}{\alpha^2+(\omega+\omega_0)^2} \tag{3.14}$$

follows. This spectrum has been depicted in Figure 3.3(b) and is called a Lorentz profile.

□

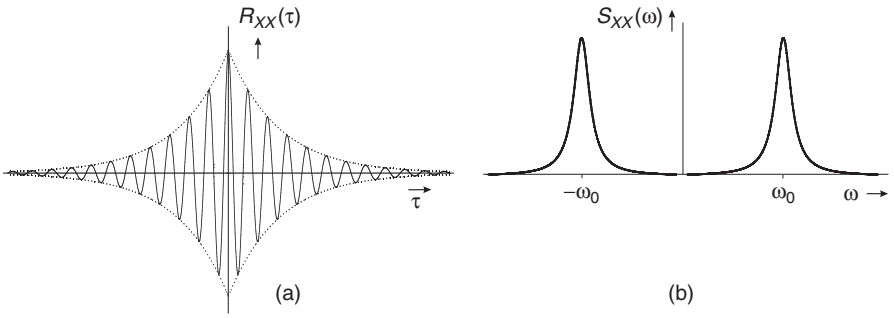


Figure 3.3 (a) The autocorrelation function and (b) the power spectrum of an oscillator

3.2 THE BANDWIDTH OF A STOCHASTIC PROCESS

The r.m.s. bandwidth W_e of a stochastic process is defined using the second normalized moment of the power spectrum, i.e.

$$W_e^2 \triangleq \frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega)\,d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega)\,d\omega} \tag{3.15}$$

This definition is, in its present form, only used for lowpass processes, i.e. processes where $S_{XX}(\omega)$ has a significant value at $\omega = 0$ and at low frequencies, and decreasing values of $S_{XX}(\omega)$ at increasing frequency.

Example 3.3:

In this example we will calculate the r.m.s. bandwidth of a very simple power spectrum, namely an ideal lowpass spectrum defined by

$$S_{XX}(\omega) = \begin{cases} 1, & \text{for } |\omega| < B \\ 0, & \text{for } |\omega| \geq B \end{cases} \tag{3.16}$$

Inserting this into the definition of Equation (3.15) yields

$$W_e^2 = \frac{\int_{-B}^{B} \omega^2\,d\omega}{\int_{-B}^{B} d\omega} = \frac{1}{3}B^2 \tag{3.17}$$

The r.m.s. bandwidth is in this case $W_e = B/\sqrt{3}$. This bandwidth might have been expected to be equal to B; the difference is explained by the quadratic weight in the numerator with respect to frequency.

□

In case of bandpass processes (see Subsection 4.4.1) the second, central, normalized moment is used in the definition

$$W_e^2 \triangleq \frac{4\int_0^\infty (\omega - \overline{\omega}_0)^2 S_{XX}(\omega)\,\mathrm{d}\omega}{\int_0^\infty S_{XX}(\omega)\,\mathrm{d}\omega} \tag{3.18}$$

where the mean frequency $\overline{\omega}_0$ is defined by

$$\overline{\omega}_0 \triangleq \frac{\int_0^\infty \omega S_{XX}(\omega)\,\mathrm{d}\omega}{\int_0^\infty S_{XX}(\omega)\,\mathrm{d}\omega} \tag{3.19}$$

the first normalized moment of $S_{XX}(\omega)$. A bandpass process is a process where the power spectral density function is confined around a frequency $\bar{\omega}_0$ and which has a negligible value (zero or almost zero) at $\omega = 0$.

The necessity of the factor of 4 in Equation (3.18) compared to Equation (3.15) is explained by the next example.

Example 3.4:

In this example we will consider the r.m.s. bandwidth of an ideal bandpass process with the power spectrum

$$S_{XX}(\omega) = \begin{cases} 1, & \text{for } |\omega - \omega_0| < \dfrac{B}{2} \text{ and } |\omega + \omega_0| < \dfrac{B}{2} \\ 0, & \text{elsewhere} \end{cases} \tag{3.20}$$

The r.m.s. bandwidth follows from the definition of Equation (3.18):

$$W_e^2 = \frac{4\int_{\omega_0 - B/2}^{\omega_0 + B/2} (\omega - \omega_0)^2\,\mathrm{d}\omega}{\int_{\omega_0 - B/2}^{\omega_0 + B/2} \mathrm{d}\omega} = \frac{1}{3}B^2 \tag{3.21}$$

which reveals that the r.m.s. bandwidth equals $W_e = B/\sqrt{3}$. Both the spectrum of the ideal lowpass process from Example 3.3 and the spectrum of the ideal bandpass process from this example are presented in Figure 3.4. From a physical point of view the two processes should

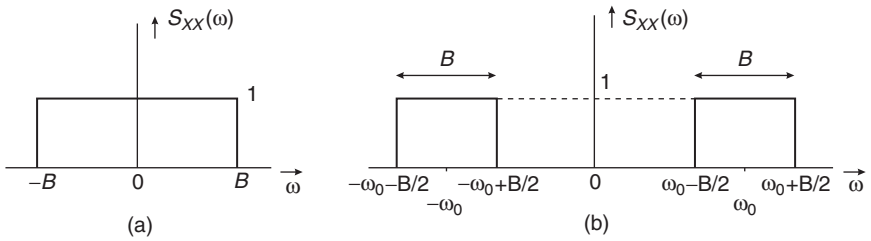


Figure 3.4 (a) Power spectrum of the ideal lowpass process and (b) the power spectrum of the ideal bandpass process

have the same bandwidth and indeed both Equations (3.17) and (3.21) have the same outcome. This is only the case if the factor of 4 is present in Equation (3.18).

□

3.3 THE CROSS-POWER SPECTRUM

Analogous to the preceding section, we can define the cross-power spectral density function, or briefly the cross-power spectrum, as the Fourier transform of the cross-correlation function

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) \exp(-\mathrm{j}\omega\tau)\, \mathrm{d}\tau \tag{3.22}$$

with the corresponding inverse transform

$$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) \exp(\mathrm{j}\omega\tau)\, \mathrm{d}\omega \tag{3.23}$$

It can be seen that the processes $X(t)$ and $Y(t)$ have to be jointly wide-sense stationary.

A physical interpretation of this spectrum cannot always be given. The function $S_{XY}(\omega)$ often acts as an auxiliary quantity in a few specific problems, such as in bandpass processes (see Section 5.2). Moreover, it plays a role when two (or even more) signals are added. Let us consider the process $Z(t) = X(t) + Y(t)$; then the autocorrelation is

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau) + R_{XY}(\tau) + R_{YX}(\tau) \tag{3.24}$$

From this latter equation the total power of $Z(t)$ is $P_{XX} + P_{YY} + P_{XY} + P_{YX}$ and it follows that the process $Z(t)$ contains, in general, more power than the sum of the powers of the individual signals. This apparently originates from the correlation of the signals. The cross-power spectra show how the additional power components P_{XY} and P_{YX} are spread over the different frequencies, namely

$$P_{XY} \triangleq \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega)\, \mathrm{d}\omega \tag{3.25}$$

and

$$P_{YX} \triangleq \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YX}(\omega)\, \mathrm{d}\omega \tag{3.26}$$

The total amount of additional power may play an important role in situations where an information-carrying signal has to be processed in the midst of additive noise or interference. Moreover, the cross-power spectrum is used to describe bandpass processes (see Chapter 5).

From Equation (3.24) it will be clear that the power of $Z(t)$ equals the sum of the powers in $X(t)$ and $Y(t)$ if the processes $X(t)$ and $Y(t)$ are orthogonal.

Properties of $S_{XY}(\omega)$ for real processes

1. $S_{XY}(\omega) = S_{YX}(-\omega) = S^*_{YX}(\omega)$ (3.27)
2. $\mathrm{Re}\{S_{XY}(\omega)\}$ and $\mathrm{Re}\{S_{YX}(\omega)\}$ are even functions of ω (3.28)

   $\mathrm{Im}\{S_{XY}(\omega)\}$ and $\mathrm{Im}\{S_{YX}(\omega)\}$ are odd functions of ω (3.29)

   where $\mathrm{Re}\{\cdot\}$ and $\mathrm{Im}\{\cdot\}$ are the real and imaginary parts, respectively, of the quantity in the braces
3. If $X(t)$ and $Y(t)$ are independent, then

   $$S_{XY}(\omega) = S_{YX}(-\omega) = 2\pi\overline{X}\,\overline{Y}\delta(\omega) \tag{3.30}$$
4. If $X(t)$ and $Y(t)$ are orthogonal, then

   $$S_{XY}(\omega) \equiv S_{YX}(\omega) \equiv 0 \tag{3.31}$$
5. If $X(t)$ and $Y(t)$ are uncorrelated, then

   $$S_{XY}(\omega) = S_{YX}(-\omega) = 2\pi\overline{X}\,\overline{Y}\delta(\omega) \tag{3.32}$$

Proofs of the properties:

1. Property 1 is proved by invoking Equation (2.48) and the definition of the cross-power spectrum

$$\begin{aligned} S_{XY}(\omega) &= \int_{-\infty}^{\infty} R_{XY}(\tau)\exp(-\mathrm{j}\omega\tau)\,\mathrm{d}\tau = \int_{-\infty}^{\infty} R_{XY}(-\tau)\exp(\mathrm{j}\omega\tau)\,\mathrm{d}\tau \\ &= \int_{-\infty}^{\infty} R_{YX}(\tau)\exp(\mathrm{j}\omega\tau)\,\mathrm{d}\tau = S_{YX}(-\omega) \end{aligned} \tag{3.33}$$

   and from this latter line it follows also that $S_{YX}(-\omega) = S^*_{YX}(\omega)$.
2. In contrast to $S_{XX}(\omega)$ the cross-power spectrum will in general be a complex-valued function. Property 2 follows immediately from the definition

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau)\cos(\omega\tau)\,\mathrm{d}\tau - \mathrm{j}\int_{-\infty}^{\infty} R_{XY}(\tau)\sin(\omega\tau)\,\mathrm{d}\tau \tag{3.34}$$

   For real processes the cross-correlation function is real as well and the first integral represents the real part of the power spectrum and is obviously even. The second integral represents the imaginary part of the power spectrum which is obviously odd.
3. From Equation (2.56) it is concluded that in this case $R_{XY}(\tau) = \overline{X}\,\overline{Y}$ and its Fourier transform equals the right-hand member of Equation (3.30).

4. The fourth property is quite straightforward. For orthogonal processes, by definition, $R_{XY}(\tau) = R_{YX}(\tau) \equiv 0$, and so are the corresponding Fourier transforms.
5. In the given situation, from Equations (2.69) and (2.72) it is concluded that $R_{XY}(\tau) = \overline{X}\ \overline{Y}$. Fourier transform theory says that the transform of a constant is a δ function of the form given by Equation (3.32).

3.4 MODULATION OF STOCHASTIC PROCESSES

In many applications (such as in telecommunications) a situation is often met where signals are modulated and synchronously demodulated. In those situations the signal is applied to a multiplier circuit, while a second input of the multiplier is a harmonic signal (sine or cosine waveform), called the carrier (see Figure 3.5). We will analyse the spectrum of the output process $Y(t)$ when the spectrum of the input process $X(t)$ is known. In doing so we will assume that the cosine function of the carrier has a constant frequency ω_0, but a random phase Θ that is uniformly distributed on the interval $(0,2\pi]$ and is independent of $X(t)$. The output process is then written as

$$Y(t) = X(t)A_0 \cos(\omega_0 t - \Theta) \tag{3.35}$$

The amplitude A_0 of the carrier is supposed to be constant. The autocorrelation function of the output $Y(t)$ is found by applying the definition to this latter expression, yielding

$$R_{YY}(t, t+\tau) = A_0^2 \mathrm{E}[X(t)\cos(\omega_0 t - \Theta)X(t+\tau)\cos(\omega_0 t + \omega_0 \tau - \Theta)] \tag{3.36}$$

At the start of this chapter we stated that we will confine our analysis to wide-sense stationary processes. We will invoke this restriction for the input process $X(t)$; however, this does not guarantee that the output process $Y(t)$ is also wide-sense stationary. Therefore we used the notation $R_{YY}(t, t+\tau)$ in Equation (3.36) and not $R_{YY}(\tau)$. Elaborating Equation (3.36) yields

$$\begin{aligned} R_{YY}(t, t+\tau) &= \frac{A_0^2}{2} R_{XX}(\tau)\ \mathrm{E}[\cos(2\omega_0 t + \omega_0 \tau - 2\Theta) + \cos\omega_0 \tau] \\ &= \frac{A_0^2}{2} R_{XX}(\tau) \left[\frac{1}{2\pi}\int_0^{2\pi} \cos(2\omega_0 t + \omega_0 \tau - 2\theta)\ \mathrm{d}\theta + \cos\omega_0 \tau\right] \\ &= \frac{A_0^2}{2} R_{XX}(\tau)\cos\omega_0 \tau \end{aligned} \tag{3.37}$$

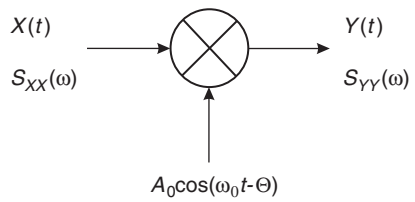


Figure 3.5 A product modulator or mixer

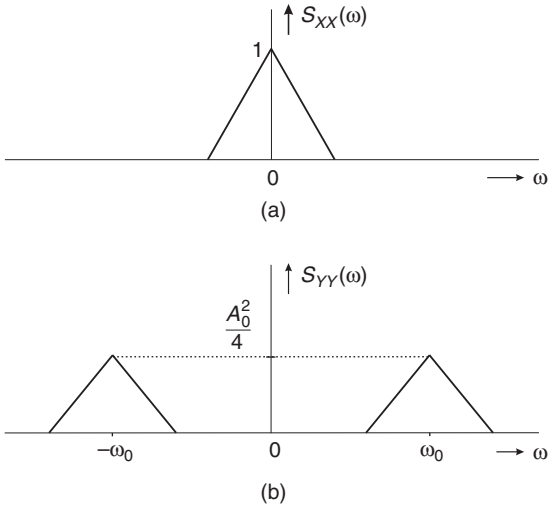


Figure 3.6 The spectra at (a) input $(S_{XX}(\omega))$ and (b) output $(S_{YY}(\omega))$ of a product modulator

From Equation (3.37) it is seen that $R_{YY}(t, t+\tau)$ is independent of t. The mean value of the output is calculated using Equation (3.35); since Θ and $X(t)$ have been assumed to be independent the mean equals the product of the mean values $\mathrm{E}[X(t)]$ and $\mathrm{E}[\cos(\omega_0 t - \Theta)]$. From Example 2.1 we know that this latter mean value is zero. Thus it is concluded that the output process $Y(t)$ is wide-sense stationary, since its autocorrelation function is independent of t and so is its mean. Transforming Equation (3.37) to the frequency domain, we arrive at our final result:

$$S_{YY}(\omega) = \frac{A_0^2}{4}[S_{XX}(\omega - \omega_0) + S_{XX}(\omega + \omega_0)] \tag{3.38}$$

In Figure 3.6 an example has been sketched of a spectrum $S_{XX}(\omega)$. Moreover, the corresponding spectrum $S_{YY}(\omega)$ as it appears at the output of the product modulator is presented. In this figure it has been assumed that $X(t)$ is a lowpass process. The analysis can, however, be applied in a similar way to processes with a different character, e.g. bandpass processes. As a consequence of the modulation we observe a shift of the baseband spectrum to the carrier frequency ω_0 and a shift to $-\omega_0$; actually besides a shift there is also a split-up. This is analogous to the modulation of deterministic signals, a difference being that when dealing with deterministic signals we use the signal spectrum, whereas when dealing with stochastic processes we have to use the power spectrum.

Example 3.5:

The method of modulation is in practice used for demodulation as well; demodulation in this way is called synchronous or coherent demodulation. The basic idea is that multiplication of

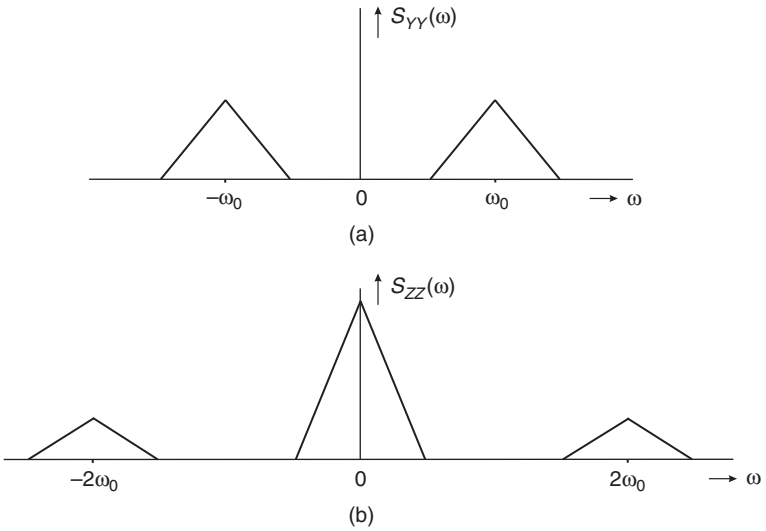


Figure 3.7 (a) The spectra of a modulated signal and (b) the output of the corresponding signal after synchronous demodulation by a product modulator

a signal with an harmonic wave (sine or cosine) shifts the power spectrum by an amount equal to the frequency of the harmonic signal. This shift is twofold: once to the right and once to the left over the frequency axis. Let us apply this procedure to the spectrum of the modulated signal as given in Figure 3.6(b). This figure has been redrawn in Figure 3.7(a). When this spectrum is both shifted to the right and to the left and added, the result is given by Figure 3.7(b). The power spectrum of the demodulated signal consists of three parts:

1. A copy of the original spectrum about $-2\omega_0$;
2. A copy of the original spectrum about $2\omega_0$;
3. Two copies about $\omega = 0$.

The first two copies may be removed by a lowpass filter, whereas the copies around zero actually represent the recovered baseband signal from Figure 3.6(a).

□

Besides modulation and demodulation, multiplication may also be applied for frequency conversion. Modulation and demodulation are therefore examples of frequency conversion (or frequency translation) that can be achieved by using multipliers.

3.4.1 Modulation by a Random Carrier

In certain systems stochastic processes are used as the carrier for modulation. An example is pseudo noise sequences that are used in CDMA (Code Division Multiple Access) systems [9]. The spectrum of such pseudo noise sequences is much wider than that of the modulating

signal. A second example is a lightwave communication system, where sources like light emitting diodes (LEDs) also have a bandwidth much wider than the modulating signal. These wideband sources can be described as stochastic processes and we shall denote them by $Z(t)$. For the modulation we use the scheme of Figure 3.5, where the sinusoidal carrier is replaced by this $Z(t)$. If the modulating process is given by $X(t)$, then the modulation signal at the output reads

$$Y(t) = X(t)\,Z(t) \tag{3.39}$$

Assuming that both processes are wide-sense stationary, the autocorrelation function of the output is written as

$$\begin{aligned} R_{YY}(t, t+\tau) &= \mathrm{E}[X(t)Z(t)\,X(t+\tau)Z(t+\tau)] \\ &= \mathrm{E}[X(t)X(t+\tau)\,Z(t)Z(t+\tau)] \end{aligned} \tag{3.40}$$

It is reasonable to assume that the processes $X(t)$ and $Z(t)$ are independent. Then it follows that

$$R_{YY}(t, t+\tau) = R_{XX}(\tau)\,R_{ZZ}(\tau) \tag{3.41}$$

which means that the output is wide-sense stationary as well. Transforming Equation (3.41) to the frequency domain produces the power spectrum of the modulation

$$S_{YY}(\omega) = \frac{1}{2\pi} S_{XX}(\omega) * S_{ZZ}(\omega) \tag{3.42}$$

where $*$ presents the convolution operation.

When $Z(t)$ has a bandpass characteristic and $X(t)$ is a baseband signal, then the modulated signal will be shifted to the bandpass frequency range of the noise-like carrier signal. It is well known that convolution exactly adds the spectral extent of the spectra of the individual signals when they are strictly band-limited. In the case of signals with unlimited spectral extent, the above relation holds approximately in terms of bandwidths [7]. This means that, for example, in the case of CDMA the spectrum of the transmitted signal is much wider than that of the information signal. Therefore, this modulation is also called the spread spectrum technique. On reception, a synchronized version of the pseudo noise signal is generated and synchronous demodulation recovers the information signal. De-spreading is therefore performed in the receiver.

3.5 SAMPLING AND ANALOGUE-TO-DIGITAL CONVERSION

In modern systems extensive use is made of digital signal processors (DSPs), due to the fact that these processors can be programmed and in this way can have a flexible functionality. Moreover, the speed is increasing to such high values that the devices become suitable for many practical applications. However, most signals to be processed are still analogue, such as signals from sensors and communication systems. Therefore sampling of the analogue signal is needed prior to analogue-to-digital (A/D) conversion. In this section we will consider both the sampling process and A/D conversion.

3.5.1 Sampling Theorems

First we will recall the well-known sampling theorem for deterministic signals [7,10], since we need it to describe a sampling theorem for stochastic processes.

Theorem 4

Suppose that the deterministic signal $f(t)$ has a band-limited Fourier transform $F(\omega)$; i.e. $F(\omega) = 0$ for $|\omega| > W$. Then the signal can exactly be recovered from its samples, if the samples are taken at a sampling rate of at least $1/T_s$, where

$$T_s = \frac{\pi}{W} \tag{3.43}$$

The reconstruction of $f(t)$ from its samples is given by

$$f(t) = \sum_{n=-\infty}^{n=\infty} f(nT_s) \frac{\sin W(t - nT_s)}{W(t - nT_s)} \tag{3.44}$$

The minimum sampling frequency $1/T_s = W/\pi = 2F$ is called the Nyquist frequency, where $F = W/(2\pi)$ is the maximum signal frequency component corresponding to the maximum angular frequency W.

The sampling theorem is understood by considering ideal sampling of the signal. Ideal sampling is mathematically described by multiplying the continuous signal $f(t)$ by an equidistant sequence of δ pulses as was mentioned in Chapter 1. This multiplication is equivalent to a convolution in the frequency domain. From Appendix G it is seen that an infinite sequence of δ pulses in the time domain is in the frequency domain an infinite sequence of δ pulses as well. A sampling rate of $1/T_s$ in the time domain gives a distance of $2\pi/T_s$ between adjacent δ pulses in the frequency domain. This means that in the frequency domain the spectrum $F(\omega)$ of the signal is reproduced infinitely many times shifted over $n2\pi/T_s$, with n an integer running from $-\infty$ to ∞. This is further explained by means of Figure 3.8. From this figure the reconstruction and minimum sampling rate is also understood; namely the original spectrum $F(\omega)$, and thus the signal $f(t)$, is recovered from the

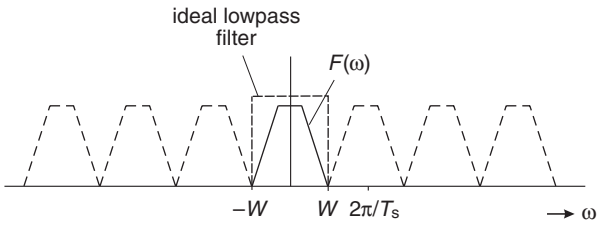


Figure 3.8 The spectrum of a sampled signal and its reconstruction

periodic spectrum by applying ideal lowpass filtering to it. However, ideal lowpass filtering in the time domain is described by a sinc function, as given in Equation (3.44). This sinc function provides the exact interpolation in the time domain. When the sampling rate is increased, the different replicas of $F(\omega)$ become further apart and this will still allow lowpass filtering to filter out $F(\omega)$, as is indicated in Figure 3.9(a). However, decreasing the sampling rate below the Nyquist rate introduces overlap of the replicas (see Figure 3.9(b)) and thus distortion; i.e. the original signal $f(t)$ can no longer be exactly recovered from its samples. This distortion is called aliasing distortion and is depicted in Figure 3.9(c).

For stochastic processes we can formulate a similar theorem.

Theorem 5

Suppose that the wide-sense stationary process $X(t)$ has a band-limited power spectrum $S_{XX}(\omega)$; i.e. $S_{XX}(\omega) = 0$ for $|\omega| > W$. Then the process can be recovered from its samples in the mean-squared error sense, if the samples are taken at a sampling rate of at least $1/T_s$, where

$$T_s = \frac{\pi}{W} \tag{3.45}$$

The reconstruction of $X(t)$ from its samples is given by

$$\hat{X}(t) = \sum_{n=-\infty}^{n=\infty} X(nT_s) \frac{\sin W(t - nT_s)}{W(t - nT_s)} \tag{3.46}$$

The reconstructed process $\hat{X}(t)$ converges to the original process $X(t)$ in the mean-squared error sense, i.e.

$$\mathrm{E}[\{\hat{X}(t) - X(t)\}^2] = 0 \tag{3.47}$$

Proof:

We start the proof by remarking that the autocorrelation function $R_{XX}(t)$ is a deterministic function with a band-limited Fourier transform $S_{XX}(\omega)$, according to the conditions mentioned in Theorem 5. As a consequence, Theorem 4 may be applied to it. The reconstruction of $R_{XX}(t)$ from its samples is written as

$$R_{XX}(t) = \sum_{n=-\infty}^{n=\infty} R_{XX}(nT_s) \frac{\sin W(t - nT_s)}{W(t - nT_s)} = \sum_{n=-\infty}^{n=\infty} R_{XX}(nT_s)\,\mathrm{sinc}[W(t - nT_s)] \tag{3.48}$$

For the proof we need two expressions that are derived from Equation (3.48). The first one is

$$R_{XX}(t) = \sum_{n=-\infty}^{n=\infty} R_{XX}(nT_s - T_1)\,\mathrm{sinc}[W(t - nT_s + T_1)] \tag{3.49}$$

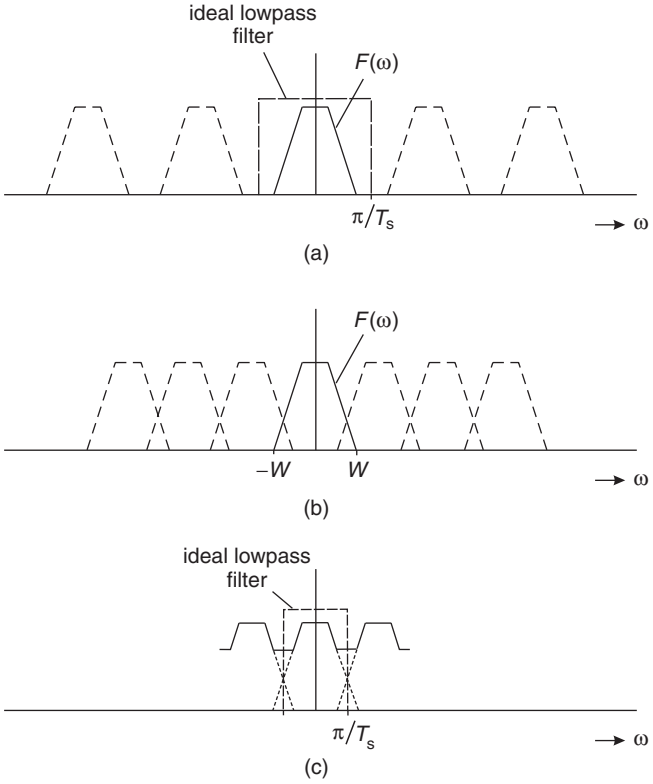


Figure 3.9 The sampled signal spectrum (a) when the sampling rate is higher than the Nyquist rate; (b) when it is lower than the Nyquist rate; (c) aliasing distortion

This expression follows from the fact that the sampling theorem only prescribes a minimum sampling rate, not the exact positions of the samples. The reconstruction is independent of shifting all samples over a certain amount. Another relation we need is

$$\begin{aligned} R_{XX}(t - T_2) &= \sum_{n=-\infty}^{n=\infty} R_{XX}(nT_s)\,\mathrm{sinc}[W(t - T_2 - nT_s)] \\ &= \sum_{n=-\infty}^{n=\infty} R_{XX}(nT_s - T_2)\,\mathrm{sinc}[W(t - nT_s)] \end{aligned} \tag{3.50}$$

The second line above follows by applying Equation (3.49) to the first line.

The mean-squared error between the original process and its reconstruction is written as

$$\begin{aligned} \mathrm{E}[\{\hat{X}(t) - X(t)\}^2] &= \mathrm{E}[\hat{X}^2(t) + X^2(t) - 2\hat{X}(t)X(t)] \\ &= \mathrm{E}[\hat{X}^2(t)] + R_{XX}(0) - 2\mathrm{E}[\hat{X}(t)X(t)] \end{aligned} \tag{3.51}$$

The first term of this latter expression is elaborated as follows:

$$\begin{aligned}\mathrm{E}[\hat{X}^2(t)] &= \mathrm{E}\left[\sum_{n=-\infty}^{n=\infty} X(nT_\mathrm{s})\,\mathrm{sinc}[W(t-nT_\mathrm{s})]\sum_{m=-\infty}^{m=\infty} X(mT_\mathrm{s})\,\mathrm{sinc}[W(t-mT_\mathrm{s})]\right]\\ &= \sum_{n=-\infty}^{n=\infty}\sum_{m=-\infty}^{m=\infty} \mathrm{E}[X(nT_\mathrm{s})X(mT_\mathrm{s})]\,\mathrm{sinc}[W(t-nT_\mathrm{s})]\,\mathrm{sinc}[W(t-mT_\mathrm{s})]\\ &= \sum_{n=-\infty}^{n=\infty}\left\{\sum_{m=-\infty}^{m=\infty} R_{XX}(mT_\mathrm{s}-nT_\mathrm{s})\,\mathrm{sinc}[W(t-mT_\mathrm{s})]\right\}\mathrm{sinc}[W(t-nT_\mathrm{s})] \qquad (3.52)\end{aligned}$$

To the expression in braces we apply Equation (3.50) with $T_2 = nT_\mathrm{s}$. This yields

$$\mathrm{E}[\hat{X}^2(t)] = \sum_{n=-\infty}^{n=\infty} R_{XX}(t-nT_\mathrm{s})\,\mathrm{sinc}[W(t-nT_\mathrm{s})] = R_{XX}(0) \qquad (3.53)$$

This equality is achieved when we once more invoke Equation (3.50), but now with $T_2 = t$.

In the last term of Equation (3.51) we insert Equation (3.46). This yields

$$\begin{aligned}\mathrm{E}[X(t)\hat{X}(t)] &= \mathrm{E}\left[X(t)\sum_{n=-\infty}^{n=\infty} X(nT_\mathrm{s})\,\mathrm{sinc}[W(t-nT_\mathrm{s})]\right]\\ &= \sum_{n=-\infty}^{n=\infty} \mathrm{E}[X(t)\,X(nT_\mathrm{s})]\,\mathrm{sinc}[W(t-nT_\mathrm{s})]\\ &= \sum_{n=-\infty}^{n=\infty} R_{XX}(t-nT_\mathrm{s})\,\mathrm{sinc}[W(t-nT_\mathrm{s})] = R_{XX}(0) \qquad (3.54)\end{aligned}$$

This result follows from Equation (3.53). Inserting Equations (3.53) and (3.54) into Equation (3.51) yields

$$\mathrm{E}[\{\hat{X}(t)-X(t)\}^2] = 0 \qquad (3.55)$$

This completes the proof of the theorem.

By means of applying the sampling theorem, continuous stochastic processes can be converted to discrete-time processes, without any information getting lost. This facilitates the processing of continuous processes by DSPs.

3.5.2 A/D Conversion

For processing in a computer or DSP the discrete-time process has to be converted to a discrete random sequence. That conversion is called analogue-to-digital (A/D) conversion. In this conversion the continuous sample values have to be converted to a finite set of discrete values; this is called quantization. It is important to realize that this is a crucial step, since this final set is an approximation of the analogue (continuous) samples. As in all

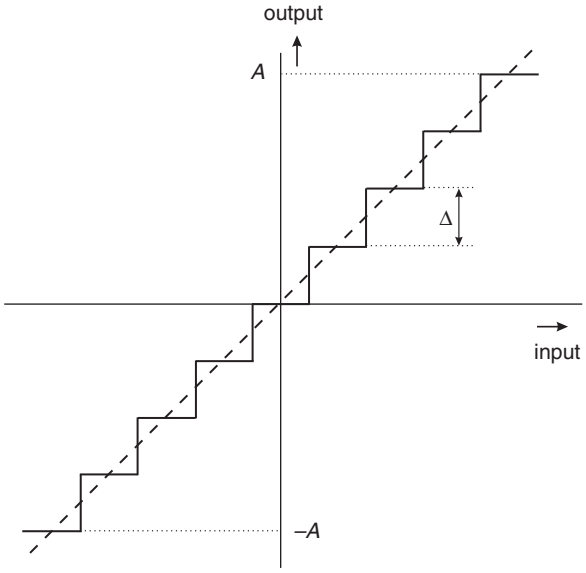


Figure 3.10 Quantization characteristic

approximations there are differences, called errors, between the original signal and the converted one. These errors cannot be restored in the digital-to-analogue reconversion. Thus, it is important to carefully consider the errors.

For the sake of better understanding we will assume that the sample values do not exceed both certain positive and negative values, let us say $|x| \leq A$. Furthermore, we set the number of possible quantization levels equal to $L + 1$. The conversion is performed by a quantizer that has a transfer function as given in Figure 3.10. This quantization characteristic is presented by the solid staircase shaped line, whereas on the dashed line the output is equal to the input. According to this characteristic the quantizer rounds off the input to the closest of the output levels. We assume that the quantizer accommodates the dynamic range of the signal, which covers the range of $\{-A, A\}$. The difference between the two lines represents the quantization error e_{q}. The mean squared value of this error is calculated as follows. The difference between two successive output levels is denoted by Δ. The exact distribution of the signal between A and $-A$ is, as a rule, not known, but let us make the reasonable assumption that the analogue input values are uniformly distributed between two adjacent levels for all stages. Then the value of the probability density function of the error is $1/\Delta$ and runs from $-\Delta/2$ to $\Delta/2$. The mean value of the error is then zero and the variance

$$\sigma_e^2 = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e_{\mathrm{q}}^2 \mathrm{d}e_{\mathrm{q}} = \frac{\Delta^2}{12} \tag{3.56}$$

It is concluded that the power of the error is proportional to the square of the quantization step size Δ. This means that this error can be reduced to an acceptable value by selecting an appropriate number of quantization levels. This error introduces noise in the quantization

process. By experience it has been established that the power spectral density of the quantization noise extends over a larger bandwidth than the signal bandwidth [11]. Therefore, it behaves approximately as white noise (see Chapter 6).

Example 3.6:

As an example of quantization we consider a sinusoidal signal of amplitude A, half the value of the range of the quantizer. In that case this range is fully exploited and no overload will occur. Recall that the number of output levels was $L + 1$, so that the step size is

$$\Delta = \frac{2A}{L} \tag{3.57}$$

Consequently, the power of the quantization error is

$$P_e = \sigma_e^2 = \frac{A^2}{3L^2} \tag{3.58}$$

Remembering that the power in a sinusoidal wave with amplitude A amounts to $A^2/2$, the ratio of the signal power to the quantization noise power follows. This ratio is called the signal-to-quantization noise ratio and is

$$\left(\frac{S}{N}\right)_{\mathrm{q}} = \frac{3L^2}{2} \tag{3.59}$$

When expressing this quantity in decibels (see Appendix B) it becomes

$$10\log\left(\frac{S}{N}\right)_{\mathrm{q}} = 1.8 + 20\log L \quad \mathrm{dB} \tag{3.60}$$

Using binary words of length n to present the different quantization levels, the number of these levels is 2^n. Inserting this into Equation (3.60) it is found that

$$10\log\left(\frac{S}{N}\right)_{\mathrm{q}} \approx 1.8 + 6n \quad \mathrm{dB} \tag{3.61}$$

for large values of L. Therefore, when adding one bit to the word length, the signal-to-quantization noise ratio increases by an amount of 6 dB. For example, the audio CD system uses 16 bits, which yields a signal-to-quantization noise ratio of 98 dB, quite an impressive value. □

The quantizer characterized by Figure 3.10 is a so-called uniform quantizer, i.e. all steps have the same size. It will be clear that the signal-to-quantization noise ratio can be substantially smaller than the value given by Equation (3.60) if the input amplitude is much smaller than A. This is understood when realizing that in that case the signal power goes down but the quantization noise remains the same.

Non-uniform quantizers have a small step size for small input signal values and this step size increases with increasing input level. As a result the signal-to-quantization noise ratio improves for smaller signal values at the cost of that for larger signal levels (see reference [6]).

After applying sampling and quantization to a continuous stochastic process we actually have a discrete random sequence, but, as mentioned in Chapter 1, these processes are simply special cases of the discrete-time processes.

3.6 SPECTRUM OF DISCRETE-TIME PROCESSES

As usual, the calculation of the power spectrum has to start by considering the autocorrelation function. For the wide-sense stationary discrete-time process $X[n]$ we can write

$$R_{XX}[m] = \mathrm{E}[X[n]\,X[n+m]] \tag{3.62}$$

The relation to the continuous stochastic process is

$$R_{XX}[m] = \sum_{m=-\infty}^{m=\infty} R_{XX}(mT_\mathrm{s})\,\delta(t - mT_\mathrm{s}) \tag{3.63}$$

From this equation the power spectral density of $X[m]$ follows by Fourier transformation:

$$S_{XX}(\omega) = \sum_{m=-\infty}^{m=\infty} R_{XX}[m]\ \exp(-\mathrm{j}\omega m T_\mathrm{s}) \tag{3.64}$$

Thus, the spectrum is a periodic function (see Figure 3.8) with period $2\pi/T_\mathrm{s}$. Such a periodic function can be described by means of its Fourier series coefficients [7,10]

$$R_{XX}[m] = \frac{T_\mathrm{s}}{2\pi}\int_{-\pi/T_\mathrm{s}}^{\pi/T_\mathrm{s}} S_{XX}(\omega)\exp(\mathrm{j}\omega m T_\mathrm{s})\,\mathrm{d}\omega \tag{3.65}$$

In particular, we find for the power of the process

$$P_X = \mathrm{E}[X^2[n]] = R_{XX}[0] = \frac{T_\mathrm{s}}{2\pi}\int_{-\pi/T_\mathrm{s}}^{\pi/T_\mathrm{s}} S_{XX}(\omega)\,\mathrm{d}\omega \tag{3.66}$$

For ease of calculation it is convenient to introduce the z-transform of $R_{XX}[m]$, which is defined as

$$\tilde{S}_{XX}(z) \triangleq \sum_{m=-\infty}^{m=\infty} R_{XX}[m]\,z^{-m} \tag{3.67}$$

The z-transform is more extensively dealt with in Subsection 4.6.2. Comparing this latter expression to Equation (3.64) reveals that the delay operator z equals $\exp(\mathrm{j}\omega T_\mathrm{s})$ and consequently

$$\tilde{S}_{XX}(\exp(\mathrm{j}\omega m T_\mathrm{s})) = S_{XX}(\omega) \tag{3.68}$$

Example 3.7:

Let us consider a wide-sense stationary discrete-time process $X[n]$ with the autocorrelation sequence

$$R_{XX}[m] = a^{|m|}, \quad \text{with } |a| < 1 \tag{3.69}$$

The spectrum expressed in z-transform notation is

$$\begin{aligned}\tilde{S}_{XX}(z) &= \sum_{m=-\infty}^{m=\infty} a^{|m|} z^{-m} = \sum_{m=-\infty}^{m=-1} a^{-m} z^{-m} + \sum_{m=0}^{m=\infty} a^m z^{-m} \\ &= \frac{az}{1-az} + \frac{z}{z-a} = \frac{1/a - a}{(1/a + a) - (z + 1/z)}\end{aligned} \tag{3.70}$$

From this expression the spectrum in the frequency domain is easily derived by replacing z with $\exp(\mathrm{j}\omega T_\mathrm{s})$:

$$S_{XX}(\omega) = \frac{1/a - a}{(1/a + a) - 2\cos(\omega T_\mathrm{s})} \tag{3.71}$$

□

It can be seen that the procedure given here to develop the autocorrelation sequence and spectrum of a discrete-time process can equally be applied to derive cross-correlation sequences and corresponding cross-power spectra. We leave further elaboration on this subject to the reader.

3.7 SUMMARY

The power spectral density function, or power spectrum, of a stochastic process is defined as the Fourier transform of the autocorrelation function. This spectrum shows how the total power of the process is distributed over the various frequencies. Definitions have been given of the bandwidth of stochastic processes. It appears that on modulation the power spectrum is split up into two parts of identical shape as the original unmodulated spectrum: one part is concentrated around the modulation frequency and the other part around minus the modulation frequency. This implies that we use a description based on double-sided spectra. This is very convenient from a mathematical point of view. From a physical point

of view negative frequency values have no meaning. When changing to the physical interpretation, the contributions of the power spectrum at negative frequencies are mirrored with respect to the y axis and the values are added to the values at the corresponding positive frequencies.

The sampling theorem is redefined for stochastic processes. Therefore continuous stochastic processes can be converted into discrete-time processes without information becoming lost. For processing signals using a digital signal processor (DSP), still another step is needed, namely analogue-to-digital conversion. This introduces errors that cannot be restored in the digital-to-analogue reconstruction. These errors are calculated and expressed in terms of the signal-to-noise ratio. Finally, the autocorrelation sequence and power spectrum of discrete-time processes are derived.

3.8 PROBLEMS

3.1 A wide-sense stationary process $X(t)$ has the autocorrelation function

$$R_{XX}(\tau) = A \exp\left(-\frac{|\tau|}{T}\right)$$

(a) Calculate the power spectrum $S_{XX}(\omega)$.
(b) Calculate the power of $X(t)$ using the power spectrum.
(c) Check the answer to (b) using Equation (3.8), i.e. based on the autocorrelation function evaluated at $\tau = 0$.
(d) Use Matlab to plot the spectrum for $A = 3$ and $T = 4$.

3.2 Consider the process $X(t)$, of which the autocorrelation function is given in Problem 2.14.

(a) Calculate the power spectrum $S_{XX}(\omega)$. Make a plot of it using Matlab.
(b) Calculate the power of $X(t)$ using the power spectrum.
(c) Calculate the mean value of $X(t)$ from the power spectrum.
(d) Calculate the variance of $X(t)$ using the power spectrum.
(e) Check the answers to (b), (c) and (d) using the answers to Problem 2.14.
(f) Calculate the lowest frequency where the spectrum becomes zero. By means of Matlab calculate the relative amount of a.c. power in the frequency band between zero and this first null.

3.3 A wide-sense stationary process has a power spectral density function

$$S_{XX}(\omega) = \begin{cases} 2, & 0 \leq |\omega| < 10/(2\pi) \\ 0, & \text{elsewhere} \end{cases}$$

(a) Calculate the autocorrelation function $R_{XX}(\tau)$.
(b) Use Matlab to plot the autocorrelation function.
(c) Calculate the power of the process, both via the power spectrum and the autocorrelation function.

3.4 Consider the process $Y(t) = A^2 \sin^2(\omega_0 t - \Theta)$, with A and ω_0 constants and Θ a random variable that is uniformly distributed on $(0, 2\pi]$. In Problem 2.8 we calculated its autocorrelation function.

(a) Calculate the power spectrum $S_{YY}(\omega)$.

(b) Calculate the power of $Y(t)$ using the power spectrum.

(c) Check the answer to (b) using Equation (3.8).

3.5 Reconsider Problem 2.9 and insert $\omega_0 = 2\pi$.

(a) Calculate the magnitude of a few spectral lines of the power spectrum by means of Matlab.

(b) Based on (a), calculate an approximate value of the total power and check this on the basis of the autocorrelation function.

(c) Calculate and check the d.c. power.

3.6 Answer the same questions as in Problem 3.5, but now for the process given in Problem 2.11 when inserting $T = 1$.

3.7 A and B are random variables. These variables are used to create the process $X(t) = A \cos \omega_0 t + B \sin \omega_0 t$, with ω_0 a constant.

(a) Assume that A and B are uncorrelated, have zero means and equal variances. Show that in this case $X(t)$ is wide-sense stationary.

(b) Derive the autocorrelation function of $X(t)$.

(c) Derive the power spectrum of $X(t)$. Make a sketch of it.

3.8 A stochastic process is given by $X(t) = A \cos(\Omega t - \Theta)$, where A is a real constant, Ω a random variable with probability density function $f_\Omega(\omega)$ and Θ a random variable that is uniformly distributed on the interval (0,2π], independent of Ω. Show that the power spectrum of $X(t)$ is

$$S_{XX}(\omega) = \frac{\pi A^2}{2} [\, f_\Omega(\omega) + f_\Omega(-\omega)]$$

3.9 A and B are real constants and $X(t)$ is a wide-sense stationary process. Derive the power spectrum of the process $Y(t) = A + B\, X(t)$.

3.10 Can each of the following functions be the autocorrelation function of a wide-sense stationary process $X(t)$?

(a) $R_{XX}(\tau) = \delta(\tau)$.

(b) $R_{XX}(\tau) = \text{rect}(\tau)$.

(c) $R_{XX}(\tau) = \text{tri}(\tau)$.

For definitions of the functions $\delta(\cdot)$, rect$(\cdot)$ and tri$(\cdot)$ see Appendix E.

3.11 Consider the process given in Problem 3.1. Based on process $X(t)$ of that problem another process $Y(t)$ is produced, such that

$$S_{YY}(\omega) = \begin{cases} S_{XX}(\omega), & |\omega| < c(1/T) \\ 0, & \text{elsewhere} \end{cases}$$

where c is a constant.

(a) Calculate the r.m.s. bandwidth of $Y(t)$.

(b) Consider the consequences, both for $S_{YY}(\omega)$ and the r.m.s. bandwidth, when $c \to \infty$.

3.12 For the process $X(t)$ it is found that $R_{XX}(\tau) = A \exp[-\tau^2/(2\sigma^2)]$, with A and σ positive constants.

(a) Derive the expression for the power spectrum of $X(t)$.

(b) Calculate the r.m.s. bandwidth of $X(t)$.

3.13 Derive the cross-power spectrum of the processes given in Problem 2.9.

3.14 A stochastic process is defined by $W(t) = AX(t) + BY(t)$, with A and B real constants and $X(t)$ and $Y(t)$ jointly wide-sense stationary processes.

(a) Calculate the power spectrum of $W(t)$.

(b) Calculate the cross-power spectra $S_{XW}(\omega)$ and $S_{YW}(\omega)$.

(c) Derive $S_{WW}(\omega)$ when $X(t)$ and $Y(t)$ are orthogonal.

(d) Derive $S_{WW}(\omega)$ when $X(t)$ and $Y(t)$ are independent.

(e) Derive $S_{WW}(\omega)$ when $X(t)$ and $Y(t)$ are independent and have mean values of zero.

(f) Derive $S_{WW}(\omega)$ when $X(t)$ and $Y(t)$ are uncorrelated.

3.15 A wide-sense stationary noise process $N(t)$ has a power spectrum as given in Figure 3.11. This process is added to an harmonic random signal $S(t) = 3\cos(8t - \Phi)$ and the sum $S(t) + N(t)$ is applied to one of the inputs of a product modulator. To the second input of this modulator another harmonic process $X(t) = 2\cos(8t - \Theta)$ is applied. The random variables Φ and Θ are independent, but have the same uniform distribution on the interval $[0, 2\pi)$. Moreover, these random variables are independent of the process $N(t)$. The output of the product modulator is connected to an ideal lowpass filter with a cut-off angular frequency $\omega_c = 5$.

(a) Make a sketch of the spectrum of the output of the modulator.

(b) Sketch the spectrum at the output of the lowpass filter.

(c) Calculate the d.c. power at the output of the filter.

(d) The output signal is defined as that portion of the output due to $S(t)$. The output noise is defined as that portion of the output due to $N(t)$. Calculate the output signal power and the output noise power, and the ratio between the two (called the signal-to-noise ratio).

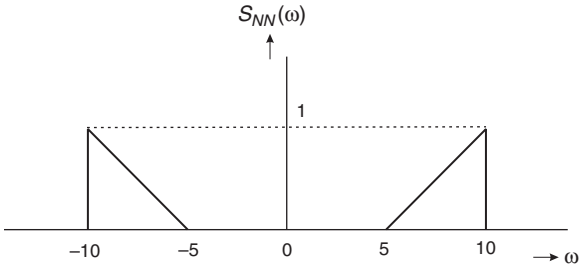


Figure 3.11

3.16 The two independent processes $X(t)$ and $Y(t)$ are applied to a product modulator. Process $X(t)$ is a wide-sense stationary process with power spectral density

$$S_{XX}(\omega) = \begin{cases} 1, & |\omega| \leq W_X \\ 0, & |\omega| > W_X \end{cases}$$

The process $Y(t)$ is an harmonic carrier, but both its phase and frequency are independent random variables

$$Y(t) = \cos(\Omega t - \Theta)$$

where the phase Θ is uniformly distributed on the interval $[0, 2\pi)$ and the carrier frequency is uniformly distributed on the interval $(\omega_0 - W_Y/2 < \Omega < \omega_0 + W_Y/2)$ and with $\omega_0 > W_Y/2 =$ constant.

(a) Calculate the autocorrelation function of the process $Y(t)$.

(b) Is $Y(t)$ wide-sense stationary?

(c) If so, determine and sketch the power spectral density $S_{YY}(\omega)$.

(d) Determine and sketch the power spectral density of the product $Z(t) = X(t)Y(t)$. Assume that $W_Y/2 + W_X < \omega_0$ and $W_X < W_Y/2$.

3.17 The following sequence of numbers represents sample values of a band-limited signal:

$$\ldots, 0, 5, 10, 20, 40, 0, -20, -15, -10, -5, 0, \ldots$$

All other samples are zero.

Use Matlab to reconstruct and graphically represent the signal.

3.18 In the case of ideal sampling, the sampled version of the signal $f(t)$ is represented by

$$f_s(t) = \sum_{n=-\infty}^{n=\infty} f(nT_s)\, \delta(t - nT_s)$$

In so-called 'flat-top sampling' the samples are presented by the magnitude of rectangular pulses, i.e.

$$f_s(t) = \sum_{n=-\infty}^{n=\infty} f(nT_s)\ \text{rect}\left(\frac{t - nT_s}{\tau_s}\right)$$

where $\tau_s < T_s$. (See Appendix E for the definition of the rect$(\cdot)$ function.) Investigate the effect of using these rectangular pulses on the Fourier transform of the recovered signal.

3.19 A voice channel has a spectrum that runs up to 3.4 kHz. On sampling, a guard band (i.e. the distance between adjacent spectral replicas after sampling) of 1.2 kHz has to be taken in account.

(a) What is the minimum sampling rate?

(b) When the samples are coded by means of a linear sampler of 8 bits, calculate the bit rate of a digitized voice channel.

(c) What is the maximum signal-to-noise ratio that can be achieved for such a voice channel?

(d) With how many dB will the signal-to-noise reduce when only half of the dynamic range is used by the signal?

3.20 A stochastic process $X(t)$ has the power spectrum

$$S_{XX} = \frac{1}{1+\omega^2}$$

This process is sampled, and since it is not band-limited the adjacent spectral replicas will overlap. If the spill-over power, i.e. the amount of power that is in overlapping frequency ranges, between adjacent replicas has to be less than 10% of the total power, what is the minimum sampling frequency?

3.21 The discrete-time process $X[n]$ is wide-sense stationary and $R_{XX}[1] = R_{XX}[0]$. Show that $R_{XX}[m] = R_{XX}[0]$ for all m.

3.22 The autocorrelation sequence of a discrete-time wide-sense stationary process $X[n]$ is

$$R_{XX}[m] = \begin{cases} 1 - 0.2|m|, & |m| \leq 4 \\ 0, & |m| > 4 \end{cases}$$

Calculate the spectrum $S_{XX}(\omega)$ and make a plot of it using Matlab.

4

Linear Filtering of Stochastic Processes

In this chapter we will investigate what the influence will be on the main parameters of a stochastic process when filtered by a linear, time-invariant filter. In doing so we will from time to time change from the time domain to the frequency domain and vice versa. This may even happen during the course of a calculation. From Fourier transform theory we know that both descriptions are dual and of equal value, and basically there is no difference, but a certain calculation may appear to be more tractable or simpler in one domain, and less tractable in the other.

In this chapter we will always assume that the input to the linear, time-invariant filter is a wide-sense stationary process, and the properties of these processes will be invoked several times. It should be stressed that the presented calculations and results may only be applied in the situation of wide-sense stationary input processes. Systems that are non-linear or time-variant are not considered, and the same holds for input processes that do not fulfil the requirements for wide-sense stationarity.

We start by summarizing the fundamentals of linear time-invariant filtering.

4.1 BASICS OF LINEAR TIME-INVARIANT FILTERING

In this section we will summarize the theory of continuous linear time-invariant filtering. For the sake of simplicity we consider only single-input single-output (SISO) systems. For a more profound treatment of this theory see references [7] and [10]. The generalization to multiple-input multiple-output (MIMO) systems is straightforward and requires a matrix description.

Let us consider a general system that converts a certain input signal $x(t)$ into the corresponding output signal $y(t)$. We denote this by means of the general hypothetical operator $\mathrm{T}[\cdot]$ as follows (see also Figure 4.1(a)):

$$y(t) = \mathrm{T}[x(t)] \tag{4.1}$$

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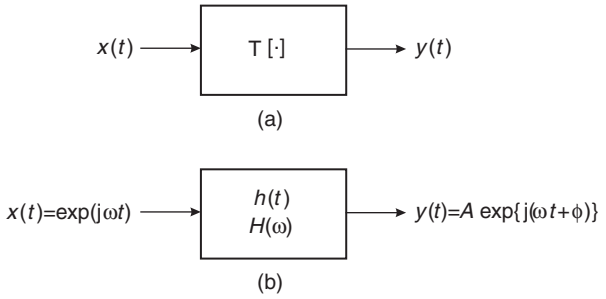


Figure 4.1 (a) General single-input single-output (SISO) system; (b) linear time-invariant (LTI) system

Next we limit our treatment to linear systems and denote this by means of $\mathrm{L}[\cdot]$ as follows:

$$y(t) = \mathrm{L}[x(t)] \tag{4.2}$$

The definition of linearity of a system is as follows. Suppose a set of input signals $\{x_n(t)\}$ causes a corresponding set of output signals $\{y_n(t)\}$. Then a system is said to be linear if any arbitrary linear combination of inputs causes the same linear combination of corresponding outputs, i.e.

$$\begin{aligned} &\text{if:} && x_n(t) \Rightarrow y_n(t) \\ &\text{then:} && \sum_n a_n x_n(t) \Rightarrow \sum_n a_n y_n(t) \end{aligned} \tag{4.3}$$

with a_n arbitrary constants. In the notation of Equation (4.2),

$$y(t) = \mathrm{L}\left[\sum_n a_n x_n(t)\right] = \sum_n a_n \mathrm{L}[x_n(t)] = \sum_n a_n y_n(t) \tag{4.4}$$

A system is said to be time-invariant if a shift in time of the input causes a corresponding shift in the output. Therefore,

$$\begin{aligned} &\text{if:} && x_n(t) \Rightarrow y_n(t) \\ &\text{then:} && x_n(t-\tau) \Rightarrow y_n(t-\tau) \end{aligned} \tag{4.5}$$

for arbitrary τ. Finally, a system is linear time-invariant (LTI) if it satisfies both conditions given by Equations (4.3) and (4.5), i.e.

$$\begin{aligned} &\text{if:} && x_n(t) \Rightarrow y_n(t) \\ &\text{then:} && x(t) = \sum_n a_n x_n(t-\tau_n) \Rightarrow y(t) = \sum_n a_n y_n(t-\tau_n) \end{aligned} \tag{4.6}$$

It can be proved [7] that complex exponential time functions, i.e. sine and cosine waves, are so-called eigenfunctions of linear time-invariant systems. An eigenfunction can physically be interpreted as a function that preserves its shape on transmission, i.e. a sine/cosine remains a sine/cosine, but its amplitude and/or phase may change. When these changes are known for all frequencies then the system is completely specified. This specification is done by means of the complex transfer function of the linear time-invariant system. If the system is excited with a complex exponential

$$x(t) = \exp(\mathrm{j}\omega t) \tag{4.7}$$

and the corresponding output is

$$y(t) = A\exp[\,\mathrm{j}(\omega t + \varphi)] \tag{4.8}$$

with A a real constant, then the transfer function equals

$$H(\omega) = \left.\frac{y(t)}{x(t)}\right|_{x(t)=\exp(\mathrm{j}\omega t)} \tag{4.9}$$

From Equations (4.7) to (4.9) the amplitude and the phase angle of the transfer function follow:

$$\begin{aligned} |H(\omega)| &= A \\ \angle H(\omega) &= \varphi \end{aligned} \tag{4.10}$$

When in Equation (4.9), ω is given all values from $-\infty$ to ∞, the transfer is completely known. As indicated in that equation the amplitude of the input $x(t)$ is taken as unity and the phase zero for all frequencies. All observed complex values of $y(t)$ are then presented by the function $Y(\omega)$. Since the input $X(\omega)$ was taken as unity for all frequencies, Equation (4.9) is in that case written as

$$H(\omega) = \frac{Y(\omega)}{1} \rightarrow Y(\omega) = H(\omega) \times 1 \tag{4.11}$$

From Fourier theory we know that the multiplication in the right-hand side of the latter equation is written in the time domain as a convolution [7,10]. Moreover, the inverse transform of 1 is a δ function. Since a δ function is also called an impulse, the time domain LTI system response following from Equation (4.11) is called the impulse response. We may therefore conclude that the system impulse response $h(t)$ and the system transfer function $H(\omega)$ constitute a Fourier transform pair.

When the transfer function is known, the response of an LTI system to an input signal can be calculated. Provided that the input signal $x(t)$ satisfies the Dirichlet conditions [10], its Fourier transform $X(\omega)$ exists. However, this frequency domain description of the signal is equivalent to decomposing the signal into complex exponentials, which in turn are eigenfunctions of the LTI system. This allows multiplication of $X(\omega)$ by $H(\omega)$ to find $Y(\omega)$, being the Fourier transform of output $y(t)$; namely by taking the inverse transform of $Y(\omega)$, the signal $y(t)$ is reconstructed from its complex exponential components. This

justifies the use of Fourier transform theory to be applied to the transmission of signals through LTI signals. This leads us to the following theorem.

Theorem 6

If a linear time-invariant system with an impulse response $h(t)$ is excited by an input signal $x(t)$, then the output is

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, \mathrm{d}\tau \tag{4.12}$$

with the equivalent frequency domain description

$$Y(\omega) = H(\omega)\, X(\omega) \tag{4.13}$$

where $X(\omega)$ and $Y(\omega)$ are the Fourier transforms of $x(t)$ and $y(t)$, respectively, and $H(\omega)$ is the Fourier transform of the impulse response $h(t)$.

The two presentations of Equations (4.12) and (4.13) are so-called dual descriptions; i.e. both are complete and either of them is fully determined by the other one. If an important condition for physical realizability of the LTI system is taken into account, namely causality, then the impulse response will be zero for $t < 0$ and the lower bound of the integral in Equation (4.12) changes into zero.

This theorem is the main result we need to describe the filtering of stochastic processes by an LTI system, as is done in the sequel.

4.2 TIME DOMAIN DESCRIPTION OF FILTERING OF STOCHASTIC PROCESSES

Let us now consider the transmission of a stochastic process through an LTI system. Obviously, we may formally apply the time domain description given by Equation (4.12) to calculate the system response of a single realization of the ensemble. However, a frequency domain description is not always possible. Apart from the fact that realizations are often not explicitly known, it may happen that they do not satisfy the Dirichlet conditions. Therefore, we start by characterizing the filtering in the time domain. Later on the frequency domain description will follow from this.

4.2.1 The Mean Value of the Filter Output

The impulse response of the linear, time-invariant filter is denoted by $h(t)$. Let us consider the ensemble of input realizations and call this input process $X(t)$ and the corresponding output process $Y(t)$. Then the relation between input and output is formally described by the convolution

$$Y(t) = \int_{-\infty}^{\infty} h(\rho) X(t-\rho)\, \mathrm{d}\rho \tag{4.14}$$

When the input process $X(t)$ is wide-sense stationary, then the mean value of the output signal is written as

$$\begin{aligned} \mathrm{E}[Y(t)] &= \mathrm{E}\left[\int_{-\infty}^{\infty} h(\rho)X(t-\rho)\,\mathrm{d}\rho\right] = \int_{-\infty}^{\infty} h(\rho)\mathrm{E}[X(t-\rho)]\,\mathrm{d}\rho \\ &= \overline{Y(t)} = \overline{X(t)}\int_{-\infty}^{\infty} h(\rho)\,\mathrm{d}\rho = \overline{X(t)}\,H(0) \end{aligned} \tag{4.15}$$

where $H(\omega)$ is the Fourier transform of $h(t)$. From Equation (4.15) it follows that the mean value of $Y(t)$ equals the mean value of $X(t)$ multiplied by the value of the transfer function for the d.c. component. This value is equal to the area under the curve of the impulse response function $h(t)$. This conclusion is based on the property of $X(t)$ at least being stationary of the first order.

4.2.2 The Autocorrelation Function of the Output

The autocorrelation function of $Y(t)$ is found using the definition of Equation (2.13) and Equation (4.14):

$$\begin{aligned} R_{YY}(t, t+\tau) &= \mathrm{E}[Y(t)\,Y(t+\tau)] \\ &= \mathrm{E}\left[\int_{-\infty}^{\infty} h(\rho_1)X(t-\rho_1)\,\mathrm{d}\rho_1 \int_{-\infty}^{\infty} h(\rho_2)X(t+\tau-\rho_2)\,\mathrm{d}\rho_2\right] \\ &= \int_{-\infty}^{\infty}\int \mathrm{E}[X(t-\rho_1)\,X(t+\tau-\rho_2)]h(\rho_1)h(\rho_2)\,\mathrm{d}\rho_1\,\mathrm{d}\rho_2 \end{aligned} \tag{4.16}$$

Invoking $X(t)$ as wide-sense stationary reduces this expression to

$$R_{YY}(\tau) = \int_{-\infty}^{\infty}\int R_{XX}(\tau+\rho_1-\rho_2)]h(\rho_1)h(\rho_2)\,\mathrm{d}\rho_1\,\mathrm{d}\rho_2 \tag{4.17}$$

and the mean squared value of $Y(t)$ reads

$$\mathrm{E}[Y^2(t)] = \overline{Y^2(t)} = R_{YY}(0) = \int_{-\infty}^{\infty}\int R_{XX}(\rho_1-\rho_2)h(\rho_1)h(\rho_2)\,\mathrm{d}\rho_1\,\mathrm{d}\rho_2 \tag{4.18}$$

From Equations (4.15) and (4.17) it is concluded that $Y(t)$ is wide-sense stationary when $X(t)$ is wide-sense stationary, since neither the right-hand member of Equation (4.15) nor that of Equation (4.17) depends on t.

Equation (4.17) may also be written as

$$R_{YY}(\tau) = R_{XX}(\tau) * h(\tau) * h(-\tau) \tag{4.19}$$

where the symbol $*$ represents the convolution operation.

4.2.3 Cross-Correlation of the Input and Output

The cross-correlation of $X(t)$ and $Y(t)$ is found using Equations (2.46) and (4.14):

$$R_{XY}(t, t+\tau) \triangleq \mathrm{E}[X(t)\, Y(t+\tau)] = \mathrm{E}\left[X(t) \int_{-\infty}^{\infty} h(\rho) X(t+\tau-\rho)\, \mathrm{d}\rho\right]$$
$$= \int_{-\infty}^{\infty} \mathrm{E}[X(t)\, X(t+\tau-\rho)]\, h(\rho)\, \mathrm{d}\rho \tag{4.20}$$

In the case where $X(t)$ is wide-sense stationary Equation (4.20) reduces to

$$R_{XY}(\tau) = \int_{-\infty}^{\infty} R_{XX}(\tau-\rho) h(\rho)\, \mathrm{d}\rho \tag{4.21}$$

This expression may also be presented as the convolution of $R_{XX}(\tau)$ and $h(\tau)$:

$$R_{XY}(\tau) = R_{XX}(\tau) * h(\tau) \tag{4.22}$$

In a similar way the following expression can be derived:

$$R_{YX}(\tau) = \int_{-\infty}^{\infty} R_{XX}(\tau+\rho) h(\rho)\, \mathrm{d}\rho = R_{XX}(\tau) * h(-\tau) \tag{4.23}$$

From Equations (4.21) and (4.23) it is concluded that the cross-correlation functions do not depend on the absolute time parameter t. Earlier we concluded that $Y(t)$ is wide-sense stationary if $X(t)$ is wide-sense stationary. Now we conclude that $X(t)$ and $Y(t)$ are jointly wide-sense stationary if the input process $X(t)$ is wide-sense stationary.

Substituting Equation (4.21) into Equation (4.17) reveals the relation between the autocorrelation function of the output and the cross-correlation between the input and output:

$$R_{YY}(\tau) = \int_{-\infty}^{\infty} R_{XY}(\tau+\rho_1) h(\rho_1)\, \mathrm{d}\rho_1 \tag{4.24}$$

or, presented differently,

$$R_{YY}(\tau) = R_{XY}(\tau) * h(-\tau) \tag{4.25}$$

In a similar way it follows by substitution of Equation (4.23) into Equation (4.17) that

$$R_{YY}(\tau) = \int_{-\infty}^{\infty} R_{YX}(\tau-\rho_2) h(\rho_2)\, \mathrm{d}\rho_2 = R_{YX}(\tau) * h(\tau) \tag{4.26}$$

Example 4.1:

An important application of the cross-correlation function as given by Equation (4.22) consists of the identification of a linear system. If for the input process a white noise process, i.e. a process with a constant value of the power spectral density, let us say of magnitude

$N_0/2$, is selected then the autocorrelation function of that process becomes $N_0\,\delta(\tau)/2$. This makes the convolution very simple, since the convolution of a δ function with another function results in this second function itself. Thus, in that case the cross-correlation function of the input and output yields $R_{XY}(\tau) = N_0\,h(\tau)/2$. Apart from a constant $N_0/2$, the cross-correlation function equals the impulse response of the linear system; in this way we have found a method to measure this impulse response.

□

4.3 SPECTRA OF THE FILTER OUTPUT

In the preceding sections we described the output process of a linear time-invariant filter in terms of the properties of the input process. In doing so we used the time domain description. We concluded that in case of a wide-sense stationary input process the corresponding output process is wide-sense stationary as well, and that the two processes are jointly wide-sense stationary. This offers the opportunity to apply the Fourier transform to the different correlation functions in order to arrive at the spectral description of the output process and the relationship between the input and output processes. It must also be stressed that in this section only wide-sense stationary input processes will be considered.

The first property we are interested in is the spectral density of the output process. Using what has been derived in Section 4.2.2, this is easily revealed by transforming Equation (4.19) to the frequency domain. If we remember that the impulse response $h(\tau)$ is a real function and thus the Fourier transform of $h(-\tau)$ equals $H^*(\omega)$, then the next important statement can be exposed.

Theorem 7

If a wide-sense stationary process $X(t)$, with spectral density $S_{XX}(\omega)$, is applied to the input of a linear, time-invariant filter with the transfer function $H(\omega)$, then the corresponding output process $Y(t)$ is a wide-sense stationary process as well, and the spectral density of the output reads

$$S_{YY}(\omega) = S_{XX}(\omega)\,H(\omega)\,H^*(\omega) = S_{XX}(\omega)\,|H(\omega)|^2 \tag{4.27}$$

The mean power of the output process is written as

$$P_Y = R_{YY}(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XX}(\omega)\,|H(\omega)|^2\,\mathrm{d}\omega \tag{4.28}$$

Example 4.2:

Consider the RC network given in Figure 4.2. Then the voltage transfer function of this network is written as

$$H(\omega) = \frac{1}{1 + \mathrm{j}\omega RC} \tag{4.29}$$

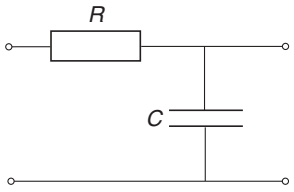


Figure 4.2 RC network

If we assume that the network is excited by a white noise process with spectral density of $N_0/2$ and taking the modulus squared of Equation (4.29), then the output spectral density reads

$$S_{YY}(\omega) = \frac{N_0/2}{1 + (\omega RC)^2} \tag{4.30}$$

For the power in the output process it is found that

$$P_Y = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{N_0/2}{1 + (\omega RC)^2}\,\mathrm{d}\omega = \frac{N_0}{4\pi RC}\arctan(\omega RC)\Big|_{-\infty}^{\infty} = \frac{N_0}{4RC} \tag{4.31}$$

□

In an earlier stage we found the power of a wide-sense stationary process in an alternative way, namely the value of the autocorrelation function at $\tau = 0$ (see, for instance, Equation (4.16)). The power can also be calculated using that procedure. However, in order to be able to calculate the autocorrelation function of the output $Y(t)$ we need the probability density function of $X(t)$ in order to evaluate a double convolution or we need the probability density function of $Y(t)$. Finding this latter function we meet two main obstacles: firstly, measuring the probability density function is much more difficult than measuring the power density function and, secondly, the probability density function of $Y(t)$ by no means follows in a simple way from that of $X(t)$. This latter statement has one important exception, namely if the probability density function of $X(t)$ is Gaussian then the probability density function of $Y(t)$ is Gaussian as well (see Section 2.3). However, calculating the mean and variance of $Y(t)$, which are sufficient to determine the Gaussian density, using Equations (4.15) and (4.28) is a simpler and more convenient method.

From Equations (4.22) and (4.23) the cross-power spectra are deduced:

$$S_{XY}(\omega) = S_{XX}(\omega)\,H(\omega) \tag{4.32}$$

$$S_{YX}(\omega) = S_{XX}(\omega)\,H(-\omega) = S_{XX}(\omega)H^*(\omega) \tag{4.33}$$

We are now in a position to give the proof of Equation (3.5). Suppose that $S_{XX}(\omega)$ has a negative value for some arbitrary $\omega = \omega_0$. Then a small interval (ω_1, ω_2) about ω_0 is found, such that (see Figure 4.3(a))

$$S_{XX}(\omega) < 0, \quad \text{for } \omega_1 < |\omega| < \omega_2 \tag{4.34}$$

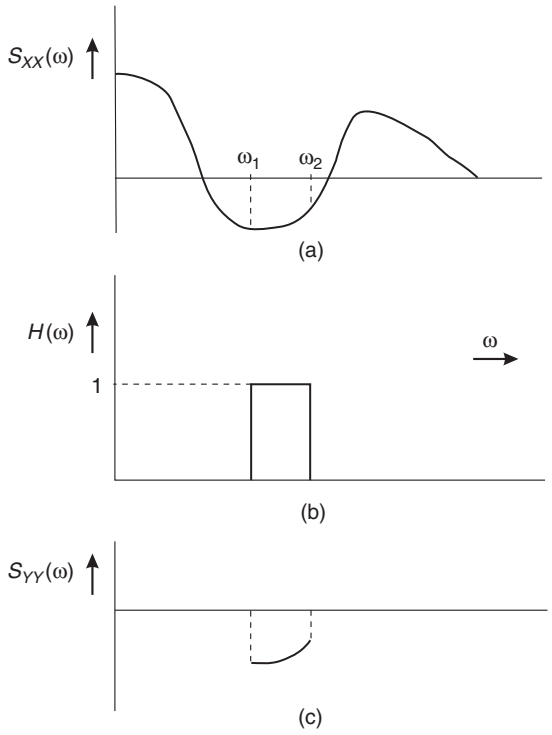


Figure 4.3 Noise filtering

Now consider an ideal bandpass filter with the transfer function (see Figure 4.3(b))

$$H(\omega) = \begin{cases} 1, & \omega_1 < |\omega| < \omega_2 \\ 0, & \text{for all remaining values of } \omega \end{cases} \tag{4.35}$$

If the process $X(t)$, with the power spectrum given in Figure 4.3(a), is applied to the input of this filter, then the spectrum of the output $Y(t)$ is as presented in Figure 4.3(c) and is described by

$$S_{YY}(\omega) = \begin{cases} S_{XX}(\omega), & \omega_1 < |\omega| < \omega_2 \\ 0, & \text{for all remaining values of } \omega \end{cases} \tag{4.36}$$

so that

$$S_{YY}(\omega) \leq 0, \quad \text{for all } \omega \tag{4.37}$$

However, this is impossible as (see Equations (4.28) and (3.8))

$$P_Y = R_{YY}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\omega)\, \mathrm{d}\omega \geq 0 \tag{4.38}$$

This contradiction leads to the conclusion that the starting assumption $S_{XX}(\omega) < 0$ must be wrong.

4.4 NOISE BANDWIDTH

In this section we present a few definitions and concepts related to the bandwidth of a process or a linear, time-invariant system (filter).

4.4.1 Band-Limited Processes and Systems

A process $X(t)$ is called a band-limited process if $S_{XX}(\omega) = 0$ outside certain regions of the ω axis. For a band-limited filter the same definition can be used, provided that $S_{XX}(\omega)$ is replaced by $H(\omega)$. A few special cases of band-limited processes and systems are considered in the sequel.

1. A process is called a lowpass process or baseband process if

$$S(\omega)\begin{cases} \neq 0, & |\omega| < W \\ = 0, & |\omega| > W \end{cases} \tag{4.39}$$

2. A process is called a bandpass process if (see Figure 4.4)

$$S(\omega)\begin{cases} \neq 0, & \omega_0 - W_1 \leq |\omega| \leq \omega_0 - W_1 + W \\ = 0, & \text{for all remaining values of } \omega \end{cases} \tag{4.40}$$

with

$$0 < W_1 < \omega_0 \tag{4.41}$$

3. A system is called a narrowband system if the bandwidth of that system is small compared to the frequency range over which the spectrum of the input process extends. A

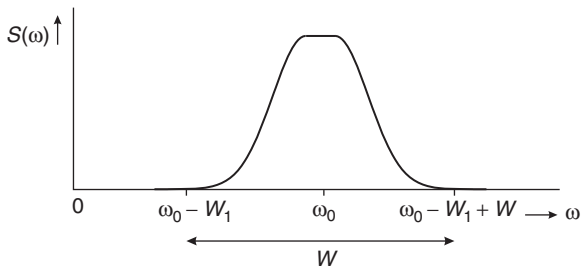


Figure 4.4 The spectrum of a bandpass process (only the region $\omega \geq 0$ is shown)

narrowband bandpass process is a process for which the bandwidth is much smaller than its central frequency, i.e. (see Figure 4.4)

$$W \ll \omega_0 \tag{4.42}$$

The following points should be noted:

- The definitions of processes 1 and 2 can also be used for systems if $S(\omega)$ is replaced by $H(\omega)$.
- In practical systems or processes the requirement that the spectrum or transfer function is zero in a certain region cannot exactly be met in a strict mathematical sense. Nevertheless, we will maintain the given names and concepts for those systems and processes for which the transfer function or spectrum has a negligibly low value in a certain frequency range.
- The spectrum of a bandpass process is not necessarily symmetrical about ω_0.

4.4.2 Equivalent Noise Bandwidth

Equation (4.28) is often used for practical applications. For that reason there is a need for a simplified calculation method in order to compute the noise power at the output of a filter. In this section we will introduce such a simplification.

To that end consider a lowpass system with the transfer function $H(\omega)$. Assume that the spectrum of the input process equals $N_0/2$ for all ω, with N_0 a positive, real constant (such a spectrum is called a white noise spectrum). The power at the output of the filter is calculated using Equation (4.28):

$$P_Y = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{N_0}{2}\,|H(\omega)|^2\,\mathrm{d}\omega \tag{4.43}$$

Now define an ideal lowpass filter as

$$H_{\mathrm{I}}(\omega) = \begin{cases} H(0), & |\omega| \le W_{\mathrm{N}} \\ 0, & |\omega| > W_{\mathrm{N}} \end{cases} \tag{4.44}$$

where W_{N} is a positive constant chosen such that the noise power at the output of the ideal filter is equal to the noise power at the output of the original (practical) filter. W_{N} therefore follows from the equation

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{N_0}{2}\,|H(\omega)|^2\,\mathrm{d}\omega = \frac{1}{2\pi}\int_{-W_{\mathrm{N}}}^{W_{\mathrm{N}}} \frac{N_0}{2}\,|H(0)|^2\,\mathrm{d}\omega \tag{4.45}$$

If we consider $|H(\omega)|^2$ to be an even function of ω, then solving Equation (4.45) for W_{N} yields

$$W_{\mathrm{N}} = \frac{\int_0^{\infty} |H(\omega)|^2 \mathrm{d}\omega}{|H(0)|^2} \tag{4.46}$$

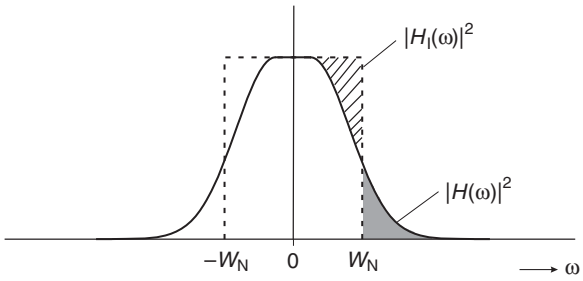


Figure 4.5 Equivalent noise bandwidth of a filter characteristic

W_N is called the equivalent noise bandwidth of the filter with the transfer function $H(\omega)$. In Figure 4.5 it is indicated graphically how the equivalent noise bandwidth is determined. The solid curve represents the practical characteristic and the dashed line the ideal rectangular one. The equivalent noise bandwidth is such that in this picture the dashed area equals the shaded area.

From Equations (4.43) and (4.45) it follows that the output power of the filter can be written as

$$P_Y = \frac{N_0}{2\pi} |H(0)|^2 W_N \tag{4.47}$$

Thus, it can be shown for the special case of white input noise that the integral of Equation (4.43) is reduced to a product and the filter can be characterized by means of a single number W_N as far as the noise filtering behaviour is concerned.

Example 4.3:

As an example of the equivalent noise bandwidth let us again consider the RC network presented in Figure 4.2. Using the definition of Equation (4.46) and the result of Example 4.2 (Equation (4.29)) it is found that $W_N = \pi/(2RC)$. This differs from the 3 dB bandwidth by a factor of $\pi/2$. It may not come as a surprise that the equivalent noise bandwidth of a circuit differs from the 3 dB bandwidth, since the definitions differ. On the other hand, both bandwidths are proportional to $1/(RC)$.

□

Equation (4.47) can often be used for the output of a narrowband lowpass filter. For such a system, which is analogous to Equation (4.43), the output power reads

$$P_Y \approx \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(0)\, |H(\omega)|^2 \, \mathrm{d}\omega \tag{4.48}$$

and thus

$$P_Y \approx \frac{S_{XX}(0)}{\pi} |H(0)|^2 W_N \tag{4.49}$$

The above calculation may also be applied to bandpass filters. Then it follows that

$$W_{\mathrm{N}} = \frac{\int_0^\infty |H(\omega)|^2 \, \mathrm{d}\omega}{|H(\omega_0)|^2} \tag{4.50}$$

Here ω_0 is a suitably chosen but arbitrary frequency in the passband of the bandpass filter, for instance the centre frequency or the frequency where $|H(\omega)|$ attains its maximum value. The noise power at the output is written as

$$P_Y = \frac{N_0}{2\pi} |H(\omega_0)|^2 W_{\mathrm{N}} \tag{4.51}$$

When once again the input spectrum is approximately constant within the passband of the filter, which often happens in narrowband bandpass filters, then

$$P_Y \approx \frac{S_{XX}(\omega_0)}{\pi} |H(\omega_0)|^2 W_{\mathrm{N}} \tag{4.52}$$

In this way we end up with rather simple expressions for the noise output of linear time-invariant filters.

4.5 SPECTRUM OF A RANDOM DATA SIGNAL

This subject is dealt with here since for the derivation we need results from the filtering of stochastic processes, as dealt with in the preceding sections of this chapter. Let us consider the random data signal

$$X(t) = \sum_n A[n] \, p(t - nT) \tag{4.53}$$

where the data sequence is produced by making a random selection out of the possible values of $A[n]$ for each moment of time nT. In the binary case, for example, we may have $A[n] \epsilon \{-1, +1\}$. The sequence $A[n]$ is supposed to be wide-sense stationary, where $A[n]$ and $A[k]$ in general will be correlated according to

$$\mathrm{E}\big[A[n] \, A[k]\big] = \mathrm{E}\,[A[n] \, A\big[n + m]\big] = \mathrm{E}\big[A[n] \, A[n - m]\big] = R[m] \tag{4.54}$$

The data symbols amplitude modulate the waveform $p(t)$. This waveform may extend beyond the boundaries of a bit interval. The random data signal $X(t)$ constitutes a cyclo-stationary process. We define a random variable Θ which is uniformly distributed on the interval $(0, T]$ and which is supposed to be independent of the data $A[n]$; this latter assumption sounds reasonable. Using this random variable and the process $X(t)$ we define the new process

$$\underline{X}(t) \triangleq X(t - \Theta) = \sum_n A[n] \, p(t - nT - \Theta) \tag{4.55}$$

Invoking Theorem 1 (see Section 2.2.2) we may conclude that this latter process is stationary. We model the process $\underline{X}(t)$ as resulting from exciting a linear time-invariant system having the impulse response $h(t) = p(t)$ by the input process

$$Y(t) = \sum_n A[n]\,\delta(t - nT - \Theta) \tag{4.56}$$

The autocorrelation function of $Y(t)$ is

$$\begin{aligned} R_{YY}(\tau) &= \mathrm{E}[Y(t)\,Y(t+\tau)] \\ &= \mathrm{E}\left[\sum_n \sum_k A[n]A[k]\,\delta(t - nT - \Theta)\,\delta(t - kT + \tau - \Theta)\right] \\ &= \sum_n \sum_k \mathrm{E}\big[A[n]\,A[k]\big]\,\mathrm{E}[\delta(t - nT - \Theta)\,\delta(t - kT + \tau - \Theta)] \\ &= \sum_n \sum_k R[k-n]\,\frac{1}{T}\int_0^T \delta(t - nT - \theta)\,\delta(t + \tau - kT - \theta)\,\mathrm{d}\theta \end{aligned} \tag{4.57}$$

For all values of T, t, τ and n there will be only one single value of k for which both $\delta(t - nT - \theta)$ and $\delta(t + \tau - kT - \theta)$ will be found in the interval $0 < \theta \leq T$. This means that actually the integral in Equation (4.57) is the convolution of two δ functions,which is well defined (see reference [7]). Applying the basic definition of δ functions (see Appendix E) yields

$$R_{YY}(\tau) = \sum_m \frac{R[m]}{T}\,\delta(\tau + mT) \tag{4.58}$$

where $m = k - n$. The autocorrelation function of $Y(t)$ is presented in Figure 4.6. Finally, the autocorrelation function of $\underline{X}(t)$ follows:

$$R_{\underline{XX}}(\tau) = R_{YY}(\tau) * h(\tau) * h(-\tau) = R_{YY}(\tau) * p(\tau) * p(-\tau) \tag{4.59}$$

The spectrum of the random data signal is found by Fourier transforming Equation (4.59):

$$S_{\underline{XX}}(\omega) = S_{YY}(\omega)\,|P(\omega)|^2 \tag{4.60}$$

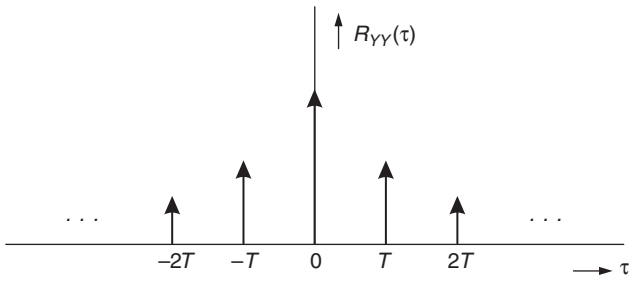


Figure 4.6 The autocorrelation function of the process $Y(t)$ consisting of a sequence of δ functions

where $P(\omega)$ is the Fourier transform of $p(t)$. Using this latter equation and Equation (4.58) the following theorem can be stated.

Theorem 8

The spectrum of a random data signal reads

$$S_{\underline{X}\underline{X}}(\omega) = \frac{|P(\omega)|^2}{T} \sum_{m=-\infty}^{\infty} R[m] \exp(\mathrm{j}\omega m T)$$
$$= \frac{|P(\omega)|^2}{T} \left\{ R[0] + 2 \sum_{m=1}^{\infty} R[m] \cos(\omega m T) \right\} \tag{4.61}$$

where $R[m]$ are the autocorrelation values of the data sequence, $P(\omega)$ is the Fourier transform of the data pulses and T is the bit time.

This result, which was found by applying filtering to a stochastic process, is of great importance when calculating the spectrum of digital baseband or digitally modulated signals in communications, as will become clear from the examples in the sequel. When applying this theorem, two cases should clearly be distinguished, since they behave differently, both theoretically and as far as the practical consequences are concerned. We will deal with the two cases by means of examples.

Example 4.4:

The first case we consider is the situation where the mean value of $\underline{X}(t)$ is zero and consequently the autocorrelation function of the sequence $A[n]$ has in practice a finite extent. Let us suppose that the summation in Equation (4.61) runs in that case from 1 to M.

As an important practical example of this case we consider the so-called polar NRZ signal, where NRZ is the abbreviation for non-return to zero, which reflects the behaviour of the data pulses. Possible values of $A[n]$ are $A[n] \in \{+1, -1\}$, where these values are chosen with equal probability and independently of each other. For the signal waveform $p(t)$ we take a rectangular pulse of width T, being the bit time. For the autocorrelation of the data sequence it is found that

$$R[0] = 1^2 \tfrac{1}{2} + (-1)^2 \tfrac{1}{2} = 1 \tag{4.62}$$
$$R[m] = 1^2 \tfrac{1}{4} + (-1)^2 \tfrac{1}{4} + (1)(-1)\tfrac{1}{4} + (-1)(1)\tfrac{1}{4} = 0, \quad \text{for } m \neq 0 \tag{4.63}$$

Substituting these values into Equation (4.61) gives the power spectral density of the polar NRZ data signal:

$$S_{\underline{X}\underline{X}}(\omega) = T \left(\frac{\sin \omega \frac{T}{2}}{\omega \frac{T}{2}} \right)^2 \tag{4.64}$$

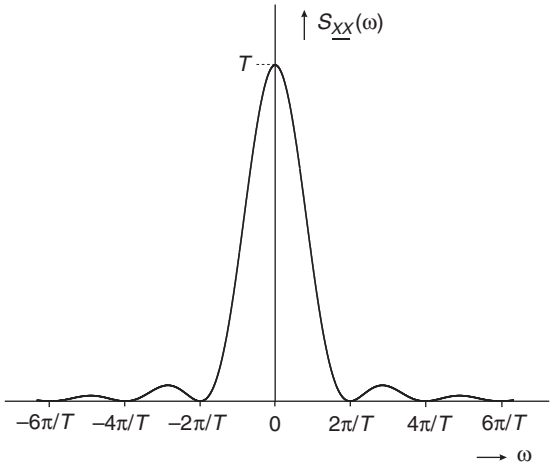


Figure 4.7 The power spectral density of the polar NRZ data signal

since the Fourier transform of the rectangular pulse is the well-known sinc function (see Appendix G). The resulting spectrum is shown in Figure 4.7.

The disadvantage of the polar NRZ signal is that it has a large value of its power spectrum near the d.c. component, although it does not comprise a d.c. component. On the other hand, the signal is easy to generate, and since it is a simplex signal (see Appendix A) it is power efficient. □

Example 4.5:

In this example we consider the so-called unipolar RZ (return to zero) data signal. This once more reflects the behaviour of the data pulses. For this signal format the values of $A[n]$ are chosen from the set $A[n] \in \{1, 0\}$ with equal probability and mutually independent. The signalling waveform $p(t)$ is defined by

$$p(t) \triangleq \begin{cases} 1, & 0 \leq t < T/2 \\ 0, & T/2 \leq t < T \end{cases} \tag{4.65}$$

It is easy to verify that the autocorrelation of the data sequence reads

$$R[m] = \begin{cases} \frac{1}{2}, & m = 0 \\ \frac{1}{4}, & m \neq 0 \end{cases} \tag{4.66}$$

Inserting this result into Equation (4.61) reveals that we end up with an infinite sum of complex exponentials:

$$S_{\underline{XX}}(\omega) = \frac{T}{4}\left(\frac{\sin \omega \frac{T}{4}}{\omega \frac{T}{4}}\right)^2 \left[\frac{1}{4} + \frac{1}{4}\sum_{m=-\infty}^{\infty} \exp(\mathrm{j}\omega m T)\right] \tag{4.67}$$

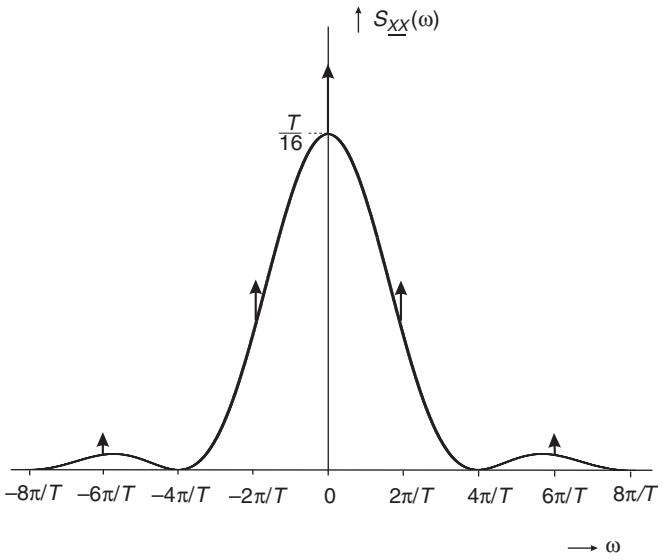


Figure 4.8 The power spectral density of the unipolar RZ data signal

However, this infinite sum of exponentials can be rewritten as

$$\sum_{m=-\infty}^{\infty} \exp(\mathrm{j}\omega mT) = \frac{2\pi}{T} \sum_{m=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi m}{T}\right) \tag{4.68}$$

which is known as the Poisson sum formula [7]. Applying this sum formula to Equation (4.67) yields

$$S_{\underline{XX}}(\omega) = \frac{T}{16}\left(\frac{\sin\omega\frac{T}{4}}{\omega\frac{T}{4}}\right)^2 \left[1 + \frac{2\pi}{T}\sum_{m=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi m}{T}\right)\right] \tag{4.69}$$

This spectrum has been depicted in Figure 4.8. Comparing this figure with that of Figure 4.7 a few remarks need to be made. First of all, the first null bandwidth of the RZ signal increased by a factor of two compared to the NRZ signal. This is due to the fact that the pulse width was reduced by the same factor. Secondly, a series of δ functions appears in the spectrum. This is due to the fact that the unipolar signal has no zero mean. Besides the large value of the spectrum near zero frequency there is a d.c. component. This is also discovered from the δ function at zero frequency. The weights of the δ functions scale with the sinc function (Equation (4.69)) and vanish at all zero-crossings of the latter.

□

Theorem 8 is a powerful tool for calculating the spectrum of all types of formats for data signals. For more spectra of data signals see reference [6].

4.6 PRINCIPLES OF DISCRETE-TIME SIGNALS AND SYSTEMS

The description of discrete-time signals and systems follows in a straightforward manner by sampling the functions that describe their continuous counterparts. Especially for the theory and conversion it is most convenient to confine the procedure to ideal sampling, i.e. by multiplying the continuous functions by a sequence of δ functions, where these δ functions are equally spaced in time.

First of all we define the discrete-time δ function as follows:

$$\delta[n] \triangleq \begin{cases} 1, & n = 0, \\ 0, & n \neq 0 \end{cases} \tag{4.70}$$

or in general

$$\delta[n-m] \triangleq \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases} \tag{4.71}$$

Based on this latter definition each arbitrary signal $x[n]$ can alternatively be denoted as

$$x[n] = \sum_{m=-\infty}^{\infty} x[m]\,\delta[n-m] \tag{4.72}$$

We will limit our treatment to linear time-invariant systems. For discrete-time systems we introduce a similar definition for linear time-invariant systems as we did for continuous systems, namely

$$\begin{aligned} &\text{if:} && x_i[n] \Rightarrow y_i[n] \\ &\text{then:} && \sum_i a_i x_i[n-m_i] \Rightarrow \sum_i a_i y_i[n-m_i] \end{aligned} \tag{4.73}$$

for any arbitrary set of constants a_i and m_i.

Also, the convolution follows immediately from the continuous case

$$y[n] = \sum_{m=-\infty}^{\infty} x[m]\,h[n-m] = \sum_{m=-\infty}^{\infty} h[n]\,x[n-m] = x[n] * h[n] \tag{4.74}$$

This latter description will be used for the output $y[n]$ of a discrete-time system with the impulse response $h[n]$, which has $x[n]$ as an input signal. This expression is directly deduced by sampling Equation (4.12), but can alternatively be derived from Equation (4.72) and the properties of linear time-invariant systems.

In many practical situations the impulse response function $h[n]$ will have a finite extent. Such filters are called finite impulse response filters, abbreviated as FIR filters, whereas the infinite impulse response filter is abbreviated to the IIR filter.

4.6.1 The Discrete Fourier Transform

For continuous signals and systems we have a dual frequency domain description. Now we look for discrete counterparts for both the Fourier transform and its inverse transform, since

then these transformations can be performed by a digital processor. When considering for discrete-time signals the presentation by means of a sequence of δ functions

$$x[n] = \sum_{n=-\infty}^{\infty} x(t)\,\delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s)\,\delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x[n]\,\delta(t - nT_s) \tag{4.75}$$

where $1/T_s$ is the sampling rate, the corresponding Fourier transform of the sequence $x[n]$ is easily achieved, namely

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]\,\exp(-\mathrm{j}\omega n T_s) \tag{4.76}$$

Due to the discrete character of the time function its Fourier transform is a periodic function of frequency with period $2\pi/T_s$. The inverse transform is therefore

$$x[n] = \frac{T_s}{2\pi}\int_{-\pi/T_s}^{\pi/T_s} X(\omega)\exp(\mathrm{j}n\omega T_s)\,\mathrm{d}\omega \tag{4.77}$$

In Equations (4.76) and (4.77) the time domain has been discretized. Therefore, the operations are called the discrete-time Fourier transform (DTFT) and the inverse discrete-time Fourier transform (IDTFT), respectively. However, the frequency domain has not yet been discretized in those equations. Let us therefore now introduce a discrete presentation of ω as well. We define the radial frequency step as $\Delta\omega \triangleq 2\pi/T$, where T still has to be determined. Moreover, the number of samples has to be limited to a finite amount, let us say N. In order to arrive at a self-consistent discrete Fourier transform this number has to be the same for both the time and frequency domains. Inserting this into Equation (4.76) gives

$$X[k] = X\left(k\frac{2\pi}{T}\right) = \sum_{n=0}^{N-1} x[n]\,\exp\left(-\mathrm{j}k\frac{2\pi}{T}nT_s\right) \tag{4.78}$$

If we define the ratio of T and T_s as $N \triangleq T/T_s$, then

$$X[k] = \sum_{n=0}^{N-1} x[n]\,\exp\left(-\mathrm{j}2\pi\frac{kn}{N}\right) \tag{4.79}$$

Inserting the discrete frequency and limited amount of samples as defined above into Equation (4.77) and approximating the integral by a sum yields

$$x[n] = \frac{T_s}{2\pi}\sum_{k=0}^{N-1} X[k]\,\exp\left(\mathrm{j}2\pi\frac{nk}{T}T_s\right)\frac{2\pi}{T} \tag{4.80}$$

or

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\ \exp\left(\mathrm{j}2\pi\frac{nk}{N}\right) \tag{4.81}$$

The transform given by Equation (4.79) is called the discrete Fourier transform (abbreviated DFT) and Equation (4.81) is called the inverse discrete Fourier transform (IDFT). They are a discrete Fourier pair, i.e. inserting a sequence into Equation (4.79) and in turn inserting the outcome into Equation (4.81) results in the original sequence, and the other way around. In this sense the deduced set of two equations is self-consistent. It follows from the derivations that they are used as discrete approximations of the Fourier transforms. Modern mathematical software packages such as Matlab comprise special routines that perform the DFT and IDFT. The algorithms used in these packages are called the fast Fourier transform (FFT). The FFT algorithm and its inverse (IFFT) are just DFT, respectively IDFT, but are implemented in a very efficient way. The efficiency is maximum when the number of samples N is a power of 2. For more details on DFT and FFT see references [10] and [12].

Example 4.6:

It is well known from Fourier theory that the transform of a rectangular pulse in the time domain corresponds to a sinc function in the frequency domain (see Appendices E and G). Let us check this result by applying the FFT algorithm of Matlab to a rectangular pulse. Before programming the pulse, a few peculiarities of the FFT algorithm should be observed. First of all, it is only based on the running variables n and k, both running from 0 to $N-1$, which means that no negative values along the x axis can be presented. Here it should be emphasized that in Matlab vectors run from 1 to N, which means that when applying this package the x axis is actually shifted over one position. Another important property is that since the algorithm both requires and produces N data values, Equations (4.79) and (4.81) are periodic functions of respectively k and n, and show a periodicity of N. Thanks to this periodicity the negative argument values can be displayed in the second half of the period. This actually means that in Figure 4.9(a) the rectangular time function is centred about zero. Similarly, the frequency function as displayed in Figure 4.9(b) is actually centred about zero as well. In this figure it has been taken that $N = 256$.

In Figure 4.10 the functions are redrawn with the second half shifted over one period to the left. This results in functions centred about zero in both domains and reflects the well-known result from Fourier theory. It will be clear that the actual time scale and frequency scale in this figure are to be determined based on the actual width of the rectangular pulse.

□

From a theoretical point of view it is impossible for both the time function and the corresponding frequency function to be of finite extent. However, one can imagine that if one of them is limited the parameters of the DFT are chosen such that the transform is a good approximation. Care should be taken with this, as shown in Figure 4.11. In this figure the rectangular pulse in the time domain has been narrowed, which results in a broader function

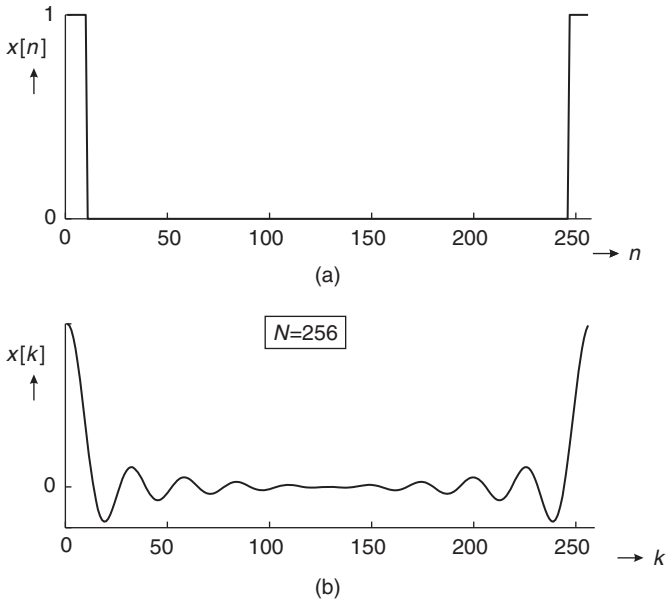


Figure 4.9 (a) The FFT applied to a rectangular pulse in the time domain; (b) the transform in the frequency domain

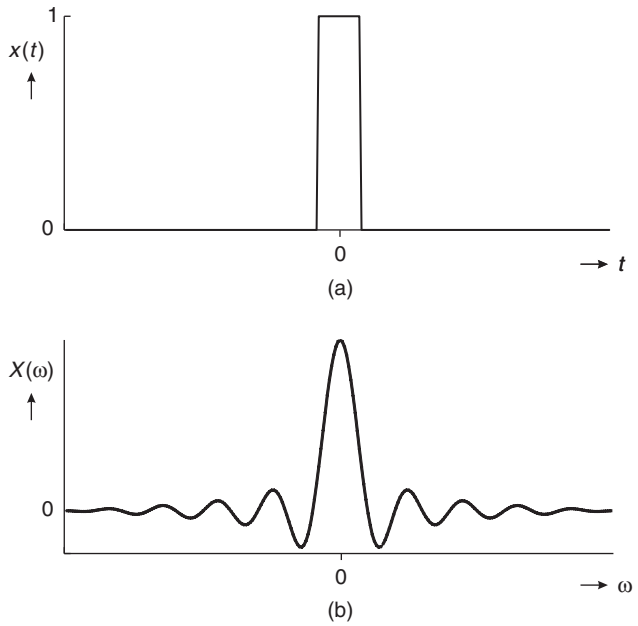


Figure 4.10 (a) The shifted rectangular pulse in the time domain; (b) the shifted transform in the frequency domain

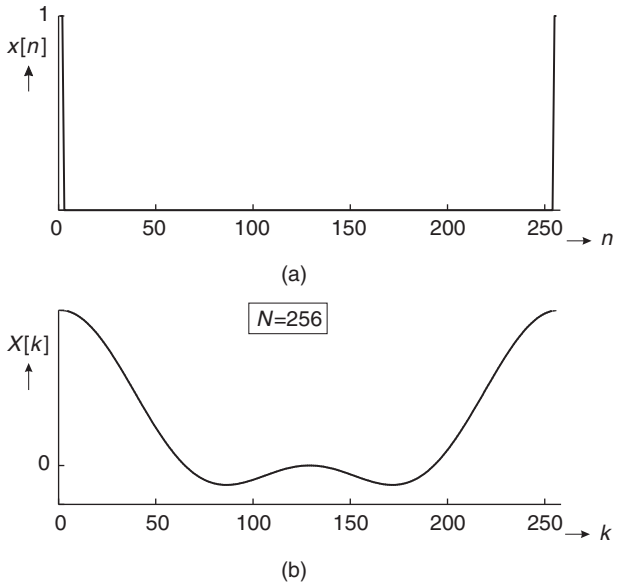


Figure 4.11 (a) Narrowed pulse in the time domain; (b) the FFT result, showing aliasing

in the frequency domain. In this case two adjacent periods in the frequency domain overlap; this is called aliasing.

Although the two functions of Figure 4.11 form an FFT pair, the frequency domain result from Figure 4.11(b) shows a considerable distortion compared to the Fourier transform, which is rather well approximated by Figure 4.9(b). It will be clear that this aliasing can in such cases be prevented by increasing the number of samples N. This has to be done in this case by keeping the number of samples of value 1 the same, and inserting extra zeros in the middle of the interval.

4.6.2 The z-Transform

An alternative approach to deal with discrete-time signals and systems is by setting $\exp(j\omega T_s) \triangleq z$ in Equation (4.76). This results in the so-called z-transform, which is defined as

$$\tilde{X}(z) \triangleq \sum_{n=-\infty}^{\infty} x[n]\, z^{-n} \tag{4.82}$$

Comparing this with Equation (4.76) it is concluded that

$$\tilde{X}(\exp(j\omega T_s)) = X(\omega) \tag{4.83}$$

Since Equation (4.82) is exactly the same as the discrete Fourier transform, only a different notation has been introduced, the same operations used with the Fourier transform are

allowed. If we consider Equation (4.82) as the z-transform of the input signal to a linear time-invariant discrete-time system, when calculating the z-transform of the impulse response

$$\tilde{H}(z) = \sum_{n=-\infty}^{\infty} h[n]\, z^{-n} \tag{4.84}$$

the z-transform of the output is found to be

$$\tilde{Y}(z) = \tilde{H}(z)\, \tilde{X}(z) \tag{4.85}$$

A system is called stable if it has a bounded output signal when the input is bounded. A discrete-time system is a stable system if all the poles of $\tilde{H}(z)$ lie inside the unit circle of the z plane or in terms of the impulse response [10]:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty \tag{4.86}$$

The z-transform is a very powerful tool to use when dealing with discrete-time signals and systems. This is due to the simple and compact presentation of it on the one hand and the fact that the coefficients of the different powers z^{-n} are identified as the time samples at nT_s on the other hand. For further details on the z-transform see references [10] and [12].

Example 4.7:

Consider a discrete-time system with the impulse response

$$h[n] = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{4.87}$$

and where $|a| < 1$. The sequence $h[n]$ has been depicted in Figure 4.12 for a positive value of a. The z-transform of this impulse response is

$$\tilde{H}(z) = 1 + az^{-1} + a^2 z^{-2} + \cdots = \frac{1}{1 - az^{-1}} \tag{4.88}$$

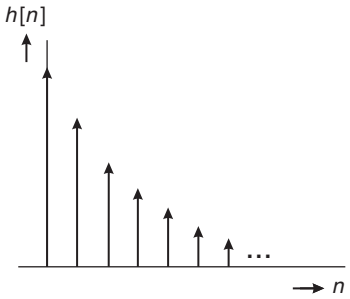


Figure 4.12 The sequence a^n with $0 < a < 1$

Let us suppose that this filter is excited with an input sequence

$$x[n] = \begin{cases} b^n, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{4.89}$$

with $|b| < 1$ and $b \neq a$. Similar to Equation (4.88) the z-transform of this sequence is

$$\tilde{X}(z) = 1 + bz^{-1} + b^2 z^{-2} + \cdots = \frac{1}{1 - bz^{-1}} \tag{4.90}$$

Then the z-transform of the output is

$$\tilde{Y}(z) = \tilde{H}(z)\,\tilde{X}(z) = \frac{1}{1 - az^{-1}} \frac{1}{1 - bz^{-1}} \tag{4.91}$$

The time sequence can be recovered from this by decomposition into partial fractions:

$$\frac{1}{1 - az^{-1}} \frac{1}{1 - bz^{-1}} = \frac{a}{a - b} \frac{1}{1 - az^{-1}} - \frac{b}{a - b} \frac{1}{1 - bz^{-1}} \tag{4.92}$$

From this the output sequence is easily derived:

$$y[n] = \begin{cases} \dfrac{a}{a - b} a^n - \dfrac{b}{a - b} b^n, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{4.93}$$

□

As far as the realization of discrete-time filters is concerned two main types are distinguished, namely the non-recursive filter structure and the recursive. The realization of the non-recursive filter is quite straightforward and the structure is given in Figure 4.13; it is also called the transversal filter or tapped delay line filter. The boxes represent delays of T_s seconds and the outputs are multiplied by a_n. The delayed and multiplied outputs are added

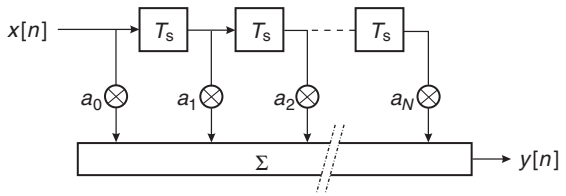


Figure 4.13 The structure of the discrete-time non-recursive filter, transversal filter or tapped delay line filter

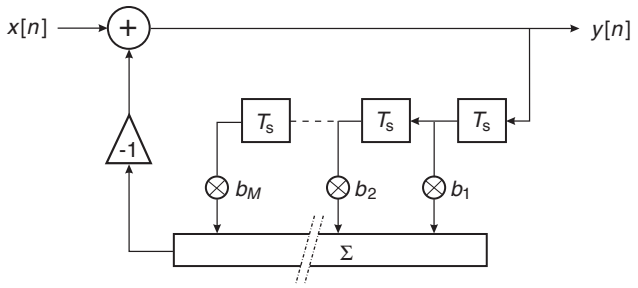


Figure 4.14 The structure of the discrete-time recursive filter

to form the filter output sequence $y[n]$. It is easy to understand that the transfer function of the filter in the z domain is described by the polynomial

$$\tilde{H}(z) = \tilde{A}(z) = a_0 + a_1 z^{-1} + \cdots + a_N z^{-N} = \sum_{n=0}^{n=N} a_n z^{-n} \tag{4.94}$$

From the structure it follows that it is a finite impulse response (FIR) filter and there is a simple and direct relation between the multiplication factors and the polynomial coefficients. An FIR filter is inherently stable.

The recursive filter is based on a similar tapped delay line filter, which is in a feedback loop depicted in Figure 4.14. The transfer function of the feedback filter is

$$\tilde{B}(z) = b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M} \tag{4.95}$$

and from the figure it is easily derived that the transfer function of the recursive filter is

$$\tilde{H}(z) \triangleq \frac{\tilde{Y}(z)}{\tilde{X}(z)} = \frac{1}{1 + \tilde{B}(z)} \tag{4.96}$$

As a rule this transfer function has an infinite impulse response, so it is an IIR filter. The stability of the recursive filter is guaranteed if the denominator polynomial $1 + \tilde{B}(z)$ in Equation (4.96) has no zeros outside the unit circle.

In filter synthesis the transfer function is often specified by means of a rational function; i.e. it is given as

$$\tilde{H}(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}} = \frac{\tilde{A}(z)}{1 + \tilde{B}(z)} \tag{4.97}$$

It can be seen that this function is realizable as the cascade of the FIR filter from Figure 4.13 and the IIR filter from Figure 4.14, as follows from Equations (4.94) to (4.96), while the same stability criterion is valid as for the IIR filter.

The Signal Processing Toolbox from Matlab comprises several commands for the analysis and design of discrete-time filters, one of which is `filter`, which calculates the output of filters described by Equation (4.97) when excited by a specified input sequence.

4.7 DISCRETE-TIME FILTERING OF RANDOM SEQUENCES

4.7.1 Time Domain Description of the Filtering

The filtering of a discrete-time stochastic process $X[n]$ by a discrete-time linear time-invariant system with the impulse response $h[n]$ is described by the convolution (see Equation (4.74))

$$Y[n] = \sum_{m=-\infty}^{\infty} X[m]\, h[n-m] = \sum_{m=-\infty}^{\infty} X[n-m]\, h[m] = X[n] * h[n] \tag{4.98}$$

It should be remembered that treatment is confined to real wide-sense stationary processes. Then the mean value of the output sequence is

$$\mathrm{E}\big[Y[n]\big] = \sum_{m=-\infty}^{\infty} \mathrm{E}\big[X[n-m]\big]\, h[m] = \overline{X} \sum_{m=-\infty}^{\infty} h[m] = \overline{X}\, \tilde{H}(1) \tag{4.99}$$

where $\tilde{H}(1)$ is the z-transform of $h[n]$ evaluated at $z = 1$, which follows from its definition (see Equation (4.84)).

The autocorrelation sequence of the output process, which can be proved to be wide-sense stationary as well, is

$$\begin{aligned} R_{YY}[m] &= \mathrm{E}[Y[n]\, Y\big[n+m\big]] \\ &= \mathrm{E}\left[\sum_{k=-\infty}^{\infty} X[n-k]\, h[k] \sum_{l=-\infty}^{\infty} X[n+m-l]\, h[l]\right] \\ &= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} \mathrm{E}[X[n-k]\, X[n+m-l]]\, h[k]\, h[l] \\ &= R_{XX}[m] * h[m] * h[-m] \end{aligned} \tag{4.100}$$

The cross-correlation sequence between the input and output becomes

$$\begin{aligned} R_{XY}[m] &= \mathrm{E}\big[X[n]\, Y[n+m]\big] \\ &= \mathrm{E}\left[X[n] \sum_{l} X[n+m-l]\, h[l]\right] \\ &= \sum_{l} \mathrm{E}\big[X[n]\, X[n+m-l]\big]\, h[l] \\ &= \sum_{l} R_{XX}[m-l]\, h[l] = R_{XX}[m] * h[m] \end{aligned} \tag{4.101}$$

In a similar way is derived

$$R_{YX}[m] = R_{XX}[m] * h[-m] \tag{4.102}$$

Moreover, the following relation exists:

$$R_{YY}[m] = R_{XY}[m] * h[-m] = R_{YX}[m] * h[m] \tag{4.103}$$

4.7.2 Frequency Domain Description of the Filtering

Once the autocorrelation sequence at the output of the linear time-invariant discrete-time system is known, the spectrum at the output follows from Equation (4.100):

$$S_{YY}(\omega) = S_{XX}(\omega)\, H(\omega)\, H^*(\omega) = S_{XX}(\omega)\, |H(\omega)|^2 \tag{4.104}$$

where the functions of frequency have to be interpreted as in Equation (4.76).

Using the z-transform, and assuming $h[n]$ to be real, we arrive at

$$\tilde{S}_{YY}(z) = \tilde{S}_{XX}(z)\, \tilde{H}(z)\, \tilde{H}(z^{-1}) \tag{4.105}$$

since for a real system $H^*(\omega) = H(-\omega)$ and consequently $\tilde{H}(z) = \tilde{H}(z^{-1})$.

Owing to the discrete-time nature of $Y[n]$ its spectrum is periodic. According to Equation (3.66) its power is denoted by

$$\begin{aligned} P_Y = \mathrm{E}\left[Y^2[n]\right] = R_{YY}[0] &= \frac{T_\mathrm{s}}{2\pi} \int_{-\pi/T_\mathrm{s}}^{\pi/T_\mathrm{s}} S_{XX}(\omega)\, |H(\omega)|^2 \, \mathrm{d}\omega \\ &= \frac{T_\mathrm{s}}{2\pi} \int_{-\pi/T_\mathrm{s}}^{\pi/T_\mathrm{s}} S_{XX}(\omega)\, |\tilde{H}(\exp(\mathrm{j}\omega T_\mathrm{s}))|^2 \, \mathrm{d}\omega \\ &= \sum_k \sum_l R_{XX}[k-l]\, h[k]\, h[l] \end{aligned} \tag{4.106}$$

The last line of this equation follows from Equation (4.100).

Example 4.8:

Let us consider the system with the z-transform

$$\tilde{H}(z) = \frac{1}{1 - az^{-1}} \tag{4.107}$$

with $|a| < 1$; this is the system introduced in Example 4.7. We assume that the system is driven by the stochastic process $X[n]$ with spectral density $\tilde{S}_{XX}(z) = 1$ or equivalently $R_{XX}[m] = \delta[m]$; later on we will call such a process white noise. Then the output spectral density according to Equation (4.105) is written as

$$\tilde{S}_{YY}(z) = \frac{z}{z-a} \frac{z^{-1}}{z^{-1}-a} = \frac{z}{(z-a)(1-az)} \tag{4.108}$$

Expanding this expression in partial fractions yields

$$\tilde{S}_{YY}(z) = \frac{1}{1-a^2}\,\frac{1}{1-az} + \frac{1}{1-a^2}\,az^{-1}\,\frac{1}{1-az^{-1}} \tag{4.109}$$

Next we expand in series the fractions with z and z^{-1}:

$$\begin{aligned}\tilde{S}_{YY}(z) &= \frac{1}{1-a^2}\left(1 + az + a^2z^2 + \cdots\right) + \frac{1}{1-a^2}\,az^{-1}(1 + az^{-1} + a^2z^{-2} + \cdots) \\ &= \frac{1}{1-a^2}\left(\sum_{n=-\infty}^{0} a^{-n}z^{-n} + \sum_{n=1}^{\infty} a^n z^{-n}\right) = \frac{1}{1-a^2}\sum_{n=-\infty}^{\infty} a^{|n|}z^{-n}\end{aligned} \tag{4.110}$$

The autocorrelation sequence of the output is easily derived from this, namely

$$R_{YY}[m] = \frac{1}{1-a^2}\,a^{|m|} \tag{4.111}$$

and in turn it follows immediately that

$$P_Y = R_{YY}[0] = \frac{1}{1-a^2} \tag{4.112}$$

The power spectrum is deduced from Equation (4.109), which for that purpose is rewritten as

$$\tilde{S}_{YY}(z) = \frac{1}{1 - a(z + z^{-1}) + a^2} \tag{4.113}$$

Inserting $z = \exp(\mathrm{j}\omega T_\mathrm{s})$ leads to the spectrum in the frequency domain

$$S_{YY}(\omega) = \tilde{S}_{YY}(\exp(\mathrm{j}\omega T_\mathrm{s})) = \frac{1}{1 - 2a\cos(\omega T_\mathrm{s}) + a^2} \tag{4.114}$$

In Figure 4.15 this spectrum has been depicted for $a = 0.8$ and a sampling interval time of 1 second. As a consequence the spectrum has a periodicity of 2π in the angular frequency domain. Clearly, this is a lowpass spectrum.

□

From this example it is concluded that the z-transform is a powerful tool for analysing discrete-time systems that are driven by a stochastic process. The transformation and its inverse are quite simple and the operations in the z domain are simply algebraic manipulations. Moreover, conversion from the z domain to the frequency domain is just a simple substitution.

Matlab comprises the procedures `conv` and `deconv` for multiplication and division of polynomials, respectively. The `filter` command in the Signal Processing Toolbox can do the same job (see also the end of Subsection 4.6.2). Moreover, the same toolbox comprises the command `freqz`, which, apart from the filter operation, also converts the result to the frequency domain.

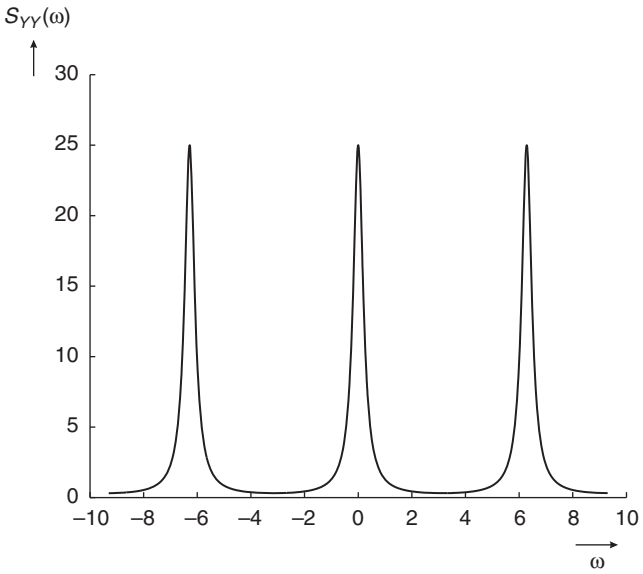


Figure 4.15 The periodic spectrum of the example with $a = 0.8$

4.8 SUMMARY

The input of a linear time-invariant system is excited with a wide-sense stationary process. In the time domain the output process is described as the convolution of the input process and the impulse response of the system, e.g. a filter. Next from this description such characteristics as mean value and correlation functions are to be determined, using the definitions given in Chapter 2. Although the expression for the autocorrelation function of the output process looks rather complicated, namely a twofold convolution, for certain processes this may result in a tractable description. Applying the Fourier transform and arriving at the power spectral density produces a much simpler expression. The probability density function of the output process can, in general, not be easily determined in analytical form from the input probability density function. There is an important exception for this; namely if the input process has a Gaussian probability density function then the output process also has a Gaussian probability density function.

The concept of equivalent noise bandwidth has been defined in order to arrive at an even more simple description of noise filtering in the frequency domain. The theory of noise filtering is applied to a specific stochastic process in order to describe the autocorrelation function and spectrum of random data signals.

Next, attention is paid to discrete-time signals and systems. Specials tools for that are dealt with. The discrete Fourier transform (DFT) and its inverse (IDFT) are derived and it is shown that this transform can serve as an approximation of the Fourier transform. Problems when applying the DFT for that purpose are indicated, including the ways used to avoid them. A closely related transform, namely the z-transform, appears to be more tractable for practical applications. The relation to the Fourier transform is quite simple. Finally, it is shown how to apply these transforms to filtering discrete-time processes by discrete-time systems.

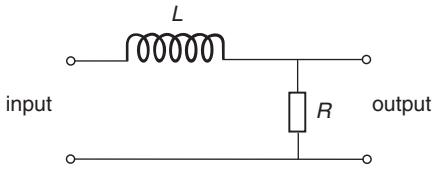


Figure 4.16

4.9 PROBLEMS

4.1 Consider the network given in Figure 4.16.

(a) Calculate the voltage transfer function. Make a plot of its absolute value on a double logarithmic scale using Matlab and with $R/L = 1$.

(b) Calculate the impulse response. Make a plot of it using Matlab.

(c) Calculate the response of the network to a rectangular input pulse that starts at $t = 0$, has height 1 and lasts for 2 seconds. Make a plot of the response using Matlab.

4.2 Derive the Fourier transform of the function

$$f(t) = \exp(-\alpha|t|)\,\cos(\omega_0 t)$$

Use Matlab to produce a plot of the transform with $\alpha = 1$ and $\omega_0 = 10$.

4.3 Consider the circuit given in Figure 4.17.

(a) Determine the impulse response of the circuit. Make a sketch of it.

(b) Calculate the transfer function $H(\omega)$.

(c) Determine and draw the response $y(t)$ to the input signal $x(t) = A\,\text{rect}\,[(t - \frac{1}{2}T)/T]$, where A is a constant. (See Appendix E for the definition of the rectangular pulse function $\text{rect}(\cdot)$).

(d) Determine and draw the response $y(t)$ to the input signal $x(t) = A\,\text{rect}\,[(t - T)/(2T)]$, where A is a constant.

4.4 A stochastic process $X(t) = A\sin(\omega_0 t - \Theta)$ is given, with A and ω_0 real, positive constants and Θ a random variable that is uniformly distributed on the interval $(0, 2\pi]$. This process is applied to a linear time-invariant network with the impulse response

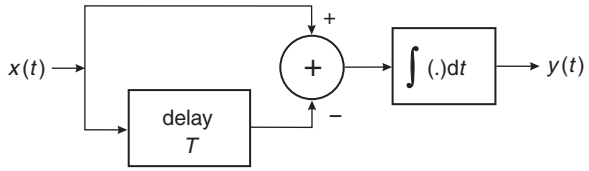


Figure 4.17

$h(t) = \mathrm{u}(t)\exp(-t/\tau_0)$. (The unit-step function $\mathrm{u}(t)$ is defined in Appendix E.) Here τ_0 is a positive, real constant. Derive an expression for the output process.

4.5 A wide-sense stationary Gaussian process with spectral density $N_0/2$ is applied to the input of a linear time-invariant filter. The impulse response of the filter is

$$h(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{elsewhere} \end{cases}$$

(a) Sketch the impulse response of the filter.

(b) Calculate the mean value and the variance of the output process.

(c) Determine the probability density function of the output process.

(d) Calculate the autocorrelation function of the output process.

(e) Calculate the power spectrum of the output process.

4.6 A wide-sense process $X(t)$ has the autocorrelation function $R_{XX}(\tau) = A^2 + B(1 - |\tau|/T)$ for $|\tau| < T$ and with A, B and T positive constants. This process is used as the input to a linear, time-invariant system with the impulse response $h(t) = \mathrm{u}(t) - \mathrm{u}(t - T)$.

(a) Sketch $R_{XX}(\tau)$ and $h(t)$.

(b) Calculate the mean value of the output process.

4.7 White noise with spectral density of $N_0/2$ V^2/Hz is applied to the input of the system given in Problem 4.1.

(a) Calculate the spectral density of the output.

(b) Calculate the mean quadratic value of the output process.

4.8 Consider the circuit in Figure 4.18, where $X(t)$ is a wide-sense stationary voltage process. Measurements on the output voltage process $Y(t)$ reveal that this process is Gaussian. Moreover, it is measured as

$$R_{YY}(\tau) = 9\exp(-\alpha|\tau|) + 25, \quad \text{where } \alpha = \frac{1}{RC}$$

(a) Determine the probability density function $f_Y(y)$. Plot this function using Matlab.

(b) Calculate and sketch the spectrum $S_{XX}(\omega)$ of the input process.

(c) Calculate and sketch the autocorrelation function $R_{XX}(\tau)$ of the input process.

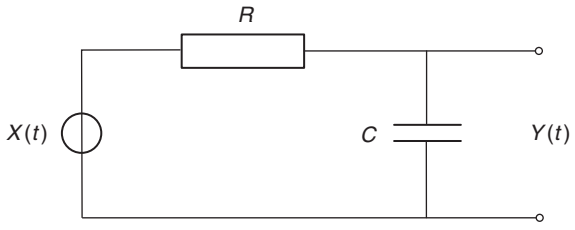


Figure 4.18

4.9 Two linear time-invariant systems have impulse responses of $h_1(t)$ and $h_2(t)$, respectively. The process $X_1(t)$ is applied to the first system and the corresponding response reads $Y_1(t)$. Similarly, the input process $X_2(t)$ to the second system results in the response $Y_2(t)$. Calculate the cross-correlation function of $Y_1(t)$ and $Y_2(t)$ in terms of $h_1(t)$, $h_2(t)$ and the cross-correlation function of $X_1(t)$ and $X_2(t)$, assuming $X_1(t)$ and $X_2(t)$ to be jointly wide-sense stationary.

4.10 Two systems are cascaded. A stochastic process $X(t)$ is applied to the input of the first system having the impulse response $h_1(t)$. The response of this system is $W(t)$ and serves as the input of the second system with the impulse response $h_2(t)$. The response of this second system reads $Y(t)$. Calculate the cross-correlation function of $W(t)$ and $Y(t)$ in terms of $h_1(t)$ and $h_2(t)$, and the autocorrelation function of $X(t)$. Assume that $X(t)$ is wide-sense stationary.

4.11 The process (called the signal) $X(t) = \cos(\omega_0 t - \Theta)$, with Θ uniformly distributed on the interval $(0, 2\pi]$, is added to white noise with spectral density $N_0/2$. The sum is applied to the RC network given in Figure 4.2.

(a) Calculate the power spectral densities of the output signal and the output noise.

(b) Calculate the ratio of the mean output signal power and the mean output noise power, the so-called signal-to-noise ratio.

(c) For what value of $\tau_0 = RC$ does this ratio become maximal?

4.12 White noise with spectral density of $N_0/2$ is applied to the input of a linear time-invariant system with the transfer function $H(\omega) = (1 + j\omega\tau_0)^{-2}$. Calculate the power of the output process.

4.13 A differentiating network may be considered as a linear time-invariant system with the transfer function $H(\omega) = j\omega$. If the input is a wide-sense stationary process $X(t)$, then the output process is $dX(t)/dt = \dot{X}(t)$. Show that:

(a) $R_{X\dot{X}}(\tau) = \dfrac{dR_{XX}(\tau)}{d\tau}$

(b) $R_{\dot{X}\dot{X}}(\tau) = -\dfrac{d^2R_{XX}(\tau)}{d\tau^2}$

4.14 To the differentiating network presented in Problem 4.13 a wide-sense stationary process $X(t)$ is applied. The corresponding output process is $Y(t)$.

(a) Are the random variables $X(t)$ and $Y(t)$, both considered at the same fixed time t, orthogonal?

(b) Are these random variables uncorrelated?

4.15 A stochastic process with the power spectral density of

$$S_{XX}(\omega) = \frac{1}{(1+\omega^2)^2}$$

is applied to a differentiating network.

(a) Find the power spectral density of the output process.

(b) Calculate the power of the derivative of $X(t)$.

4.16 Consider the stochastic process

$$Y(t) = \int_{t-T}^{t+T} X(\xi)\,\mathrm{d}\xi$$

where $X(t)$ is a wide-sense stationary process.

(a) Design a linear time-invariant system that produces the given relation between its input process $X(t)$ and the corresponding output process $Y(t)$.

(b) Express the autocorrelation function of $Y(t)$ in terms of that for $X(t)$.

(c) Express the power spectrum of $Y(t)$ in terms of that for $X(t)$.

4.17 Reconsider Problem 4.16.

(a) Prove that the given integration can be realized by a linear time-invariant filter with an impulse response $\mathrm{rect}[t/(2T)]$.

(b) Fill an array in Matlab with a harmonic function, let us say a cosine. Take, for example, 10 cycles of $1/(2\pi)$ Hz and increments of 0.01 for the time parameter. Plot the function.

(c) Generate a Gaussian noise wave with mean of zero and unit variance, and of the same length as the cosine using the Matlab command `randn`, add this wave to the cosine and plot the result. Note that each time you run the program a different wave is produced.

(d) Program a vector consisting of all ones. Convolve this vector with the cosine plus noise vector and observe the result. Take different lengths, e.g. equivalent to $2T = 0.05$, 0.10 and 0.20. Explain the differences in the different curves.

4.18 In FM detection the white additive noise is converted into noise with spectral density

$$S_{NN}(\omega) = \omega^2$$

and assume that this noise is wide-sense stationary. Suppose that the signal spectrum is

$$S_{XX}(\omega) = \begin{cases} \dfrac{S_0}{2}, & |\omega| \le W \\ 0, & |\omega| > W \end{cases}$$

and that in the receiver the detected signal is filtered by an ideal lowpass filter of bandwidth W.

(a) Calculate the signal-to-noise ratio.

In audio FM signals so-called pre-emphasis and de-emphasis filtering is applied to improve the signal-to-noise ratio. To that end prior to modulation and transmission the audio baseband signal is filtered by the pre-emphasis filter with the transfer function

$$H(\omega) = 1 + \mathrm{j}\frac{\omega}{W_\mathrm{p}}, \qquad W_\mathrm{p} < W$$

At the receiver side the baseband signal is filtered by the de-emphasis filter such that the spectrum is once more flat and equal to $S_0/2$.

(b) Make a block schematic of the total communication scheme.

(c) Sketch the different signal and noise spectra.

(d) Calculate the improvement factor of the signal-to-noise ratio.

(e) Evaluate the improvement in dB for the practical values: $W/(2\pi) = 15\,\text{kHz}$, $W_\text{p}/(2\pi) = 2.1\,\text{kHz}$.

4.19 A so-called nth-order Butterworth filter is defined by the squared value of the amplitude of the transfer function

$$|H(\omega)|^2 = \frac{1}{1 + (\omega/W)^{2n}}$$

where n is an integer, which is called the order of the filter. W is the -3 dB bandwidth in radians per second.

(a) Use Matlab to produce a set of curves that present this squared transfer as a function of frequency; plot the curves on a double logarithmic scale for $n = 1, 2, 3, 4$.

(b) Calculate and evaluate the equivalent noise bandwidth for $n = 1$ and $n = 2$.

4.20 For the transfer function of a bandpass filter it is given that

$$|H(\omega)|^2 = \frac{1}{1 + (\omega - \omega_0)^2} + \frac{1}{1 + (\omega + \omega_0)^2}$$

(a) Use Matlab to plot $|H(\omega)|^2$ for $\omega_0 = 10$.

(b) Calculate the equivalent noise bandwidth of the filter.

(c) Calculate the output noise power when wide-sense stationary noise with spectral density of $N_0/2$ is applied to the input of this filter.

4.21 Consider the so-called Manchester (or split-phase) signalling format defined by

$$p(t) = \begin{cases} 1, & 0 \le t < T/2 \\ -1, & T/2 \le t < T \end{cases}$$

where T is the bit time. The data symbols $A[n]$ are selected from the set $\{1, -1\}$ with equal probability and are mutually independent.

(a) Sketch the Manchester coded signal of the sequence 1010111001.

(b) Calculate the power spectral density of this data signal. Use Matlab to plot it.

(c) Discuss the properties of the spectrum in comparison to the polar NRZ signal.

4.22 In the bipolar NRZ signalling format the binary 1's are alternately mapped to $A[n] = +1$ volt and $A[n] = -1$volt. The binary 0 is mapped to $A[n] = 0$ volt. The bits are selected with equal probability and are mutually independent.

(a) Sketch the bipolar NRZ coded signal of the sequence 1010111001.

(b) Calculate the power spectral density of this data signal. Use Matlab to plot it.

(c) Discuss the properties of the spectrum in comparison to the polar NRZ signal.

4.23 Reconsider Example 4.6. Using Matlab fill in an array of size 256 with a rectangular function of width 50. Apply the FFT procedure to that. Square the resulting array and subsequently apply the IFFT procedure.

(a) Check the FFT result for aliasing.

(b) What in the time domain is the equivalence of squaring in the frequency domain?

(c) Check the IFFT result with respect to your answer to (b).

Now fill another array of size 256 with a rectangular function of width 4 and apply the FFT to it.

(d) Check the result for aliasing.

(e) Multiply the FFT result of the 50 wide pulse width that of the 4 wide pulse and IFFT the multiplication. Is the result what you expected in view of the result from (d)?

4.24 In digital communications a well-known disturbance of the received data symbols is the so-called intersymbol interference (see references [6], [9] and [11]). It is actually the spill-over of the pulse representing a certain bit to the time interval assigned to the adjacent pulses that represent different bits. This disturbance is a consequence of the distortion in the transmission channel. By means of proper filtering, called equalization, the intersymbol interference can be removed or minimized. Assume that each received pulse that represents a bit is sampled once and that the sampled sequence is represented by its z-transform $\tilde{R}(z)$. For an ideal channel, i.e. a channel that does not produce intersymbol interference, we have $\tilde{R}(z) = 1$.

Let us now consider a channel with intersymbol interference and design a discrete-time filter that equalizes the channel. If the z-transform of the filter impulse response is denoted by $\tilde{F}(z)$, then for the equalized pulse the condition $\tilde{R}(z)\,\tilde{F}(z) = 1$ should be satisfied. Therefore, in this problem the sequence $\tilde{R}(z)$ is known and the sequence $\tilde{F}(z)$ has to be solved to satisfy this condition. It appears that the Matlab command `deconv` is not well suited to solving this problem.

(a) Suppose that $\tilde{F}(z)$ comprises three terms. Show that the condition for equalization is equivalent to

$$\begin{bmatrix} r[0] & r[-1] & r[-2] \\ r[1] & r[0] & r[-1] \\ r[2] & r[1] & r[0] \end{bmatrix} \cdot \begin{bmatrix} f[-1] \\ f[0] \\ f[1] \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

(b) Consider a received pulse $\tilde{R}(z) = 0.1z + 1 - 0.2z^{-1} + 0.1z^{-2}$. Design the equalizer filter of length 3.

(c) As the quality factor with respect to intersymbol interference we define the 'worst case interference'. It is the sum of the absolute signal samples minus the desired sample value 1. Calculate the output sequence of the equalizer designed in (b) using `conv` and calculate its worst-case interference. Compare this with the unequalized worst-case interference.

(d) Redo the equalizer design for filter lengths 5 and 7, and observe the change in the worst-case interference.

4.25 Find the transfer function and filter structure of the discrete-time system when the following relations exist between the input and output:

(a) $y[n] + 2y[n-1] + 0.5y[n-2] = x[n] - x[n-2]$

(b) $4y[n] + y[n-1] - 2y[n-2] - 2y[n-3] = x[n] + x[n-1] - x[n-2]$

(c) Are the given systems stable?

Hint: use the Matlab command `roots` to compute the roots of polynomials.

4.26 White noise with spectral density of $N_0/2$ is applied to an ideal lowpass filter with bandwidth W.

(a) Calculate the autocorrelation function of the output process. Use Matlab to plot this function.

(b) The output noise is sampled at the time instants $t_n = n\pi/W$ with n integer. What can be remarked with respect to the sample values?

4.27 A discrete-time system has the transfer function $\tilde{H}(z) = 1 + 0.9z^{-1} + 0.7z^{-2}$. To the input of the system the signal with z-transform $\tilde{X}(z) = 0.7 + 0.9z^{-1} + z^{-2}$ is applied. This signal is disturbed by a wide-sense stationary white noise sequence. The autocorrelation sequence of this noise is $R_{NN}[m] = 0.01\,\delta[m]$.

(a) Calculate the signal output sequence.

(b) Calculate the autocorrelation sequence at the output.

(c) Calculate the maximum value of the signal-to-noise ratio. At what moment in time will that occur?

Hint: you can eventually use the Matlab command `conv` to perform the required polynomial multiplications. In this way the solution found using pencil and paper can be checked.

4.28 The transfer function of a discrete-time filter is given by

$$\tilde{H}(z) = \frac{1}{1 - 0.8z^{-1}}$$

(a) Use Matlab's `freqz` to plot the absolute value of the transfer function in the frequency domain.

(b) If the discrete-time system operates at a sampling rate of 1 MHz and a sine wave of 50 kHz and an amplitude of unity is applied to the filter input, compute the power of the corresponding output signal.

(c) A zero mean white Gaussian noise wave is added to the sine wave at the input such that the signal-to-noise ratio amounts to 0 dB. Compute the signal-to-noise ratio at the output.

(d) Use the Matlab command `randn` to generate the noise wave. Design and implement a procedure to test whether the generated noise wave is indeed approximately white noise.

(e) Check the analytical result of (c) by means of proper operations on the waves that are generated by Matlab.